

**PSYCHOLOGICAL ASSESSMENT
OF CHILDREN AND ADOLESCENTS
IN THE CLINICAL SETTING**

Editors

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**PSYCHOLOGICAL ASSESSMENT OF CHILDREN AND
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DEDICATION

*To our dear friend
and colleague*

Anna Oommen

PREFACE TO THE FIRST EDITION

The monograph is a product of years of clinical experience of the editors and research carried out by the postgraduate scholars. It is unique as it deals with assessment as well as intervention for attention and learning problems. The title is broad in scope while the coverage is narrow because of practical and logistic constraints. The focus is on the assessment aspects of psychopathology, attention, specific learning difficulties, memory and remediation.

The monograph is aimed at helping teachers, psychologists working in schools and Child Guidance Clinics, and teachers working in resource rooms for remedial work. The monograph also highlights the need to work with the child as a whole person.

The monograph would make good reading for postgraduates in clinical and educational psychology.

August 2002

Prof. M. Gourie-Devi
Director/Vice Chancellor
NIMHANS, Bangalore

PREFACE TO THE SECOND EDITION

The monograph has been well received by practising Psychologists, postgraduates in Clinical Psychology as well as by school teachers. The tests have been widely used for identification of behavioural/emotional problems and learning disabilities.

The content of the second edition remains the same with the chapters slightly reorganized.

I wish many more psychologists will find the monograph useful in their work with children.

December 2006

Prof. D. Nagaraja
Director/Vice Chancellor
NIMHANS, Bangalore

PREFACE TO THE THIRD EDITION

The monograph has been well received by practicing psychologists, postgraduates in clinical psychology, as well as by special educators and teachers. The tests have been widely used for identification of behavioral problems and learning disabilities.

The third edition has been expanded with new chapters related to phonological skills in specific learning disability and statistical properties in research on SLD. This edition also has a new chapter on a semi-projective test called The NIMHANS Sentence Completion Test for Children and Adolescents. Considering the paucity of tests for exploring adjustment of children, this test would be useful for mental health professionals dealing with children and adolescents.

I hope this edition would reach many more clinical psychologists and help in their assessment and intervention with children with psychological and learning problems.

February 2011

Prof. Satish Chandra
Director/Vice Chancellor
NIMHANS, Bangalore

PREFACE TO THE FOURTH EDITION

Welcome to the fourth edition of Psychological Assessment of Children and Adolescents in the Clinical Setting, known fondly to many as SLD (Specific Learning Disability) book. It is indeed heartening to know that the previous editions of the monograph were well received by practicing psychologists, special educators, teachers and post-graduates in clinical psychology. The tests in the monograph are widely used for assessment of learning disabilities, behavioural problems and adjustment of children and adolescents. It is also important to note that one of the tests in the monograph is recommended for assessment of learning disabilities by the Government of India in a gazette notification.

In the fourth edition, NIMHANS Index for SLD is revised to include tests for assessing learning disabilities of older children. The new tests in this edition encompass 'My Experience Scale' and 'Resilience Tool'.

I hope this edition would reach many more clinical psychologists in the country and help in their assessments and intervention with children and adolescents.

August 2020

Prof. B.N. Gangadhar
Director
NIMHANS, Bengaluru

ACKNOWLEDGEMENT

During a departmental presentation in September 2001, the then Director Dr. Gourie-Devi and Academic Council members suggested that the tools of assessment developed in the department could be published in the form of a monograph. Subsequently, the proceedings of a departmental workshop on psychological assessment of children in clinical setting, in addition to other relevant material culminated in publication of the first edition of this monograph. In presenting the fourth edition, our grateful thanks are due to all the contributors, and to the children who participated in the study. We fondly remember Dr. Anna Oommen who actively contributed to the first and second editions. The positive feedbacks received from all those who used the monograph are gratefully acknowledged.

We thank Dr. Manpreeth Kaur, Dr. Nalini N. R., Dr. Dhritimaan, Ms. Teena Paul, Ms. Ruchi, Mrs. Shubha Bhat, Ms. Ahil, Dr. Nithya Poornima and the management team of Sham School for their support. Timely editorial and proofreading assistance by Ms. Tejashwini A. and Dr. Prabhu Dev M. from the Publication Department, NIMHANS are sincerely acknowledged.

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August 2020

**Uma H.
Malavika Kapur**

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CHAPTER 1

INTRODUCTION

Uma Hirisave & Malavika Kapur

Psychological assessment is the process by which the clinical psychologist observes and evaluates the child's psycho-social problems, capabilities and limitations of the child. Three important questions asked are: (1) How is the child's functioning impaired and what are the consequences (2) What are the social factors that appear to maintain the problem? and (3) What therapy is best suited for the child? Psychological assessment focuses on description and analysis of the performance of a particular child in the problem situation. In order to achieve this proficiency, one also requires clinical competence.

For a comprehensive understanding of the child's problems, psychological tests can be used in addition to case history, Clinical interviews, behavioural observation and school reports. The tests provide objective information which can be used to formulate several hypotheses to explain the child's problems. It needs to be emphasized that as the psychological tests are one of the sources of information about the individual child, the test findings should not be viewed as an end point of psychological assessment, but rather should be used to interpret the clinical information in a more comprehensive manner.

When a child is referred for testing, several points need to be considered. The reasons for referral should be very clear. Even if the referral question is general, it is important to know what exactly is the need. The selection of the test depends on the nature of the clinical problems.

The examiner needs to be flexible in the clinical situation. For eg., if the child is very shy or non-compliant, then special procedures of psychological assessment should be used. There are several factors that can affect the child's performance in testing. These are child's physical and mental health problems, child's ability to communicate, and comprehend the instructions, motivation, anxieties about being tested, length and difficulty level of the test items and finally the rapport between the child and the examiner. These factors could lead to inconsistencies in the test performance.

Since the child is still developing, one should be extremely careful while interpreting the results. When giving feedback to the parent about the test results one has to be cautious and compassionate.

Broadly speaking, there are two types of tests, norm-referenced and criterion referenced. For a norm referenced test such as an intelligence test, usually data from the same normative group is taken for comparison, whereas in a criterion referenced test, items represent achievement of skills. For eg., a battery for assessing specific learning disabilities.

Different tests used in the clinical setting include tests for cognitive functions, rating scales and projective techniques. The commonly used tests of cognitive functions are tests of attention, memory, intelligence and scholastic abilities. Rating scales are used for assessing the severity of behavior problems and for evaluating the effectiveness of a particular intervention technique for a specific behaviour problem. Projective techniques such as Children's Apperception Test, Draw-A-Person Test, Raven's Controlled Projection Test and the Rorschach Inkblot Test provide

data about the intra psychic and interpersonal stresses. Rorschach Inkblot Test occasionally helps in confirming the diagnosis. In this compilation, the developmental psychopathology checklist which gives a comprehensive clinical profile of the child and sentence completion test for assessment of adjustment are discussed.

An important point to be kept in mind is the limitations of the tests. Firstly, since the child's performance can vary at different points, any generalizations based on the test data alone can further complicate matters for the child. Second, when tests are used routinely, the information is not very useful. Third, even though the test results help in planning intervention, the outcome does not depend on test findings only but on many factors in the child's environment. Fourth, as the priority is given to the clinical needs of the individual child, the basic purpose of the test is not to provide answers to research questions. However, test findings can help in planning research. Finally, there is a need to evaluate the existing tests for their adequacy and develop new tests suited to our culture.

To sum up, the psychological assessment using tests can contribute to the understanding of the child, provided the tests are used and interpreted properly. The test data should not be taken as an alternative to the clinical information but should be understood as a way of complementing the information obtained from clinical interviews with the child and parents.

It is essential to obtain a profile of the child using Developmental psychopathology checklist and Sentence completion test along with the information obtained from the tests of Attention, Memory and Specific learning disability index. Assessment of resilience and experiences would lead to a comprehensive understanding of the child's strengths and develop an effective intervention plan.

CHAPTER 2

DEVELOPMENTAL PSYCHOPATHOLOGY CHECKLIST FOR CHILDREN (DPCL)

**Malavika Kapur, Ida Barnabas, Reddy
M.V., Joe Rozario & Uma Hirisave**

This is a preliminary report on the construction of a Developmental Psychopathology Checklist (DPCL) for children in clinical settings in India. Cluster analysis was carried out on relevant demographic, psychosocial psychopathological parameters reported in 221 children attending a psychiatric facility. The seven clusters that emerged were : hyperkinesis, autism, conduct disorder, learning disorder, hysteria, emotional disorder and psychoses. The checklist with 124 items, had satisfactory inter rater reliability and validity. The research and clinical implications of Developmental Psychopathology Check List (DPCL) are discussed in this chapter.

The tools developed in the west, for assessment of behaviour problems and temperament, have been translated and tried out on normal and disturbed children by Malhotra et al (1986) and Malhotra and Malhotra (1988), in Indian setting. They found it more practical to have a single version of Child Behaviour Check List (Achenbach and Edelbrock, 1983) across age and gender, unlike the original versions.

The need to develop tools of assessment to study children has been a concern of psychologists for several decades. The earlier trend of using "watered down" versions of assessment tools used with adults for children has rightly given way to tools exclusively developed for children. Yet, tools which are constructed in the developmental context for behaviour

problems have been very few. The versions of Child Behavior Check List (CBCL) developed by Achenbach and Edelbrock (1983) are methodologically sound as they take into account a variety of clinical syndromes, differential effects of age and gender, with data gathered from multiple sources. Problems encountered in the use of CBCL in India as reported by Shenoy (1992) have been inapplicability of competency items and cut-off point, as well as large number of items.

Werry (1985) has described three major approaches to diagnosis in child psychiatry. These are medical, dimensional and ideographic. The present study follows the dimensional approach, where it is assumed that communal or nomothetic elements of psychopathology are reduceable to a number of dimensions or clusters along which every child can be placed. In addition, the present checklist goes beyond psychopathology to lend itself to an ideographic approach by providing information on developmental and family history, interpersonal relationships, stressors, temperament and supportive factors, to enable the clinician to adopt therapeutic strategies best suited for the child.

The study of child psychopathology from a developmental perspective ideally takes into account the nature of specific items suitable for the setting, age and gender differences, developmental history, temperament along with psychosocial correlates such as family interaction, stressor and social supports. In the clinical settings in India, there is a need to develop a screening tool to assess psychopathology in children, which is comprehensive, yet developmental in perspective and can be used with relatively little training. Such a tool can also form the basis for developing strategies for the management of a child. Thus, the DPCL is designed to bridge the gap between clinical practice and research.

The DPCL has 124 items and six sub sections. The sub sections are: (i) developmental history, (ii) developmental problems, (iii) psychopathology, (iv) psychosocial factors; (v) temperament, (vi) social supports and assets of the child.

Procedure

The steps involved in the development of the DPCL were: (i) item selection (ii) sample selection (m) cluster analysis (iv) reliability and (v) validation.

i) Item Selection

Items were selected based on research data on hyperkinesis by Oommen (1990), disorders of emotion and conduct by Daniel (1989), hysteria by Uma (1988), scholastic backwardness by John (1989), pervasive developmental disorders by Vythilingam (1991) and psychoses by Reddy (1991). Thus, the items considered sensitive enough to identify disorders in the Indian setting were specially compiled. A group of 100 children below the age of 16 years presenting at the child psychiatric clinic were assessed on the DPCL by four different investigators. Aspects of clarity, simplicity and capacity of the items to elicit symptoms in a wide range of psychiatric disorders (excluding mental retardation) were considered and items were modified, deleted or added, dependent on the findings.

ii) Sample Selection

A total of 221 child psychiatric cases were seen at the Child and Adolescent Psychiatry Unit at NIMHANS, Bengaluru. These cases were reported at the inpatient and outpatient facilities and had a wide range

of disorders. Mental retardation was an exclusion criteria. Children below 16 years were seen. In addition, 25 cases were seen by the independent raters as a reliability exercise and 45 cases were assessed on CBCL (Achenbach & Edelbrock 1983) and DPCL for validation purposes.

Description of the Check List (DPCL)

a) Developmental History (Items 1-10) : This consists of pre, peri and post-natal problems, possible brain injury, sensori-motor handicaps, developmental delays in motor, language, cognitive, emotional and social development.

b) Developmental Problems (Items 11-28) : This section elicits developmental problems such as clumsiness, breath holding spells, habits, speech and language problems, feeding, elimination, sleeping and sexual problems.

c) Psychopathology (Items 29-78) : Under this, externalizing problems of hyperactivity and conduct disturbance, scholastic problems, internalizing problems of the emotional, somatic and neurotic kind and psychoses are elicited.

d) Psychosocial Factors (Items 79-101) : In this section, the family history of illnesses, interactions within the family, child rearing practices, child's relationships with parents and siblings and relationships in the school setting are explored.

e) Temperamental Profile (Items 102-118) : This is a simple version with three major categories with special focus on the resilient or competent child. The dimensions measured are manageability, trust, dependence, sleep, appetite, activity level, emotionality, sociability and aggression.

The dimensions go beyond the western conceptualization of temperament and comprises the ancient Indian model of "trigunas".

f) Social supports and assets of the child (Items 119-124) : In this section details regarding the presence of helpful people within and outside the family, friends and special interests, talents and hobbies the child may have are elicited.

iii) Cluster Analysis

The cluster analytic approach is most suited to deal with classification problems, particularly in the confirmation or otherwise, of the traditional nosological groups of childhood disorders. An attempt has been made in the present study to demonstrate the utility of the DPCL for developing profiles of child and adolescent psychiatric disorders. A sample of 221 cases of a Child Guidance Population formed the material for the study. The seven commonly used hierarchical methods each with Euclidian distance and standardized Euclidian distance measures were employed using 78 (Items 1-78) symptoms of DPCL. The classification suggested by the Ward method (1963) was selected for further analysis. The application of Mojena's rule (1977) determined that there were seven optimum number of clusters present in the hierarchical classification. Two K-means Algorithms partitioning methods, namely the Forgy (1965) method and its variant proposed by Jancey (1966) were employed, using the centroids of the seven clusters obtained by the Ward method. This was carried out as there was no reallocation facilities in the hierarchical methods and poor initial classification may have cut through natural structure in the data. The partition classification obtained by the Forgy method was selected since the occurrence of two DPCL symptoms in the Jancey classification had no clinical meaningfulness. In addition, the

demographic variables would have better discrimination powers in the Forgy classification than that of Ward classification.

The seven clusters were named as follows :

Cluster 1	Emotion disorder	(ED) -	13.6 percent
Cluster 2	Hyperkinesis	(Hyp) -	10.4 percent
Cluster 3	Psychoses	(Psy) -	16.7 percent
Cluster 4	Learning disorder	(LD) -	16.7 percent
Cluster 5	Hysteria	(Hys) -	18.6 percent
Cluster 6	Conduct disorder	(CD) -	20.8 percent
Cluster 7	Autism	(Aut) -	3.2 percent

Description of the disorder and their association with developmental and psychosocial correlates :

The cluster analysis revealed that the frequency of occurrence of symptoms other variables in the seven clusters were statistically significant in relation to the various items of the check list as described below.

Cluster 1 : Emotion Disorder (ED)

The children manifested day dreaming, being withdrawn, talking very little, worrying, anxious, shyness, clinging, crying easily as the main features in the ED cluster and dizziness, fits and pulling sensations as somatic symptoms of the hysteria cluster. The child rearing practice was characterized by over indulgence and over expectation by the parents.

Cluster 2 : Hyperkinesia (Hyp)

In the demographic section, mother's education, urban residence and being the only child were significant variables. In the developmental history-section, difficult pregnancy of the mother, post-natal problems in the child, presence of epilepsy/head injury/infection, poor hearing, delays in gross and fine motor movements, speech/language, emotional social development and poor self-help skills were evident. The group had self-help developmental problems of clumsiness, speech and articulation, echolalia, receptive and expressive language difficulties, deviant language, inability to relate to people and to play with children. In the psychopathology section, restlessness and impulsivity in addition to being clinging were present. They had family history of mental retardation. Temperamentally, they were difficult to manage, dependent, not dependable, neither trusting nor trustworthy, not moral, manifesting high activity, low persistence and physical aggression.

Cluster 3 : Psychoses (Psy)

The symptoms manifested in this psychopathology section were hallucinations, catatonic features, delusions, talking and laughing to self, poor appetite and sleep, incontinence, loss of interest, psychomotor retardation, depression, pressure of ideas, irritability and elation. The significant background variables were presence of precipitating factors and poverty.

Cluster 4 : Learning disorder (LD)

The demographic data revealed the families to be of professional background. Developmental delays were present in gross motor, fine motor, and speech / language development, with speech and articulation

problems. Psychopathology section revealed poor attention, poor school performance with reading, writing, arithmetic difficulties and poor memory. Social supports were significantly evident with helpful persons and friends in the environment.

Cluster 5 : Hysteria (Hys)

The psychopathology section revealed somatic symptoms consisting of aches, pains, tiredness, fainting spells, fits and pulling sensations. Social support of friends was evident.

Cluster 6 : Conduct disorder (CD)

Significant demographic factor was the male gender. Developmental history had epilepsy, head injury / infection; psychopathology section had poor attention, distractibility, restlessness, impulsivity, stubbornness, disobedience, disruptiveness, quarrelsomeness, aggression, temper tantrums, truancy, lying and stealing as main manifestations in the CD sub-section. They also had school refusal, poor school performance, reading, writing and arithmetic difficulties in the learning disorder sub-section. There was family history of epilepsy and speech problems. In the interpersonal realm, sibling rivalry was evident. Temperamentally they were difficult to manage, undependable, insensitive to others, not trusting, and not moral. They were characterized by high activity, low persistence, emotionality, with physical and verbal aggression.

Cluster 7 : Autism (Aut)

Demographic data revealed mother's educational level and being the only child as significant. The developmental history had poor hearing. Developmental delays were present in gross and fine motor movements,

speech / language and emotional-social development, along with poor self-help skills. The developmental disorders were clumsiness, tics / mannerisms, speech / articulation problems, stuttering / stammering, elective mutism, receptive and expressive language difficulties, inability to relate to people and to play with children, along with feeding problems, impulsivity and primary enuresis and encopresis. There was family history of alcoholism. Temperamentally they were dependent and undependable.

To sum up, developmental history of illness affecting the central nervous system (CNS) and developmental delays and disorders were present in hyperkinesis, autism and learning disorders. In ED and hysteria only symptoms were manifest. In the CD cluster, developmental history of CNS damage, hyperkinetic and LD symptoms too were present in addition to CD symptoms. Temperamental difficulties were prominent in hyperkinesis and CD. Psychoses was a syndrome with a severe range of symptoms with precipitating factors and presence of poverty. The study revealed that though there were seven clusters, there was a good deal of overlap amongst them. In clinical practice, this overlap has to be recognized as it has important implications in planning strategies of management.

The demographic correlates of the seven clusters

Table 1

			ED (23)	Hyp (23)	Psy (37)	LD (37)	Hys (41)	CD (46)	AU (7)	
1.	Age (in years)	Mean SD	11.5 3.31	7.7 2.61	14.0 1.85	10.4 3.18	11.1 2.89	10.3 2.88	5.3 3.61	F= 17.159*
2.	Sex (Males)		56.7%	78.3%	62.2%	81.1%	43.9%	87.0%	57.1%	X ² = 23.235**
3.	Class (No. of Years)	Mean SD	5.7 2.78	1.9 1.75	6.2 2.66	4.1 3.0	5.0 2.48	3.9 2.35	1.0 2.45	F= 10.941**
4.	Father's Age (Years)	Mean SD	42.3 6.66	38.9 5.29	44.6 8.14	41.7 5.44	44.1 6.14	42.1 7.88	36.0 3.70	F= 3.249**
5.	Mother's Age (Years)	Mean SD	35.1 6.05	33.2 5.14	35.6 6.99	34.2 6.99	36.0 6.07	33.7 7.34	31.3 5.01	F= 1.282
6.	Father's Education (in Years)	Mean SD	7.19 5.38	11.7 4.56	6.2 5.46	10.6 5.62	8.4 5.94	8.8 5.37	10.9 6.36	F= 3.409**
7.	Mother's Education (in Years)	Mean SD	6.5 5.38	11.0 5.39	3.5 4.60	9.0 4.65	5.7 5.87	5.8 5.59	9.6 7.0	F= 6.651**
8.	Locality (Urban)		63.3%	91.3%	48.7%	75.7%	70.7%	73.9%	57.1%	X ² = 14.827*
9.	Family Occupation (Professional & Service)		33.3%	56.5%	27.0%	62.2%	43.9%	45.7%	67.1%	X ² = 12.568
10.	Sibling Pattern (only child)		3.3%	34.8%	2.7%	10.8%	9.8%	15.2%	42.9%	X ² = 22.315**

The above table showed some interesting developmental trends. These were:

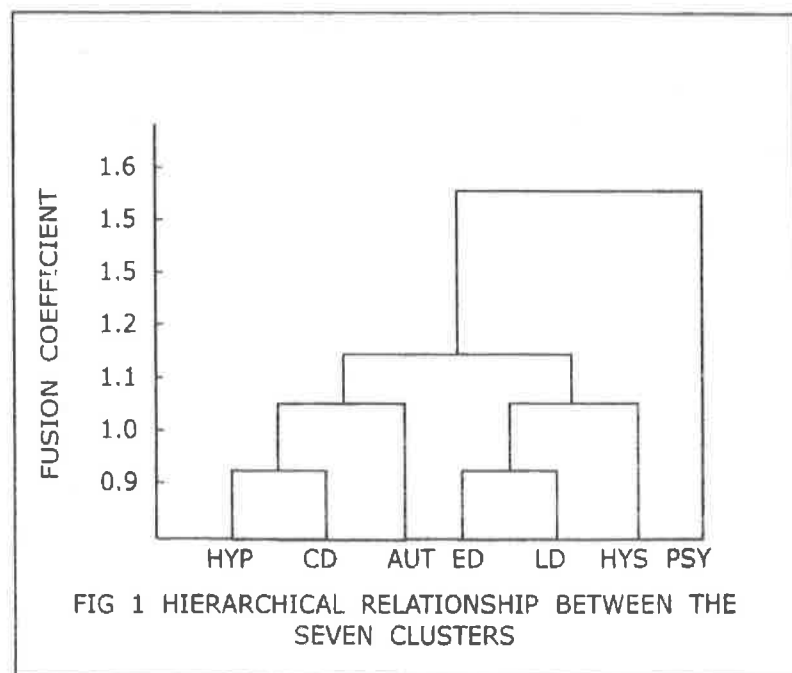
i) Age trends : Autism and psychoses were at the extreme ends of the age spectrum, supporting a bimodal distribution of severe psychiatric problems in infancy and childhood. Next to the autistic children, the hyperkinetics were the youngest with the mean age of 7.7 years. While in the other disorders the mean ages were between 10 and 11.5 years.

ii) Gender trends : Male predominance occurred consistently, with CD, LD and hyperkinesis (87.1, 81.0 and 78.3 percents respectively); psychoses

with 62.2 percent, autism with 57.1 percent and ED with 56.7 percent. The only exception was hysteria with 43.9 percent in males.

iii) Parental characteristics : Father's age was significant and was in the range of 36.0 and 44.10 years in the seven groups. Parental educational levels too were significant with higher mean years of schooling in the parents of hyperkinetics, learning disordered and autistic children in comparison to the others. Most of them were urban families with professional background.

iv) Sibling status : Sibling status of being the only child was found in hyperkinesis and autistic groups. This appeared to suggest that these disorders with early onset of severe behavioural problems possibly discouraged the parents from having more children. The hierarchical relationship is displayed in the dendrogram in Fig.1.



The dendrogram highlighted the following :

Psychoses formed an independent cluster, ED, LD and hysteria were close to each other, while on the other hand hyperkinesis, autism and CD were placed close together. Cluster analysis using sociodemographic and psychosocial variables in addition to items related to psychopathology offered the following insights.

Hyperkinesis and autism shared demographic characteristics of sibling status of being the only child and having mothers with higher education, developmental delays and problems with early onset. Conduct disorder occurred predominantly in males and shared a large number of symptoms with hyperkinesis and LD. Temperamental difficulties of similar nature were present in children with CD who also had problems with teachers, peers and sibling rivalry, with inconsistency in disciplining by the parents. Hysteria and ED had in common some somatic complaints. In addition, ED had symptoms of being socially withdrawn, anxious and worrying with over expecting / over indulgent parents while children with hysterical symptoms had adequate social support. The LD group shared with hyper kinetics developmental delays and attentional problems and in addition had social support. Unlike CD and hyper kinetics the LD, ED and hysteria groups had no temperamental difficulties.

In India, in a ICMR multicentre study (1984), cluster analysis of 434 cases yielded six clusters. These were : Psychoses, hyper kinetics anxiety, conduct disorder, hysteria and scholastic problems. In the present study, an autism cluster emerged because of inclusion of parameters of development.

iv) Reliability

Twenty five child psychiatric cases were rated by two independent raters who were qualified clinical psychologists (IB & JR). Kappa, a chance corrected scaled reliability measure was employed. The Kappa coefficient was used for item analysis and revealed poor consistency between the two raters for 4 items ranging from -0.11 to +1.00. These were : mothers health during pregnancy and emotional and social development of the child in the section on developmental history, feeding problems in the section on developmental disorders and withdrawal in the psychoses subsection of the Psychopathology section. Consequently these items were modified to render them more understandable. The reliability for the entire check list was tested by employing Interclass Correlation Co-efficient (ICC) via analysis of variance. The ICC was 0.965 (significant at 0.001 level).

v) Validation

The DPCL was subjected to external criteria of validation against the CBCL.

Table 2

Comparison of 45 cases on Child Behaviour Check List (CBCL) and Developmental Psychopathology Check List (DPCL)

		CBCL				DPCL		
		I	E	LD	ED	Hys	Hyp	CD
CBCL	I	1.000	0.671	0.006	0.291*	0.138	0.228	0.167
	E	0.671	1.000	0.123	0.137	-0.046	0.431**	0.598**
DPCL	LD	0.006	0.123	1.000	0.357**	-0.060	0.373**	0.444**
	ED	0.291	0.137	0.357	1.000	0.295*	0.169	0.114
	Hys	0.138	-0.046	-0.060	0.295	1.000	-0.249	-0.229
	Hyp	0.228	0.431	0.373	0.169	-0.249	1.000	0.471**
	CD	0.167	0.598	0.444	0.114	-0.229	0.471	1.000

* Significant at 0.05 level

* Significant at 0.01 level

The correlation coefficient was applied between the two broad band spectrums of Externalizing and Internalizing disorders of the CBCL with 5 subsections of the DPCL which were expected to be similar to Externalizing and Internalizing dimensions.

The table shows significant positive correlation between emotion disorder (DPCL) and Internalizing disorder of CBCL. Hyperkinesis and conduct disorder (DPCL) and Externalizing disorder (CBCL) have similar positive correlation.

In addition, emotion disorder had positive and significant correlation with hysteria, in the DPCL, but not Internalizing disorder of CBCL of which somatic symptoms formed a sub scale. This appears to suggest that though both hysteria and somatization are internalizing disorders, the two are distinct entities. This highlights the need to view hysteria as a distinct syndrome in the Indian setting.

Learning disorder which emerged as an independent cluster in the present study, had low correlation with Externalizing and Internalizing disorders of the CBCL. On the DPCL it had high positive correlation with emotion disorder on one hand and hyperkinesis and conduct disorder on the other. Learning disorder once again is a syndrome to reckon with in the Indian setting.

In addition, external validation of the check list is substantiated in three findings already described. These are :

i) The demographic data showed significant relationship between the seven clusters and ages of the child and the father, education of the parents, urban residence and sibling status.

ii) The cluster analysis of all the items inclusive of developmental history, psychopathology and psychosocial correlates showed up in the respective clusters in keeping with the general trend in research findings in the Indian settings.

iii) The relationship with DPCL and the sub categories are validated as positive correlation was obtained between emotional disorder and hysteria; learning disorder, equally with emotional disorder, hyperkinesis and conduct disorder, confirming the clinical observations.

b) Internal validation of the seven clusters : Of the 78 items of the psychopathology section of the check list used for identification of the clusters in the present classification, 68 binary variables acquired significant discriminatory power (Aldenderfer and Blaschfield, 1984).

Table 2a
Frequency of occurrence in percentages associated with seven clusters in the developmental problems / disorders section

SI. No.	Developmental problems/disorders	ED (30)	HYP (23)	LD (37)	PSY (37)	HYS (41)	CD (46)	AUT (07)	Total (221)
1.	Problems of Pregnancy in Mother	10.0	47.8*	2.7	27.0	12.2	23.9	42.9	19.9
2.	Post Natal problem of Child	6.7	34.8*	2.7	16.2	7.3	8.7	28.6	11.7
3.	Epilepsy, Head Injury infections	26.7	52.2*	5.4	21.6	12.2	65.2	14.3	29.9
4.	Poor Vision	3.3	8.7	0.0	10.8	0.0	6.5	14.3	5.0
5.	Poor hearing	0.0	17.4*	2.7	5.4	0.0	4.4	28.6*	5.0
6.	Gross Motor	0.0	39.1*	2.7	29.7*	2.4	2.2	42.9*	11.8
7.	Fine Motor	0.0	43.5*	2.7	32.4*	7.3	2.2	57.1*	14.0
8.	Speech/Language	13.3	87.0*	5.4	37.8*	7.3	10.9	57.1*	23.5

Table 2a
Frequency of occurrence in percentages associated with seven clusters in the developmental problems / disorders section

SI. No.	Developmental problems/disorders	ED (30)	HYP (23)	LD (37)	PSY (37)	HYS (41)	CD (46)	AUT (07)	Total (221)
9.	Emotional Social	3.3	73.9*	2.7	18.9	7.3	4.3	57.1*	15.8
10.	Self Help	6.7	65.2*	5.4	29.7	9.8	19.6	100.0	22.6
11.	Clumsiness	10.0	34.8*	5.4	16.2	0.0	15.2	85.7*	14.5
12.	Breath Holding	13.3	13.0	0.0	8.1	0.0	2.2	14.3	5.4
13.	Tics, Mannerisms	0.0	8.7	0.0	8.1	0.0	2.2	42.9*	4.1
14.	Speech / Articulation	3.3	56.5*	5.4	35.1*	7.3	13.0	57.1*	19.0
15.	Stuttering Stammering	10.0	13.0	0.0	16.2	2.4	8.7	42.9*	9.1
16.	Elective Mutism	3.3	4.3	0.0	2.7	0.0	2.2	28.6*	2.7
17.	Echolalia	0.0	26.1*	0.0	5.4	0.0	0.0	14.3	4.1
18.	Language expressive	10.0	56.5*	2.7	27.0	0.0	6.5	85.7*	16.3
19.	Language receptive	0.0	21.7*	0.0	8.1	0.0	2.1	57.1*	5.9
20.	Language deviant	0.0	21.7*	0.0	2.7	0.0	0.0	14.3	3.2
21.	Inability to relate to people	30.0	69.6*	5.4	18.9	0.0	13.0	100.0*	21.3
22.	Inability to play with children	23.3	60.9*	5.4	24.3	0.0	30.4	85.7*	23.5
23.	Feeding problems	20.0	30.4	0.0	27.0	7.3	13.0	85.7*	17.2
24.	Enuresis (primary)	0.0	13.0	0.0	5.4	4.9	4.4	71.4*	6.3
25.	Enuresis (secondary)	16.7	17.4	13.5	16.2	12.2	21.7	42.9*	17.2
26.	Encopresis	6.7	13.0	0.0	2.7	0.0	8.7	71.4*	6.8
27.	Sleep problems	10.0	13.0	0.0	5.4	7.3	13.0	0.0	7.7
28.	Sexual problems	0.0	4.4	5.4	5.4	0.0	2.1	0.0	2.7

Table 2b

Frequency of occurrence in percentages associated with seven clusters in the Psychopathology section

SI. No.	Developmental problems/disorders	ED (30)	HYP (23)	LD (37)	PSY (37)	HYS (41)	CD (46)	AUT (07)	Total (221)
29.	Poor attention	10.0	95.6*	10.8	67.6*	4.9	73.9*	57.1	42.5
30.	Distractible	6.7	100.0*	8.1	54.0	0.0	73.9*	57.1	38.9
31.	Restless	3.3	91.3*	8.1	16.2	4.9	69.6*	57.1	31.2
32.	Impulsive	3.3	73.9*	5.4	10.8	0.0	60.9*	71.4*	25.8
33.	Stubborn	36.7	47.8	16.2	24.3	24.4	93.5*	28.6	41.6
34.	Disobedient	40.0	56.5	21.6	37.8	17.1	93.5*	42.9	45.3
35.	Disruptive	23.3	47.8	2.7	13.5	4.9	76.1*	57.1	29.4
36.	Quarrelsome	43.3	39.1	13.5	16.2	7.3	82.6*	0.0	33.5
37.	Temper Tantrums	30.0	43.5	10.8	13.5	9.8	65.2*	14.3	28.5
38.	Truancy	3.3	4.4	8.1	5.4	4.9	63.0*	0.0	17.2
40.	Lying and Stealing	20.0	13.0	10.8	2.7	4.9	65.2*	0.0	20.8
41.	School refusal	30.0	34.8	24.3	32.4	12.2	52.2*	0.0	30.30
42.	Poor school performance	46.7	69.6	21.6	94.6*	19.5	78.3*	28.6	53.8
43.	Reading difficulty	36.7	39.1	5.4	70.3*	2.4	54.4*	0.0	33.5
44.	Writing difficulty	43.3	39.1	5.4	97.3*	4.9	60.9*	0.0	40.7
45.	Arithmetic difficulty	53.3	39.1	13.5	86.5*	9.8	69.6*	0.0	44.3
46.	Forgetfulness, poor memory	40.0	52.2	10.8	78.4*	7.3	45.7	0.0	36.7
47.	Day dreaming	23.3*	8.7	8.1	8.1	0.0	8.7	14.3	9.1
48.	Withdrawn	60.0*	17.4	27.0	16.2	12.2	10.9	14.3	22.2

SI. No.	Developmental problems/disorders	ED (30)	HYP (23)	LD (37)	PSY (37)	HYS (41)	CD (46)	AUT (07)	Total (221)
49.	Talks little	70.0*	34.8	27.0	29.7	9.8	10.9	14.3	27.2
50.	Worrying	40.0*	4.4	21.6	5.4	12.2	6.5	0.0	14.0
51.	Anxious	60.0*	30.4	16.2	8.1	7.3	17.4	0.0	20.4
52.	Shyness / timidity	70.0*	30.4	13.5	16.2	17.1	17.4	12.3	24.9
53.	Fearful	76.7*	13.0	21.6	24.3	19.5	21.7	57.1	29.4
54.	Clinging	50.0*	56.2*	13.5	18.9	7.3	21.7	42.9	25.3
55.	Cries easily	76.7*	47.8	16.2	29.7	19.5	34.8	42.9	35.3
56.	OCD	10.0	8.7	13.5	5.4	4.9	0.0	14.3	6.8
57.	Dizziness	16.7*	0.0	8.1	0.0	17.1*	0.0	0.0	6.8
58.	Aches & Pains	10.0	0.0	5.4	0.0	29.3*	4.4	0.0	8.6
59.	Tiredness	3.3	0.0	0.0	0.0	4.9	0.0	0.0	1.4
60.	Stomach ache	0.0	0.0	2.7	0.0	2.4	4.4	0.0	1.8
61.	Fainting Spells	6.7	4.4	0.0	2.7	26.9*	6.5	0.0	8.1
62.	Fits	30.0*	0.0	2.7	2.7	22.0*	21.7	0.0	13.6
63.	Pulling sensation	26.7*	0.0	0.0	2.7	34.2*	0.0	0.0	10.4
64.	Chronic physical illness	0.0	0.0	0.0	2.7	7.3	4.3	0.0	2.7
65.	Physical handicaps	0.0	0.0	2.7	0.0	4.9	2.2	0.0	1.8
66.	Hallucinations	13.3	0.0	59.5*	2.7	4.9	10.9	0.0	15.4
67.	Catatonic feature	0.0	0.0	5.4*	0.0	2.4	0.0	0.0	1.4
68.	Delusions	0.0	0.0	43.2*	0.0	2.4	6.5	0.0	9.1
69.	Talking / laughing to self	3.3	17.4	70.3*	8.1	4.9	15.2	14.3	19.9

SI. No.	Developmental problems/disorders	ED (30)	HYP (23)	LD (37)	PSY (37)	HYS (41)	CD (46)	AUT (07)	Total (221)
70.	Poor-appetite	10.0	0.0	70.3*	5.4	0.0	4.4	14.3	15.4
71.	Poor sleep	16.7	0.0	89.2*	5.4	0.0	6.5	14.3	19.9
72.	Incontinence	0.0	0.0	37.8*	0.0	2.4	4.4	14.3	8.1
73.	Loss of interest	16.7	0.0	73.0*	8.1	2.4	6.5	14.3	18.1
74.	Psychomotor Retardation	20.0*	0.0	35.1*	0.0	0.0	0.0	14.3	9.1
75.	Depression	20.0	0.0	59.5*	2.7	2.4	0.0	0.0	13.6
76.	Pressure of ideas	0.0	4.4	48.6*	5.4	2.4	4.4	0.0	10.9
77.	Irritable	3.3	8.7	83.8*	8.1	7.3	15.2	14.3	21.7
78.	Elated	0.0	0.0	29.7*	0.0	2.4	4.4	0.0	6.3

* Significant at 0.05 level

Table 2c
Frequency of occurrence in percentages associated with seven clusters in the family history of illness section

SI. No.	Family History	ED	HYP	LD	PSY	HYS	CD	AUT	Total
1.	Mental Illness	16.7	26.1	21.6	27.0	9.8	10.9	42.9	18.6
2.	Alcoholism	23.3	21.7	29.7	16.2	19.5	26.1	57.1*	24.0
3.	Epilepsy	20.0	34.8	8.1	8.1	19.5	39.1*	14.3	21.3
4.	Learning problems	10.0	8.7	10.8	10.8	12.2	17.4	0.0	11.8
5.	Bed wetting	6.7	4.4	10.8	0.0	7.3	8.7	0.0	6.3
6.	Speech problems	3.3	4.4	5.4	2.7	4.9	13.0	0.0	5.9
7.	Mental retardation	6.7	21.7*	5.4	5.4	4.9	6.5	14.3	7.7

Table 2d
Frequency of occurrence in percentages associated with seven clusters in the Psychosocial stressors section

SI. No.	Psychosocial Stressors (STS)	ED (30)	HYP (23)	LD (37)	PSY (37)	HYS (41)	CD (46)	AUT (07)	Total (221)
08.	Precipitating factors	6.7	13.4	21.6*	2.7	9.8	13.0	14.3	11.3
09.	Problems with parents	13.3	21.7	21.6	8.1	9.8	26.1	28.6	17.2
10.	Sibling rivalry	36.7	39.1	24.3	13.5	19.5	41.3*	14.3	28.1
11.	Marital disharmony	30.0	17.4	16.2	8.1	19.5	30.4	28.6	20.8
12.	Punitiveness	36.7	47.8	27.0	24.3	26.8	47.8	28.6	34.4
13.	Over expectation	26.7*	26.1	8.1	8.1	17.1	2.2	0.0	12.7
14.	Over involvement	13.3	30.4*	8.1	10.8	9.8	15.2	14.3	13.6
15.	Over indulgence	20.0*	8.7	2.7	5.4	9.8	8.7	0.0	8.6

Sl. No.	Psychosocial Stressors (STS)	ED (30)	HYP (23)	LD (37)	PSY (37)	HYS (41)	CD (46)	AUT (07)	Total (221)
16.	Indifference	23.3	30.4	2.7	10.8	22.0	30.4	42.9	20.4
17.	Inconsistent disciplining	30.0	43.5	8.1	24.3	24.4	50.0*	28.6	29.9
18.	Multiple care taking	30.0	39.1	21.6	24.4	24.4	34.8	42.9	29.0
19.	Single parent	6.7	4.4	2.7	2.7	12.2	13.0	14.3	7.7
20.	Change in school / medium / subject	33.3	17.4	27.0	29.7	36.6	37.0	28.6	31.2
21.	Problem with teachers	23.3	43.5	10.8	27.0	17.0	41.3*	28.6	26.7
22.	Problems with peers	33.3	47.8	18.9	29.7	9.8	52.2*	28.6	31.2
23.	Poverty etc	13.3	4.4	24.3*	2.7	12.2	15.2	0.0	12.2

* Significant at 0.05 level

Table 2e
Frequency of occurrence in percentages associated with seven clusters in the temperament section

Sl. No.	Temperament Item	ED	HYP	LD	PSY	HYS	CD	AUT	Total
24.	Easy to manage	36.7	91.3*	43.2	48.7	24.4	87.0*	85.7	55.2
25.	Independence	43.3	65.2*	40.5	48.7	17.1	41.3	100.0*	42.5
26.	Dependability	40.0	82.6*	46.0	43.2	22.0	69.6*	100.0*	50.7
27.	Sensitivity to others	56.7	56.5	40.5	35.1	34.2	69.6*	57.1	48.9
28.	Sensitivity (self)	66.7	60.9	51.4	37.8	36.6	63.0	85.7	52.9
29.	Trusting	16.7	60.9*	29.7	40.5	9.8	58.7*	57.1	36.2
30.	Trustworthy	13.3	65.2*	32.4	35.1	17.1	60.9*	57.1	37.6
31.	Moral	26.7	73.9*	24.3	35.1	9.8	21.7	14.3	18.6
32.	Sleep pattern	20.0	17.4	21.6	21.6	9.8	21.7	14.3	18.6
33.	Appetite	26.7	26.1	24.3	29.7	34.2	39.1	42.9	31.2
34.	Activity	30.0	87.0*	35.1	59.5	19.5	73.9*	71.4	50.2
35.	Persistence	26.7	65.2*	29.7	29.7	19.5	63.0*	28.6	38.0
36.	Emotionality	66.7	73.9	64.9	21.6	48.8	73.9*	57.1	57.5
37.	Sociability (family)	30.0	30.4	32.4	32.4	14.6	34.8	28.6	29.0
38.	Sociability (others)	60.0	65.2	40.5	46.0	29.3	45.7	42.9	45.7
39.	Aggressiveness (verbal)	50.0	34.8	35.1	21.6	41.5	84.8*	0.0	45.3
40.	Aggressive (physical)	36.7	65.2*	37.8	27.0	9.8	84.8*	57.1	43.9

Note : Higher scores indicate severity of temperamental problem

Implications

1. The DPCL is the first ever check list to include items of background variables in the developmental context to psychopathology. These background variables are often underutilized in the taxonomic efforts, despite the empirical evidence to support their inclusion into a single tool of assessment. Thus, the present check list is truly developmental in perspective.

2. The DPCL is validated against a well known western tool with established reliability and validity. Yet it goes beyond CBCL, by providing anchor points to syndromes relevant in the Indian context and possibly elsewhere too. The check list includes autism, developmental disorder, late onset childhood psychoses, hysteria and learning disorder apart from disorders of emotion, conduct and hyperkinesis.

3. A unique and simple temperament subsection based on ancient Indian thought, has a provision of biological, psychological and social dimensions, in addition to studying competency.

4. It is a simple, brief, yet comprehensive tool that can be used across disorders, age and gender.

5. It can be used for purposes of diagnoses and planning of interventions.

Research:

The checklist has been used in research for screening children with anxiety symptoms in the school community setting (Selvi 2013, Ahil 2019), to classify temperamental profiles of adolescents with psychiatric disorders in the clinical setting (Bhati 2015). Recently it has been used as a screening tool for emotional problems in a study on group therapy for adolescents in a psychiatric outpatient setting (Ninan 2019). There is a need for research on a large sample of normal child population across age, gender and settings.

DEVELOPMENTAL PSYCHOPATHOLOGY CHECK-LIST FOR CHILDREN (DPCL)

Malavika Kapur, Ida Barnabas, Reddy, M.V., Joe Rozario & Uma Hirisave

A DEVELOPMENTAL HISTORY	No	Yes / Present
1. Did the mother, before, during or just after child birth, suffer from any problems like illness or difficult labour	0	1
2. Did the child have any serious illness soon after birth, if yes, specify.	0	1
3. Has the child had epilepsy, head injury, infections or any other serious illness ? (If yes, specify)	0	1
4. Does the child have any problems in seeing? (If yes, describe)	0	1
5. Does the child have any problems in hearing? (If yes, describe)	0	1
6. Between the ages of 1 and 3, could the child walk, climb, throw a ball ? (If not, mark as present)	0	1
7. Could the child by the age of 3 cut paper, thread beads? (If not, mark as present)	0	1
8. Between the ages of 1 and 3, could the child speak in small sentences ? (If not, mark as present)	0	1
9. Could the child between the age of 1 and 3 years show appropriate emotional expression in relation to parents and others and did he / she enjoy playing with other children? (If not, mark as present)	0	1
10. Could the child between the age of 3 and 5 years, feed, wash and dress him / herself ? (If not, mark as present)	0	1
B. DEVELOPMENTAL PROBLEMS		
Currently or in the past, has there been a problem :		
11 Of dropping things, falling or tripping frequently.	0	1

11. For brief periods when the child cried continuously to the extent of holding breath, become stiff, and turning blue in the face.	0	1
12. Of making odd or funny, repeated movements of the face, body, arms and legs.	0	1
13. In pronouncing words clearly (for example : 'Labbit' for 'Rabbit')	0	1
15. In speaking, as stammering or stuttering Currently does the child have the problems of :	0	1
16. Not talking at all and remaining mute, despite knowing how to speak, in some situations and to some people.	0	1
17. Repeating the words spoken by others exactly in the manner it was heard, without appearing to understand the meaning.	0	1
18. Appearing to understand what is being said but seemingly not being able to answer.	0	1
19. Of not being able to understand what is being spoken	0	1
20. Appearing to understand and know how to speak, but speaking in a manner which other people find difficult to understand, and refusing to use gestures to convey his/her needs.	0	1
21. Not being able to relate to people	0	1
22. Not being able to play with other children	0	1
23. Feeding such as overeating, under eating, food fads or fussy eating habits, and eating non-edible things such as mud (if present, specify)	0	1
23. Wetting clothes or bed from a very early age.	0	1
25. Resuming wetting of the clothes or bed, after being dry earlier on.	0	1
26. Of soiling of the clothes with stools, or has constipation (if present, specify)	0	1

27. Sleeping problems such as sleep-walking, sleep talking, teeth grinding, nightmares, etc, (If present, specify)	0	1
28. Masturbating or any other sexual problems (which is indulged in public)	0	1

C. PSYCHOPATHOLOGY

(The items below are marked as being present only when they occur often or most of the time but not when they occur sometimes) Does the child have the problems of :

29. Poor attention	0	1
30. Destructibility - (if the child is doing a task and someone enters the room, or he hears a sound does he easily get distracted by this)	0	1
31. Inability to sit in a place, and always moving around.	0	1
32. Acting without thinking, like while crossing the road not looking out for the traffic.	0	1
33. Stubbornness	0	1
34. Disobedience	0	1
35. Often interrupting others games, talk, being disruptive while playing, or breaking / throwing things frequently.	0	1
36. Quarrelsomeness and fighting		
37. Aggression as seen by hitting, biting and pinching others (with/ without provocation).	0	1
38. Getting very angry, crying a lot, rolling on the ground and continuing to be so for a long time, when his/her demands are not met.	0	1
39. Going to school and coming back on time, but actually does not attend the school.	0	1
40. Indulging in lying and cheating.	0	1

41. Refusing to go to school and staying back home for a duration of weeks or months.	0	1
42. Poor school performance	0	1
43. Reading difficulty	0	1
44. Difficulty in writing	0	1
45. Difficulty in arithmetic	0	1
46. Forgetfulness or poor memory	0	1
47. Day dreaming	0	1
48. Being very quiet and reserved (withdrawn).	0	1
49. Talking very little even with family members.	0	1
50. Worrying	0	1
51. Anxiousness and nervousness	0	1
52. Shyness and timidity	0	1
53. Is fearful of animals/people/situations	0	1
54. Clinging	0	1
55. Crying easily	0	1
56. Expressing that certain thoughts come to his/her mind repeatedly, the child says he/she as unable to control the thoughts although the Child recognizes that these thoughts are meaningless and these thoughts interfere with his/her daily activities. The child may go on doing a particular thing over and over again such as washing hands or checking which may reduce anxiety. If the items 57 to 63 are marked as present, consider dissociative / conversion disorder where necessary after ruling out organic conditions such as epilepsy by a clinician.	0	1
57. Complaining of dizziness or giddiness.	0	1
58. Complaining of aches and pains.	0	1
59. Complaining of / appearing to be always tired.	0	1

60. Complaining of stomach ache.	0	1
61. Fainting spells	0	1
62. Attacks of jerky movements and unconsciousness.	0	1
63. Complaining of pulling sensation of the limbs.	0	1
64. Chronic physical illness (specify, if present)	0	1
65. Physical handicaps (specify, if present) (the items 66 to 67 are to be marked as present even if it has occurred more than once in the past or present, the diagnosis of BPAD & Schizophrenia to be made by a clinician where necessary. Currently or in the past, has there been a problem:	0	1
66. Of hearing voices and seeing things when no one was around.	0	1
67. Of maintaining postures, being stiff, over long periods of time (if present, describe).	0	1
68. Of saying that he/she was a great person, or a bad person, or that he/she was being harmed by other people without a real basis for such beliefs.	0	1
69. Of talking and laughing to self.		
70. Of very poor appetite, sometimes leading to loss of weight.	0	1
71. Of poor sleep/disturbed sleep.	0	1
72. Of wetting and soiling during illness and was unaware of it.	0	1
73. Of loss of interest in play and daily activities.	0	1
74. Of moving and responding unusually slowly.	0	1
75. Of being depressed, sad and dull.	0	1
76. Of talking much more or faster than he/she normally used to.	0	1

77. Of being irritable 0 1
78. Of being unusually cheerful and happy (others, if any) 0 1

Note : Details of technology use to be obtained.

D) Psychosocial Factors Family History of:

79. Any one having mental illness 0 1
80. Any one taking alcohol excessively 0 1
81. Any one having epilepsy 0 1
82. Any one having problems in reading, writing or arithmetic 0 1
83. Any one having bed-wetting problem 0 1
84. Any one having speech problems 0 1
85. Any one being very dull or mentally retarded 0 1

E) Interaction in the Family (Stressors)

86. Have there been any precipitating events at the time of onset of the problems (specify, if present) Is there any evidence of: 0 1
87. Problems with parents (if present, describe). 0 1
88. Sibling rivalry (jealously of brothers and sisters) 0 1
89. Marital disharmony (parents fight a lot) 0 1
90. Punitiveness: Parents frequently resort to hitting, beating or punishing the child. 0 1
91. Over expectations: The parents expect from the child beyond his /her abilities, especially in school performance. 0 1
92. Over involvement: The parents are involved with all the child's activities to the extent that he/she does not do anything on his/her own. 0 1
93. Over indulgence: The parents meet all the demands of the child, whether reasonable or not. 0 1

94. Indifference: The parents are not bothered about the child's physical or psychological needs. 0 1
95. Inconsistent disciplining: The parents do not agree about the way to discipline the child. 0 1
96. Multiple care taking: The child is brought up by a number of adults in the family. 0 1
97. Single parent: The child has been cared for by a single parent. 0 1
98. Any change of school, medium or specific subjects or teachers, leading to unhappiness in the child. 0 1
99. The child complaining of problems with teachers. 0 1
100. The child having problems in playing, mixing or socialising with other children. 0 1
101. The child having problems such as poverty and other stressors, not covered in the above section if present, specify. 0 1

F) Temperament Profile

Descriptions of some aspects of the child's nature or temperament are given, and each description has three options to choose from. Encircle the options which fit the child best. If the description is not applicable it may be mentioned, especially for younger children.

	S	R	T	Not Applicable
a. Psychosocial				
102. Easy to manage	Mostly	Not at all	Somewhat	
103. Independent (can manage himself/ herself)	Mostly	Somewhat	Not at all	
104. Dependable	Mostly	Somewhat	Not at all	

105. Sensitive (to others needs, emotions)	Mostly	Somewhat	Not at all
106. Sensitive (only about one self)	Somewhat	Mostly	Not at all
107. Trusting	Mostly	Somewhat	Not at all
108. Trustworthy	Mostly	Somewhat	Not at all
109. Moral (discriminates between good & bad. Knows it is bad to hit others, steal, etc.)	Mostly	Somewhat	Not at all

b. Bio-social

110. Sleep	Moderate	Too little	Too much
111. Appetite	Moderate	Too little / Variable	Too little
112. Activity	Moderate	Too much	Too little
113. Emotionality	Cheerful	Angry/tense	Dull (non-reactive)
114. Persistence	Good	Variable	Too little
115. Sociability with family members	Adequate	Mostly	not at all
116. Sociability with others outside	Adequate	Mostly	Not at all
117. Aggression (verbal)	Not at all	Mostly	Somewhat
118. Aggression (physical)	Not at all	Mostly	Somewhat
TOTAL	S =	R =	T = NA=

NOTE: Each column total shows predominance of Satvik, Rajasik or Tamasik temperament in comparison to each other. The total consists of encircled options under each category. More of satvik temperament means better the temperament.

I) HELPFUL FACTORS FOR MANAGEMENT

119. Does the child have any helpful person at home/outside: somebody with whom the child is attached to, who helps the child, takes him/ her out, buys the child gifts? (if yes, describe)	0	1
120. Does the child have friends in the neighbourhood or school? (if yes, describe)	0	1
121. Does the child have interest in drawing, painting, games, music, etc.? (if yes, describe)	0	1
122. Does the child have any special talents? (if yes, describe)	0	1
123. Is the child good at sports? (if yes, describe)	0	1
124. Is the child creative, can put together common place objects in a new fashion, or make objects with hands? (if yes, describe)	0	1

NOTE : High score indicates more helpful factors.

G. Other observations, if any:

H. Summary:

A. Development history	1-10
B. Development problems including autism	11-28
C. Psychopathology	
i) Hyperkinesis (ADHD)	29-32
ii) Conduct problems	33-40
iii) Learning problems	41-46
iv) Emotional problems	47-55
v) Obsessive-compulsive symptoms	56
vi) Somatic symptoms (consider dissociative / conversion disorder where necessary)	57-63
vii) Psychotic symptoms	66-78
D. Psychosocial factors (includes family history & stressors)	79-101
E. Temperamental profile	102-118
F. Helpful factors for management	119-124

RECORD SHEET

DEVELOPMENTAL PSYCHOPATHOLOGY CHECK-LIST FOR CHILDREN

Malavika Kapur, Ida Barnabas, Reddy, M.V., Joe Rozario & Uma Hirisave
NIMHANS, Bengaluru - 560 029.

Sl.No : _____ Date: _____

Name of the Child : _____ Class : _____

Age : _____ Sex: _____

Mother tongue: _____ Languages Spoken : _____

Medium of Instruction: _____ Handedness: _____

Details regarding the family: _____ Age : _____ Education : _____ Occupation : _____

Child lives with: Father Mother Others _____

Number of siblings and the order of birth: 0 0 0 0 0 0 0 0 0 [] []
[] [] (mark 0 for girls and [] for boys, darken the circle 0 for the index child)

A. Developmental History	Consanguinity: If present, specify:	
	Urban	Rural
1. Pre-peri-post natal problems of Mother	No	Yes/Present
2. Post natal problems of child	0	1
3. Epilepsy, head injury, infections	0	1
4. Poor vision	0	1
5. Poor hearing	0	1
6. Gross- motor	0	1
7. Fine- motor	0	1

8. Speech/ language	0	1
9. Emotional-social	0	1
10. Self help	0	1
B. Developmental Problems		
11. Clumsiness	0	1
12. Breath holding	0	1
13. Tics, mannerisms	0	1
14. Speech/ articulation	0	1
15. Stuttering / stammering	0	1
16. Elective mutism	0	1
17. Echolalia	0	1
18. Language (expressive)	0	1
19. Language (receptive)	0	1
20. Language (deviant)	0	1
21. Inability to relate to people	0	1
22. Inability to play with children	0	1
23. Feeding problems	0	1
24. Enuresis (primary)	0	1
25. Enuresis (secondary)	0	1
26. Encopresis	0	1
27. Sleeping	0	1
28. Sexual problems	0	1
c. Psychopathology		
29. Poor attention	0	1
30. Distractible	0	1

31.	Restless / Overactive	0	1	Attention-Deficit Hyperactivity problems
32.	Impulsive	0	1	
33.	Stubbornness	0	1	
34.	Disobedient	0	1	
35.	Disruptive	0	1	
36.	Quarrelsome	0	1	
37.	Aggressive	0	1	
38.	Temper tantrums	0	1	
39.	Truancy	0	1	
40.	Lying & stealing	0	1	
41.	School refusal	0	1	
42.	Poor school performance	0	1	
43.	Reading difficulty	0	1	
44.	Writing difficulty	0	1	
45.	Arithmetic difficulty	0	1	
46.	Forgetfulness, poor memory	0	1	
47.	Day dreaming	0	1	
48.	Withdrawn/Avoids going to school	0	1	
49.	Talks little	0	1	
50.	Worrying	0	1	
51.	Anxious	0	1	
52.	Shyness / timidity	0	1	

53. Fearful	0	1	Emotional problems
54. Clinging	0	1	
55. Cries easily	0	1	
56. Obsessive and compulsive symptoms	0	1	Obsessive and com- pulsive symptoms
57. Dizziness	0	1	
58. Aches and pains	0	1	
59. Tiredness	0	1	
60. Stomach ache	0	1	
61. Fainting spells	0	1	
62. Fits	0	1	Somatic symptoms
63. Pulling sensation	0	1	
64. Chronic physical illness	0	1	
65. Physical handicaps	0	1	
66. Hallucinations	0	1	
67. Catatonic features	0	1	
68. Delusions	0	1	
69. Talking/ laughing to self	0	1	
70. Poor appetite	0	1	
71. Poor sleep	0	1	
72. Incontinence	0	1	
73. Loss of interest	0	1	
74. Psychomotor retardation	0	1	
75. Depression	0	1	
76. Pressure of Ideas	0	1	
77. Irritable	0	1	Psychotic symptoms
78. Elated	0	1	

D. Psychosocial Factors

Family history of		
79. Mental illness	0	1
80. Alcoholism	0	1
81. Epilepsy	0	1
82. Learning problems	0	1
83. Bed wetting	0	1
84. Speech problems	0	1
85. Mental retardation	0	1
86. Precipitating factor	0	1
87. Problems with parents	0	1
88. Sibling rivalry	0	1
89. Marital disharmony	0	1
90. Punitiveness	0	1
91. Over expectation	0	1
92. Over involvement	0	1
93. Over indulgence	0	1
94. Indifference	0	1
95. Inconsistent disciplining	0	1
96. Multiple care taking	0	1
97. Single parent	0	1
98. Change in school/ medium/ subjects	0	1
99. Problem with teachers	0	1
100. Problem with peers	0	1
101. Poverty, etc.	0	1

E. Temperamental dimensions

S R T NA

102. Management				
103. Independence				
104. Dependability				
105. Sensitivity (others)				
106. Sensitivity (self)				
107. Trusting				
108. Trustworthy				
109. Moral				
110. Sleep pattern				
111. Appetite				
112. Activity				
113. Emotionality				
114. Persistence				
115. Sociability (family)				
116. Sociability (others)				
117. Aggressive (verbal)				
118. Aggressive (physical)				

F. Helpful factors

119. Helpful person	0	1
120. Friends	0	1
121. Interests	0	1
122. Talents	0	1
123. Sports	0	1
124. Creativity	0	1

Sl. No.	Summary	Total Score	Cut Off	Obtained score
1.	Dev. History	10	3	-
2.	Dev. Problems / disorders	18	5	-
3.	ADHD	4	2	-
4.	CP	8	4	-
5.	LP	6	3	-
6.	EP	9	3	-
7.	OC symptoms	1	1	-
8.	Som symptoms	9	3	-
9.	Psychotic symptoms	13	3	-
10.	Family history	7	-	-
11.	Stressors	16	-	-
12.	Temperament (S/R/T/NA)		-	

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CHAPTER 3

ASSESSMENT OF ATTENTION AND THE PROFILE OF HYPERKINETIC CHILDREN WITH ATTENTION DEFICIT HYPERACTIVITY DISORDER ON PSYCHOLOGICAL ASSESSMENT

Anna Oommen

Attention refers to the selective aspects of perception which function so that at any instant a person focuses on certain features of the environment, to the exclusion of other features. It controls the flow of information processing. The concept of attention includes three functions, selection, capacity and sustained concentration.

In a Child Guidance Clinic or a Child Mental Health Unit, the complaint we most often hear from parents is that their child is doing poorly in school. There could be several reasons for this, each of which need to be explored in order to arrive at the right diagnosis and in order to plan the correct management strategy. Few of the reasons could be:

- Poor intellectual functioning
- Sensory deficits
- Poor comprehension of language
- Poor motivation
- Learning disability
- Attention deficit
- Emotional disorder
- Family pathology

Inability to sustain attention is often the cause of poor academic performance. Parents complain that the child just doesn't sit in one place for long, he is restless and fidgety all the time. Teachers complain that the child does not follow instructions in the class, disturbs other children and is impulsive.

The most frequently cited primary or core symptoms of ADHD include chronic hyperactivity, short attention span, marked distractibility, emotional lability and impulsivity, all of which are likely to be characterised by cross-situational and cross-temporal variability. Among the secondary symptoms are deficits in academic performance, low self-esteem, sleep-related problems, difficulties in social interaction particularly with peers, fluctuations in performance and aggression. As in most childhood behaviour disorders, ADHD is more common in boys than in girls, with sex ratios ranging from 5:1 to 9:1.

The onset is usually in childhood. Looking at the course of the disorder, overactivity is usually the first symptom to remit and destructibility the last. In adolescence, Children often continue to have poor academic performance, difficulties with peers, poor self-esteem and often antisocial behaviour. Etiological factors explored include genetic factors, neurological factors, pre-natal complications, bio-chemical abnormalities, lead levels in the blood, dietary factors and psychological factors.

Now, is this attention problem situational or pervasive? Is it only in particular places or in the presence of particular people that the child behaves this way? Is the child just seeking more attention from significant others?

What kind of treatment do you plan for this child? Behavioural lines of management, with attention enhancing strategies and correct use of reinforcements? Medicines? Family therapy? Intervention at the school level?

A detailed assessment of the child's attention and other psychological functions would help to answer these questions and would definitely help to plan management. If the child does have an attention deficit, the way one administers the other psychological tests and the interpretations one makes of the results would also change. The following tests were used in the assessment of hyperkinetic children in a study by Oommen (1991). The sample consisted of 123, 5-10 year old boys - 20 diagnosed as Simple Hyperkinesis (on ICD-9), 30 Hyperkinetic Conduct disorder, 30 Conduct disorder and 43 Normal controls matched to age, family status, religious background and parents' age.

I. Colour Cancellation Test (Kapur, 1974)

The Colour Cancellation Test, used to measure attention, concentration and impulsivity, consists of a sheet with 150 coloured dots — red, blue, yellow, black and grey, each occurring 30 times. The dots are arranged equidistant from each other in a random order.

The subject is first asked to name or match the coloured dots in order to test his colour vision. Even if the child is unable to name the colours he should atleast be able to match the dots that look similar.

There are two parts of the test. Simple Colour Cancellation, where the subject is instructed to cross out all the black dots with the pencil provided, as fast as he can. The examiner can demonstrate on one or

two dots. The examiner starts the stop-watch as soon as the subject starts cancelling and says 'stop' and removes the sheet at the end of one minute. For the Complex Colour Cancellation task, the subject is instructed to cross out all the yellow and red dots within the time limit of one minute. Note whether he proceeds systematically, column by column, horizontally, vertically or at random. Was the child able to concentrate on the task or was he easily distracted?

Variations of this test include picture cancellation, letter cancellation, number cancellation and symbol cancellation.

In Oommen's study, the number of correct dots cancelled, the number of wrong dots cancelled and the number of missing dots were recorded. The boys with hyperkinetic disorder crossed out significantly fewer number of correct dots. They were impulsive, crossing out more wrong dots and missing out more correct dots, when compared to the normal children. (Oommen, 1993).

Note: The CCT forms the backcover of this monograph and can be copied and laminated for repeat use.

II. Bender Visuo-Motor Gestalt Test (BGT) (Koppitz, 1975)

The BGT, a test of visual-motor integration, developed by Lauretta Bender (1938), consists of 9 cards with abstract designs on them. The subject is required to copy the designs one at a time, on a blank sheet of paper.

Koppitz has evolved a scoring system, standardized for ages 5-10 years, at 6-month intervals. Each design is scored, where appropriate, for distortion of shape, for rotation of the whole design or part of it, for

failure to integrate the parts of the design and for perseveration. There are a total of 30 scoring items. As the BGT scores are for imperfections, a high score indicates a poor BGT performance, while a low score reflects a good test performance.

The BGT developmental score is then converted to an age equivalent, which is compared to the subject's chronological age and his level of maturation is interpreted. A revised version of BGT is available.

Hyperkinetic boys made more errors of Distortion, Rotation, Integration and Perseveration, earning a lower age equivalent than the normal boys of the same age (Oommen, 1993).

III. Seguin Form Board (Cattell, 1953)

The Seguin Form Board consists of a large wooden board from which ten geometrical shapes are cut. The subject is instructed to replace the blocks in their right places as fast as he can. Three trials are given and the time taken is noted. The total timing of the three trials and the shortest timing is taken and the subject's mental age calculated.

The visuo-motor co-ordination of the subject is assessed on this test. Timings on the three trials also show us whether the child is able to learn from experience and better his performance. The examiner also observes whether the child has performed by insight or by trial and error.

The total timing taken by the hyperkinetic boys was longer than that taken by the normal boys, giving the former group a lower mental age (Oommen, 1993).

IV. Wechsler Intelligence Scale for Children (WISC) (Wechsler, 1949)

The WISC, designed for children aged 5 years to 15 years consists of 12 tests, divided into two sub-groups, verbal and performance.

The verbal tests include General Information, Comprehension, Arithmetic, Similarities, Vocabulary and Digit Span. The Picture Completion, Picture Arrangement, Block Design, Object Assembly, Coding and Maze tests constitute the performance portion of the WISC.

There are different profiles seen among children with ADHD of Some researchers write about the ACID profile or Freedom from Distractibility factor score. Children with ADHD do poorly on the Arithmetic, Coding, Information and Digit Span sub tests, which are considered to be especially sensitive to deficits in attention, concentration, and alertness to the environment.

Most studies show children with ADHD to have lower IQ scores than matched normal children, as in this study. On discriminant function analysis, the picture completion test stood out as the most powerful discriminator of all the WISC subtests, between ADHD and normal children (Oommen, 1997). This test calls for visual identification of familiar objects, forms and living things and the further capacity to identify and isolate essential from non-essential characteristics.

The Indian version the WISC namely Malin's Intelligence Scale for Indian Children, WISC-III UK and WISC-IV are also available.

V. Porteus Maze Test (Porteus, 1965)

This test consists of 11 mazes arranged in order of increasing difficulty for children aged 3 through 14 years. The child is asked to draw a line between the lines carefully without touching the sides. He is also instructed not to lift his pencil off the paper once he begins drawing.

The manual gives details of the instructions to be given, how many trials to be given for each maze and the type of errors to be scored. The qualitative scoring as distinct from the mental age scoring, is concerned with errors in drawing or execution rather than in planning. The score reveals any haphazard, impulsive or over confident habits or action, often seen with Children with ADHD The higher the Q score, the poorer the record.

In this study, failures were recorded often for Children with ADHD boys resulting in the test being terminated with few mazes being attempted. (Oommen, 1993).

VI. Tapping Test (Stevens et al., 1970)

The tapping test consists of a wooden board with two metal plates at either end. The subject is instructed to tap the metal plate with a stylus, with his dominant hand and non-dominant hand on alternate trials. A time limit of 15 seconds for each trial is maintained and number of taps recorded on an electric counter.

The tapping test measures motor co-ordination and speed. Boys with ADHD were noted to be clumsy, being unable to maintain their speed and co ordination of taps.

VII. Reaction Time Test (Woodworth & Schlosberg, 1976)

Reaction time is the period between the onset of the stimulus and the beginning of the response. The RT apparatus was first invented in 1850 by the famous physiologist Helmholtz. The Reaction Time Test consists of the Simple Reaction Time Test and the Choice Reaction Time Test. In

the former, the subject is instructed to release the button his finger is on, as soon as he sees the red light (visual stimuli) or hears a tone (auditory stimuli). Ten visual trials and ten auditory trials are given and the reaction time recorded on a chronoscope. In the Choice RT task, two coloured lights or two sounds are presented and the subject is instructed to respond only to one. Reaction time and the number of wrong responses made are recorded.

The mean reaction time recorded for the boys with ADHD using both visual and auditory stimuli, was longer than that of the normals.

However the two groups did not differ on the number of wrong responses made, that is, releasing the finger at the exposure of the wrong stimulus. The boys with ADHD would often fiddle with other knobs on the board and their attention had to be often drawn back to the task at hand (Oommen, 1993).

Similar results have been reported previously wherein children with ADHD are found to make more inappropriate responses, respond to the warning signal and respond more than once to the reaction signal.

VIII. Conners Abbreviated Rating Scale (CARS) (Conners, 1973)

This scale consists of 10 overlapping parent and teacher items from the 39-Item Conners Teacher Rating Scale. The 10 items were selected from those most often checked by teachers and they have proven to be reliable in identifying hyperactive children. It can also be used as a pre- and post-treatment assessment scale.

The items are rated on a four-point scale with the following categories: Not at all (Score=0), Just a little (Score:1), Pretty much (Score=2) and Very much (Score=3).

The cut-off score for classifying a child as hyperactive is a mean item rating of 1.5 or a total score of 15 on 10 items. In this study, the boys with ADHD got higher scores on the CARS when rated by both their parents and the investigator.

Of all the tools employed, the CARS was found to best discriminate between children with ADHD and normal children. The other tests were the BGT, WISC (particularly the Picture completion sub test) and the Tapping Test, measuring spatial perception and organisation, intelligence and motor speed and co-ordination.

Interview with the parents and the child, physical examination, information from the school teachers together with psychological assessment helps in planning a comprehensive management programme. The following guidelines could be used. It is called ADHD rating scale: home version as well as teachers version are frequently used for diagnosing children with ADHD.

Management of Children with Attention deficits.

The treatment children for ADHD behaviour is complex and multidimensional and might include Medical Management, Behaviour Modification, Parental Counselling and Educational Intervention. Criteria to be kept in mind are:

- Individualized programme specific to each child.
- ADHD is often referred to as a Variability Disease. These children have incredible fluctuations in their performance. On some days they will be able to complete all the tasks you assign them. On other days they are unable to do any work. So your behavioural and work expectations should be within the child's capabilities.
- Parents, teacher and therapist must work together.

Interventions for Children with ADHD (for teachers or parents)

1. Reduce distractions in study area (clear study table, no colourful posters on the walls).
2. Structure the task so that it is possible to proceed in a step-by-step manner.
3. Set short-term goals (Write answer to one question or do 2 arithmetic sums).
4. Break long assignments into smaller parts so that child can see an end to the work.
5. Seat child near good role model or 'study buddy'.
6. Use some physical contact with the child, which may be a hand on the shoulder, or directing of arm and hand movements.
7. Avoid teaching at a frustration level. Teach at a tolerance level at which it is easy for the child to work and at a level that is challenging for the child.
8. Be sure the child completely understands what is being taught.
9. Teach when the child is well motivated and ready to work.
10. Shorten work/study periods to coincide with span of attention; use timer.
11. Instruct child in self-monitoring.
12. Pair written instructions with oral instructions.
13. Cue child to stay on task, i.e. provide signal.

1. Provide "Seat breaks" i.e., run errands, clean the blackboard, collect the books.
2. Allow child to stand at times while working.
3. Use a multisensory teaching approach with specialized teaching materials.
4. They get bored easily so keep up their interest and motivation with adequate and immediate reinforcements.
5. Give frequent feedback of child's performance ("Good, keep it up").
6. Be firm, do not allow the child to escape a task that you know he is capable of performing.
7. Be consistent. Do not alternate between giving in to the child and being firm about completing a goal.
8. Ignore minor impulsive behaviour.
9. Use time-out for misbehaviour.
10. Acknowledge positive behaviour.
11. Help child organize his daily routine.
12. Look for signs of stress build up and provide encouragement or reduce work load to alleviate pressure and avoid temper outbursts.

Attention enhancing strategies.

- Stringing beads : big to small
- Matching, sorting
- Join-the-dot games
- Finger dexterity games

- Colouring within lines
- Mazes
- Painting within lines
- Cutting Pictures and sticking them
- Jig-saw puzzles
- Scanning pictures, numbers
- Letter cancellation
- Hocus-focus (find differences)
- CI Reaction time apparatus to teach child to withhold responses.
- Watching T.V. / listening to stories and narrating them.

Finger / Motor Co-ordination Activities

- Plasticine / clay play
- Scissor work - Scrap book
- Hopping games
- Skipping
- Rangoli work
- Swimming
- Dancing
- Putting coins / buttons into slot
- Nuts and bolts
- Tapping board
- Finger puppets

Other Management Strategies

- 'Stop, look and listen method' (Meichenbaum and Goodman, 1971): Children are taught to recognise problem situations to give self instructions, self-monitor and self-reinforce.
- Cognitive behavioural techniques (Oberoi and Kapur, 1995)
- Neuropsychological remediation programme - focusing on attentional deficits of resource allocation, vigilance and perceptual sensitivity and prefrontal deficits of perseveration and susceptibility to interference. (Agarwal and Rao, 1997).
- Compliance Training
- Parent Management Training (Barkley, 1949, Basu and Deb, 1996, Rejani, 2005)
- Neuro feedback training, Training in brain exercises for frontal lobe deficits, Mindfulness based procedures.

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CHAPTER 4

ASSESSMENT OF SPECIFIC LEARNING DISABILITIES

Annie John, Joe Rozario, Anna Oommen and Uma Hirisave

The term Learning Disabilities emerged from a need to identify and serve students who continually fail, in school yet are not intellectually challenged.

The concept of learning disabilities has undergone distinct phases of development in history. The study of learning disabilities began with the works of Strauss and his colleagues in the early 1940s. The term they used for these children was "brain-injured" children. They established seven criteria to classify these children. They were: perceptual disorders, perseveration, conceptual or thinking disorders, behavioural disorders, soft neurological signs, a history of neurological impairment and no history of mental retardation.

In 1960s a shift in terminology occurred with the introduction of the term "Minimal Brain Dysfunction (MBD)". The term brought in the concept of minor brain injury and linked it with learning problems. MBD was popularized by Clements who was the project director of Task Force I which was appointed to study the characteristics of these brain impaired individuals and suggest identification procedures. They identified ten markers - hyperactivity, perceptual motor impairments, emotional lability, general orientation deficits, disorders of attention, impulsivity, disorders of memory and thinking, disorders of speech and hearing and soft neurological signs. Samuel Kirk coined the term 'Learning disability' and put it in use around 1962. Under Kirk's leadership the National Advisory Committee on handicapped children submitted the first definition. Since then a number of definitions have been proposed but none of them was totally acceptable. According to DSM V "Specific Learning Disability refers to the difficulties in learning and using academic skills like reading /

understanding / spelling / written expression / and mathematical reasoning which are not due to intellectual disability, neurological disorder, psychosocial adversity and inadequate / lack of proficiency in the language of academic instruction persistent for atleast 6 months despite provision for intervention (American Psychiatric Association, 2013).

ICD-10 defines Specific Developmental Disorder of Scholastic Skill Disorder (SDDSS) is a developmental disorder that comprise of group of disorder with significant impairment in learning and scholastic skills that are not due to mental retardation, lesser intelligence, neurological deficits, sensory problem or emotional disturbance. (World Health Organisation, 2004).

Characteristics of learning disabilities:

1. Discrepancy factor-A discrepancy exists between a child's ability and academic performance across one or all skill areas.
2. Academic learning difficulty-Academic problems exist in the area of reading, writing, spelling and mathematics.
3. Perceptual disorder-Perceptual problems include inability to recognize, discriminate and interpret sensation. It can be in the area of auditory channel or/and visual channel.
4. Metacognitive deficits-Metacognitive abilities involve the ability to use self regulatory mechanisms such as planning moves, evaluating effectiveness of ongoing activities, checking the outcome and remediating the errors.
5. Memory problems-Most students complain of poor memory. Students with learning disabilities fail to use strategies that nondisabled students readily use.
6. Phonological deficits-students with reading, spelling difficulties often have poor phonological skills.

7. Motor disorders: Children with this problem are clumsy, unable to skip or hop. Some exhibit fine motor difficulties such as in cutting with scissors.
8. Attention problems and hyperactivity: Many researchers have documented that many learning disabled children have short attention span, distractibility and impulsivity. Some are restless and hyperactive.
9. Social emotional problems: The prevalence rate of psychological disturbance is high among learning disabled children compared to the normal population. Continuous failures in academics result in poor self concept and self esteem.

Causes of learning disabilities :

There is no single cause to explain the origin of learning disabilities. A variety of factors have been linked with its occurrence.

- * Genetic: It is found that learning disabilities run in families.
- * Language: Poor phonological processing is seen in children with reading & spelling disabilities.
- * Biochemical: Possible chemical imbalance that lead to inefficient or faulty brain functioning.
- * Biological: Head trauma that causes brain injuries, foetal distress during birth.
- * Unfavourable uterine environment: Drugs, smoking, alcohol that adversely affect the brain's development in the uterus.
- * Environmental deprivation: A lack of stimulation at critical times of development that prevents optimal brain function.

These are some of the common correlates. There are also other factors, like the problem of laterality, size of the left hemisphere and the size of the neurons in the left hemisphere.

NIMHANS INDEX FOR SPECIFIC LEARNING DISABILITIES

The battery was originally developed by John (1989) for the purpose of her doctoral thesis. The battery then consisted of Bender Gestalt Test, Minnesota Percepto-Diagnostic Test, tests of reading, writing, comprehension, spelling and tests of arithmetic. This battery was administered to a sample of 50, eight to twelve-year-old clinic children who presented with scholastic backwardness and also to a control group of 50 children with average to above average scholastic achievement. The results revealed that the profile of clinic children was significantly different from that of normal children in all the areas assessed.

Later the battery was routinely used with the children at the Child and Adolescent Mental Health Unit, NIMHANS (Presently named Child and Adolescent Psychiatry) for confirmation of specific learning disabilities. The profile derived was used for drawing up a remedial plan for learning disabled children. In the year 1992, Kapur et al compiled the following tests into a battery and named it NIMHANS Index for SLD. The Index comprised of the following tests:

- a) Attention Test (Number cancellation)
- b) Language Test (Reading, Writing, Spelling and Comprehension)
- c) Arithmetic (Addition, Subtraction, Multiplication, Division and Fractions)
- d) Visuo-motor Skill (The Bender Gestalt Test and the Developmental Test of Visuo-Motor Integration)
- e) Memory (Auditory and Visual)

The NIMHANS index for SLD II did not have norms for certain subtests. Hence the second revision of the test was carried out (Hirisave, U., Kapur, M., John, A., & Rozario, J, 2019). The norms were collected from children studying in an English medium school, Bengaluru catering to lower middle and middle socio-economic status. The details of the revision are given in the following table.

Test	Second revision- modifications carried out
1. Brigance diagnostic system	No modifications
2. SLD level I	No modifications. It is suggested that when children aged 6-7 years pass all the items and if they are able to read and write simple words, SLD level II may be administered (Subtests appropriate for other children). Other tests such as phoneme segmentation and non sense word fluency may be administered.
3. SLD Level II-Attention	No new norms were collected. It is suggested that interpretation should be based on the total time, number of correct answers, number of errors (omission and commission) as well as approach to the task.
4. SLD Level II- Reading 1 st to 5 th standard	No new norms were collected. The paragraphs were found to be appropriate when tested on a small sample of children randomly selected from respective classes. The number of words in the paragraphs was reduced.
5. SLD Level II- Reading comprehension 1 st to 5 th standard	No modifications were made.
6. Reading and reading comprehension 6 th to 9 th standard	Norms were collected from the sample of 20 children from class 6 th to 9 th standard. These children were reported to have adequate intellectual functions by their teachers.

7. 10 th standard reading and reading comprehension	Norms were not collected for this group of children. It is suggested that the children should be tested based on the respective curriculum.
8. Listening comprehension for all standards (1 st to 9 th standard)	When children have severe reading difficulties, reading comprehension may be poor. Hence listening comprehension should be tested. The examiner should read the paragraph and ask the child to answer the questions orally.
9. Spelling for all standards	The norms were collected from the sample of children studying in 1 st to 10 th standard. These children were in the age group 6-16 years. Both boys and girls were included. The Schonell test of spelling was administered to the class as a whole. The sample size varied across the classes. The numbers of participants were 36 in 1 st standard, 29 in 2 nd standard, 28 in 3 rd standard, 31 in 4 th standard, 38 in 5 th standard, 25 in 6 th standard, 25 in 7 th standard, 35 in 8 th standard, 30 in 9 th standard and 34 in 10 th standard.
10. Arithmetic test 1 st to 5 th standard	No new norms were collected for 1 st to 4 th standard arithmetic sums. Small samples of children across 1 st to 4 th standard were tested and they were able to solve 2-3 sums appropriate for their respective classes. Norms were collected for fractions test from a sample of children studying in 5 th , 6 th , 7 th , and 8 th standard. There were 15 children in 5 th and in 8 th standard respectively and 18 children in 6 th and 7 th standard respectively.
11. Mathematics Test 8 th to 9 th standard	A new test was prepared to test mathematics skills. A sample of 37 children in 8 th standard and 30 children in 9 th standard were administered a list of 65 sums in a group session for each class. These sums were taken from the initial

	chapters of maths text book- 6 th and 7 th standard (state syllabus).
12. Visuo-Motor perception test (BGT)	No new norms were collected. For older children aged 16-18 years, it is suggested to use scoring procedure given by Bender L.
13. DTVM	No new norms were collected.
14. Memory test (BVRT)	No new norms were collected. When visuo-motor perception is poor, it is suggested not to administer this test.
15. Auditory memory test	No new norms were collected.

NOTE: It is suggested to administer neuro-psychological battery when there are difficulties in perception, attention or memory.

Performance of normal children on the following tests are given in Tables (1-34)

- I. Brigance Diagnostic System of Basic Skills (Table 1a & 1b)
- II. NIMHANS Index for Learning Disability (Level - I) (Tables 2a & 2b)
 - a) Visuo Motor skills (Copying of three geometrical figures (Table 2b)
 - b) Writing of Capital Letters (Table 2b)
 - c) Writing of Small Letters (Table 2b)
 - d) Writing of an alphabet preceeding a specified alphabet (Table 2b)
 - e) Writing of an alphabet succeeding a specified alphabet (Table 2b)
 - f) Writing of numbers serially (1-20) (Table 2b)
 - g) Writing of number preceeding a specified number (Table 2b)
 - h) Writing of a number succeeding a specified number (Table 2b).
Other tests of the battery are used with children with learning difficulties/learning disabilities for planning remediation. Performance of normal children on these subtests are yet to be obtained. These are:-
 - i) Colour Cancellation Test
 - ii) Visual discrimination
 - m) Visual memory
 - iv) Auditory discrimination
 - v) Auditory memory
 - vi) Speech/language
 - vii) Knowledge of sounds of alphabets.

- III. NIMHANS Index for learning disability Level-II (Revised Tables 2c-34)
 - a) Number Cancellation (Tables 2c,3b,4,9,15,16)
 - b) Reading of English passages of I-IX grade levels (Tables 2c,3b,8,14,22)
 - c) Spelling of English words of I-X Std (Tables 23-32)
 - d) Reading comprehension of English passages of I-IX grade levels (Tables 2c, 3b, 8, 15, 21,22)
 - e) Copying & written expression of English passages I-IX Grade levels (Tables 23-32)
 - f) Arithmetic subtest and Mathematics Test sums (Simple digit addition, Simple digit subtraction, Simple digit multiplication, Simple digit division, graded addition, graded subtraction, graded multiplication, graded division, addition of fractions, subtraction of fractions, multiplication of fractions, division of fractions and Mathematics Test (Tables 2c, 3b, 6, 7, 11, 12, 13, 18, 19, 20, 33, 34)
 - g) Visual spatial abilities are assessed on Bender Gestalt Test (BGT) Tables 3c, 5, 10, 17) (Koppitz, 1974)

Other tests of the battery routinely used are Malin's Intelligence Scale for Indian Children (MISIC) (Malin, 1969), WISC-III UK-Modified (Panicker, 2006), WISC-IV, Developmental Test of Visual Motor Integration (DTVMI) (Beery and Buktenica, 1967), Bender Gestalt Test (Koppitz - 1974) Benton Visual Retention Test (BVRT, Benton, 1953), Auditory memory tests. Except writing, memory and mathematics tests, other test materials are to be obtained from publishers of the tests.

A child who has average / above average intelligence but functioning at two grades below his age appropriate grade can be considered to have specific learning disability. This could be either in reading, writing, spelling, arithmetic or mixed.

In the Indian context it is important to note that the quality of education the children receive may have an impact on their academic skills. A study of reading, writing and arithmetic skills was conducted in two types of schools. Two schools in the urban slums, with 25 children from one and 29 in another and one school from the city of Bengaluru catering to middle class families, with 35 children were studied. All the children were in the fifth standard. They were tested on the Third standard level of the NIMHANS SLD Index. The study revealed that only 15% of the 54 slum children had class appropriate academic skills, while in the school catering to middle income group, it was 66%. It is worth noting that nearly one third of the children from middle income homes had deficits in academic skills. This seems to suggest that certain inadequacies exist even in these schools. The study indicates that effective educational inputs are required in both kinds of schools though to a varying extent (Srinath, S. Personal Communication).

A study was conducted to understand the cognitive and phonological deficits in children with specific reading and spelling disorder (Ralte, 2012). The participants included regular school going children between the ages of 8 to 10 years, who were studying in the 3rd and 4th standards. These children were screened on tests such as, Raven's coloured progressive matrices test, Bender Gestalt test and Spelling, Reading and Arithmetic subtests of NIMHANS Index for Specific Learning Disabilities. There were 8 children with reading and spelling disorder and 6 children without any reading or spelling problem.

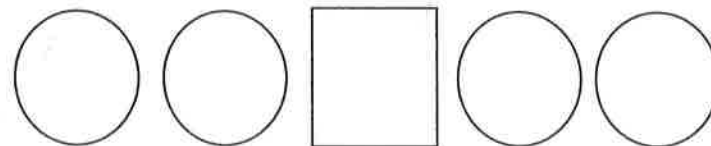
The tools used to assess cognitive and phonological skills were Wechsler Intelligence Scale-III (Indian Version), Working memory subtests from NIMHANS Neuropsychological Battery for children, Elision Test, Word Attack and Word Identification (Subtests from WRMT-R).

The results indicated significant difference in verbal intelligence, verbal and visual working memory and Phonological skills between the 2 groups indicating that children with reading and spelling difficulties had deficits in these abilities.

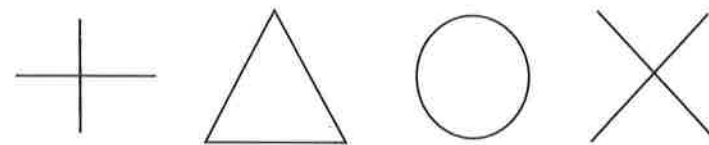
LIST OF BASIC SKILLS AS GIVEN BY BRIGANCE, A.H. (1977)

It is important to test basic skills of young children before administering SLD Level I and II, Some of the basic skills listed by Brigance are given below:

1. Colour recognition (asked to name colours) Record the performance.
2. Visual Discrimination (asked to point out the figure that is different.) Sample item:



3. Visuo-motor skills (Asked to draw some designs). Sample item:
Record the performance.



4. Visual memory (Asked to reproduce symbols from memory). Sample items:



5. Body image (asked to draw a picture of a person)
6. Gross motor co-ordination (asked to walk with both feet on the line, asked to jump, hop, skip).
7. Identification of body parts (asked to identify body parts).

8. Directional/ positional skills (Comprehension of instructions. Sample item: "Stand up" "Put your hands over your eyes" "Show your right hand").
9. Fine motor skills (asked to button, put on shirt, use scissors, tie shoe lace).
10. Verbal fluency (Child's ability to converse).
11. Verbal directions (Asked to remember and follow verbal directions. Sample items: "Clap your hands twice", "Stand-up and touch your head", "Go to the black board, write your name on the board, then look out of the window")
12. Articulation of sounds (Asked to name the pictures of objects with names which have different initial sounds. Sample item: Apple, Ball, Men, Key, Dog, Rabbit).
13. Personal data response (Asked to respond to questions such as "What is your name?", "How old are you?", "How many sisters do you have?").
14. Sentence memory (Asked to repeat sentences from memory).
15. Counting (Asked to count by rote).
16. Alphabets (Asked to recite alphabets).
17. Numeral recognition (Asked to read numerals 1-10).
18. Numerical comprehension (asked to show/specific number of fingers)
19. Recognition of lower case letters (asked to name lower case letters. Sample item : o d g b p)
20. Recognition of upper case letters (asked to name upper case letters. Sample Item: O A D G B P)
21. Writing names (asked to write his/her name).
22. Numbers in sequence (asked to write numbers 1-10 in sequence).
23. Lower case letters by dictation (asked to write dictated lower case letters)
24. Upper case letters by dictation (asked to write dictated upper case letters)

The reader may obtain more sub items for testing each skills by referring to the original Brigance Manual.

Interpretation

Most children in the age range of 3-5 years are able to pass item nos. 1-18. Some children may have difficulty in item nos. 19-24. This indicates that the child has difficulties in these skills (and not to be diagnosed as learning disabled). Such children are able to master these skills with inputs.

If a child aged 5-7 years is unable to do the items on the list, it indicates that the child is "at risk" for learning disabilities. He/She needs adequate inputs both at school and home to acquire these skills. If a child above 7 years is unable to pass the items, he is referred to as being learning-disabled and requires remedial training.

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(Hirisave, U., Kapur, M., John, A., Rozario, J., & Oommen, 2019)

Since MISIC does not have norms for children above 15 years, Wechsler's Adult Performance Intelligence Scale (WAPIS) may be considered for older children. When WISC III Indian adaptation, is not available, WISC IV may be considered for the assessment of Intelligence of children. At NIMHANS, the procedure for arriving at severity is as follows:

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(LEVEL-1 and LEVEL-2)

General Abilities :

MISIC PROFILE

SCALES

VERBAL IO =

PERFORMANCE IQ =

FULL SCALE IQ =

* also known as intelligence scale for Indian Children.

LEVEL - I

YOUNGER STUDENTS (5-7 Years)

ATTENTION

The ability of the child to focus his attention on a given job can be tested by using :-

1. Simple Colour Cancellation test.
2. Double Colour Cancellation test.

For Simple Colour Cancellation, request the child to cancel out any one colour from the colour cancellation sheet, within the time limit of one minute.

For the double colour cancellation, request the child to cancel out any two colours.

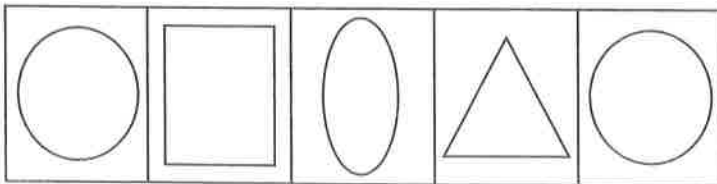
The back cover of the book contains the colour cancellation sheet.

Directions :

"I am going to show you a sheet which has different coloured dots. I want you to cancel out all the black and yellow colours as fast as you can". If the child still doesn't understand, you can demonstrate it.

VISUAL DISCRIMINATION

Directions - "Mark the one that looks the same as the first one." Three or more errors suggest difficulty with visual discrimination for children aged 5 or older.



1.	M	T	M	P	O
2.	3	2	3	5	8
3.	ll	M	N	U	ll
4.	b	D	P	b	G
5.	ME	SHE	WE	HE	ME
6.	SHIP	SNIP	SLIP	SHIP	SKIP
7.	ARE	AIR	FIRE	ARE	ART
8.					

Supplement your test findings with your behavioural observation/ information obtained from parents.

- May have difficulty in matching shapes, geometric forms.
- Loses place frequently when copying.
- Has difficulty in recognizing objects when parts are missing.
- Makes poorly formed or reversed letters.

VISUAL MEMORY - Directions

"I am going to show you a row of pictures. After I cover the pictures, I want you to tell me each time exactly what pictures you saw. Try to tell me in the same order they were in".

For children aged 5, recalling 3 items in any order within a row suggests adequate visual memory,

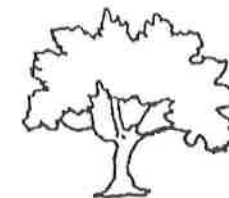
For children aged 6 and older, recalling four items in any order within a row suggests adequate visual memory.

Exposure time for each row

Row	1 & 2	:	2 Sec
	3	:	3 Sec
	4	:	4 Sec
	5	:	5 Sec

Response

- 1.
- 2.
- 3.
- 4.
- 5.



Supplement your test findings with behavioural observation/information obtained from parents.

- the child may recognize the symbol when shown but unable to recall.
- the child may remember all the parts seen but got them in wrong sequence.

AUDITORY DISCRIMINATION

Directions

The child should not see the examiner's face. This is to avoid guessing from lip reading.

"I am going to tell you some words and you tell me if they are same or different".

If the child makes a mistake, a second presentation could be made of the same words. If he fails again it can be counted as an error. Five or more uncorrected errors by children aged 6 or older suggest difficulty with auditory discrimination and possible problems with learning through a phonetic approach.

1. shine - sign
2. tin - thin
3. pine - pine
4. mob - mop
5. ship - sip
6. goal - coal

7. run - run
8. bus - buzz
9. lake - lake
10. and - and
11. moon - noon
12. mud - mug
13. bill - mill
14. and - and
15. kill - till
16. drink - drink
17. mash - mush
18. deck - dock
19. band - band
20. Slid - Slad

Total errors

Supplement your test findings with your behavioural observation/information obtained from parents.

- Cannot tell when the sounds are the same or when pitch or intensity is changed.
- Confuses similar sounding letters like d and t, b and p.

Auditory Behaviour

1. Has difficulty in paying attention to the speaker.
2. Is a poor listener.
3. Disturbed by background sound/ noise, specifically.
 - i) Speech; (ii) Whispering; (m) Shuffling papers, feet etc.; (iv) Play ground noise; (v) noise from other class rooms or halls and (vi) others.
4. Poor comprehension of speech in the presence of competing sounds such as: (i) T.V./Radio broadcast; (ii) background communication.
5. Misunderstands verbal instruction.
6. Asks to repeat verbal instructions.
7. Slow or delayed response to verbal stimuli.
8. Relies heavily on visual cues.

AUDITORY MEMORY

Directions

"I am going to read some sentences to you. When I finish, just repeat what I said".

1. The boy has a ball.
2. I like to sit in the car.
3. Last Sunday we went for a movie.
4. Ramu is going to the market.
5. I like to go to school.

6. Sita is dancing.
7. I like to eat sweets.
8. It is raining heavily

One or more errors in each of the eight sentences suggests difficulty with auditory memory in children aged 5.

A child of 6 years can repeat atleast five sentences without any error. Supplement your test findings with behavioural observations/information obtained from parents.

- may have difficulty in following a sequence of directions at home or in school.
- able to understand and to recognise words but has difficulty retrieving them.
- parents complain that what they told him yesterday, has been forgotten today. .

SPEECH AND LANGUAGE

Verbal language expression - Directions

Pointing to the picture tell the child, "Tell me all you can about this". "Tell me what it is, and what it does or what you can do with it".

- List the number of responses.
- For children ages 4-6 less than four total responses (atleast one per item) suggests problems with verbal language expression.
- For aged 7 and older under 6 responses suggests difficulty in the area of verbal language expression.



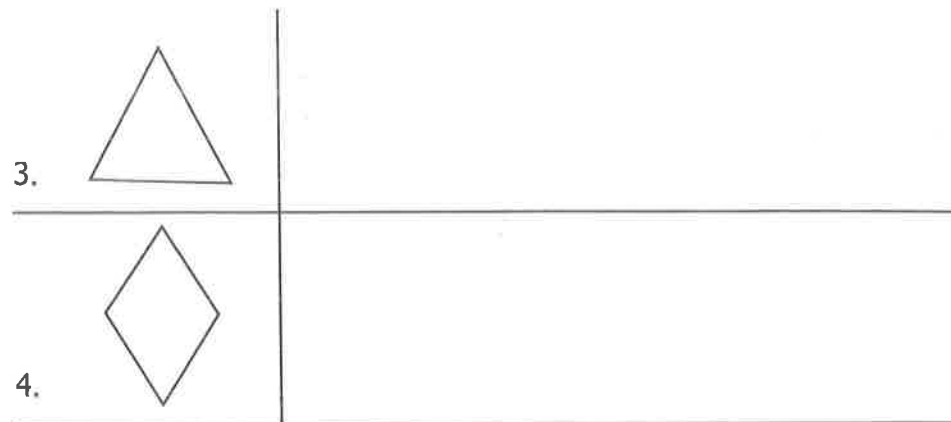
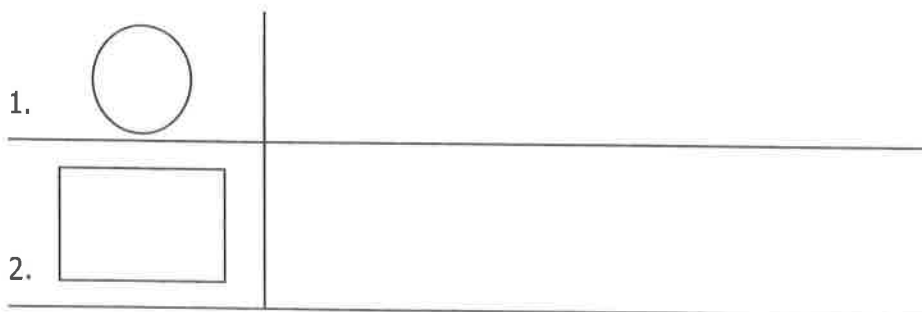
Supplement your findings with behavioural observation / information obtained from parents.

- doesn't speak
- speaks only in monosyllables
- has problems with articulation
- lacks ability to describe persons, places, etc.
- doesn't comprehend simple sentences

VISUAL - MOTOR SKILLS

Directions

The child is asked to copy a design exactly the way he or she sees it. The child is given three chances, but only the best effort is counted. Difficulty with these designs after age seven indicates a need for a programme of comprehensive visual- motor activities to develop eye-hand readiness skills.



Normative Data

1. A child 3 years of age should be able to make a single circle.
2. A square shape is normative for age 4.
3. Triangle is normative for 5 1/2 to 6 years age.
4. The diamond is for children after age 6 to 7 years.

Supplement your test findings with behavioural observation/information obtained from the parents.

- poor in games and appears clumsy or uncoordinated
- unable to catch or throw a ball
- loses balance easily
- unable to walk steadily

WRITING SKILLS

Write the Alphabets

Capital letters

Small letters

Write the letter that comes before and after

Before		After
-	C	-
-	K	-
-	T	-
-	M	-
-	R	-

Remarks :

Numbers

Write the numbers up to 20

Write the number that comes before and after

Before		After
-	8	-
-	17	-
-	25	-
-	39	-
-	63	-

Remarks :

Reading of alphabets :

Dictation of alphabets :

Knowledge of sounds of alphabets & vowels :

Level I Profile -

Note :

1. Additional tests such as Diebels (Copyrighted test of Dynamic measurement group) phoneme Segmentation Fluency and Nonsense word fluency may be administered.
2. Children aged 6-7 years (studying in I standard) are able to read and write simple words then the SLD - level - II (I standard tests of reading, comprehension, spelling, written expression, copying, single digit addition & subtractions) may be administered.

INTERPRETATION

Most children in the age range of 5-7 years are able to perform adequately on the tests. If a child aged 5-7 years is not able to perform adequately, it indicates a 'difficulty' not 'disability'. If a child aged above 8 years is unable to perform adequately on Level-I, it indicates a learning disability.

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CHILD AND ADOLESCENT MENTAL HEALTH UNIT
SLD LEVEL – I (For 5 to 7 years)

1) ATTENTION

TEST	No. of correct	Errors	Time
Single colour cancellation			60 secs
Double colour cancellation			120 secs

2) VISUAL DISCRIMINATION

								Total correct
Right								
Or	1	2	3	4	5	6	7	8
wrong								

Remarks

3) VISUAL MEMORY

	Exposure Time	No. of figures Recalled
A. Trial (2 figures)	2"	
1. Two figures	2"	
2. Threes figures	3"	
3. Four figures	4"	
4. Five figures	5"	

Remarks

4) AUDITORY DISCRIMINATION

Item no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Right or Wrong														
Item no.	15	16	17	18	19	20	Total correct							
Right or Wrong														

Remarks :

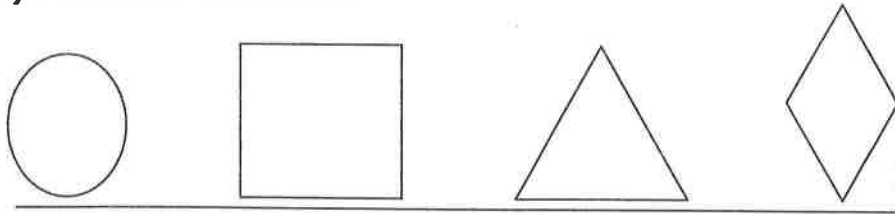
5) AUDITORY MEMORY

Record the response verbatim

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.

Remarks:

6) VISUAL MOTOR SCREEN



Remarks

7) SPEECH AND LANGUAGE

Record the responses verbatim

- i) Car
- ii) Ball
- iii) Bucket

Remarks

8) ALPHABETS

CAPITAL LETTERS

SMALL LETTERS

Write the letter that comes before and after

Before		After
-	C	-
-	K	-
-	T	-
-	M	-
-	R	-

Remarks:

9) PHONIC SKILLS - Check whether child is able to articulate the sound corresponding to each alphabet.

Note: Additional test such as Diebels (copyrighted test of dynamic measurement group, 2007). Phoneme segmentation fluency and Nonsense word fluency may be administered.

10) NUMBERS

Write numbers up to 20

Write the numbers that come before and after

Before

After

- 8
- 17
- 25
- 39
- 63

Remarks:

PROFILE OF THE CHILD :-

(Qualitative evaluation by the examiner of the Child's performance)

Above Average _____

Average _____

Below Average _____

General abilities	Attention	Discriminat ion	Memory	Motor skills	Discriminat ion	Memory	Alphabets	Expressive speech	Phonic skills	Numbers
		Visual	Auditory		Language		Arithmetic			

Note: If children aged 6-7 years -studying in 1st standard are able to read and write simple words, then the SLD level II (1st standard tests of reading, comprehension, spelling, written expression, copying, single digit addition and subtraction) may be administered

Level II

OLDER STUDENTS (8 to 16 years)

ATTENTION

Number cancellation test can be used to test the student's ability to sustain attention.

Instruction

"Of the several numbers given below, I want you to cancel all 3s and 4s. Do it as quickly as possible". (If a child finds the task difficult then instruct him / her to cancel any one number).

Note down the time taken, number of right numbers cancelled and number of wrong numbers cancelled.

Score = No. of right numbers cancelled - no. of wrong numbers cancelled.

Comment on the approach : Systematic or haphazard.

READING

Instructions for Reading Tasks: "I want you to read this passage loudly and clearly". (After the child reads, for e.g. 1st standard paragraph, ask comprehension questions from 1st standard paragraph.

If the child reads 1st standard paragraph adequately, then go on to 2nd standard paragraph reading and 2nd standard comprehension and so on)

Paragraph for Reading – Standard - I

THIS IS A BIG TREE. IT IS A MANGO TREE.

THIS IS A ROSE PLANT. THIS IS A ROSE.

THIS IS A LEAF. IT IS SMALL.

THESE ARE ROSE PLANTS. THESE ARE ROSES.

THESE ARE RED ROSES.

THESE ARE YELLOW ROSES.

Paragraph for Reading – Standard - II

Once there were many rats in the house. One day a big cat came into the house. She killed many rats. A brown rat was the leader of the rats. Rats wanted to kill the cat. One small rat said "I want to pull the tail of the cat with my teeth". The leader said "We can all kill the cat without any problem".

Paragraph for Reading - Standard - III

King Krishnadevaraya had a jester called Tenali Rama. Tenali Rama was from a place called Tenali in Andhra Pradesh. One day the king

told Tenali Rama "I want to give you a horse. Take good care of it. I am going to send an officer after some time to see the horse".

Tenali Rama took the horse with him. He wanted to have some fun out of this. So he tied the horse in a small room with only one window to it. Every day Tenali Rama gave his horse a handful of hay. After a few days the horse became very hungry. He was always looking out of the window for some more hay.

Paragraph for Reading – Standard - IV

Once a monkey lived on a tree near the bank of a river. There was a crocodile in the river. One day the crocodile came near the tree and saw the monkey eating delicious fruit. The crocodile was hungry. "My friend, I am hungry. Please give me some fruit to eat".

The monkey was moved at the request of the crocodile and threw some fruits to the crocodile. The crocodile ate the fruits and thanked the monkey. Very soon both of them became good friends. They ate fruits together. One day the monkey suggested to the crocodile to take some fruits to his wife who lived on the other bank of the river. The crocodile agreed to take some fruits to his wife. The crocodile's wife was very happy after eating the fruits.

Paragraph for Reading – Standard -V

This story is based on a famous English play called "Merchant of Venice".

Ananda was a merchant from Bengaluru. He had a friend called Bharat. Bharat needed money to marry a beautiful lady called Parvathi. So, he asked Ananda to help him. Ananda borrowed fifty thousand rupees

from an old merchant called Shrikanth. Shrikanth was a cruel man. He asked Ananda to sign a bond before giving the money. According to the bond Ananda had to give a piece of flesh from his body if he did not repay the money within three months. Ananda was confident of returning the money. But he was not able to give back the money. Shrikanth filed a case against Ananda claiming a piece of flesh from Ananda's body. The Judge wanted to help Ananda. He instructed Shrikanth to take a piece of flesh from Ananda's body without a drop of blood. Since this was not possible, Ananda's life was saved.

Paragraph for Reading – Standard -VI

Once there lived a wood-cutter named Alibaba. He was a poor man. Every morning he went to the forest to cut wood. He returned in the evening and sold the wood in the market. One day while he was cutting the wood in the forest, he heard some sound. Men came riding on horses. Alibaba saw them and was frightened. He thought they were robbers. He hid himself among the branches of a tree. Forty men led by their leader came riding on horses.

The forty men and their leader got off their horses and tied them to a tree. Their leader went near a cave and said "open sesame". The door of the cave opened. All of them went in. After some time they came out. Now the leader said "close sesame", and the door closed behind him. They went away. Alibaba waited till they were out of sight. Then he came down from the tree. He went near the cave and said "open sesame". The door opened. He went inside. He saw heaps of gold coins and precious diamonds inside the cave.

Paragraph for Reading – Standard -VII

One hot summer morning a forest officer was driving through a thick forest. He spotted from a distance a group of animals drinking water at

a large water pool. There were spotted deer drinking water and enjoying themselves. The forest officer was attracted by this beautiful sight. As he was trying to adjust his camera lens, one of the deer saw him and let out a deafening yell in terror and disappeared into the forest. All the animals understood that there was danger to their lives and ran away. In a minute all of them disappeared. The mere sight of a man was a danger signal to them. He drove further in the forest and saw some rare trees such as sandalwood, rosewood, ebony and mahogany. He also saw several monkeys, squirrels and colourful birds. He wished to see tigers and elephants. To his luck a herd of elephants walked past his vehicle. The officer took some pictures of elephants. As it was getting dark, he carefully drove back without disturbing the peaceful atmosphere of the forest.

Paragraph for Reading – Standard - VIII

It was my second year at boarding-school, and I was sitting on platform No. 8 at Ambala station, waiting for the north-bound train. I think I was about 12 at that time. My parents considered me old enough to travel alone and I had arrived by bus at Ambala early in the evening: now there was a wait till midnight before my train arrived. Most of the time, I had been pacing up and down the platform, browsing at the book-stall, or feeding broken biscuits to stray dogs; trains came and went, and the platform would be quiet for a while and then when a train arrived, it would be an inferno of shouting, agitated human bodies. As the carriage doors opened, a wave of people would sweep down upon the nervous little ticket- collector at the gate; and every time this happened, I would be caught in the rush and swept outside the station. Now tired of this game, I sat down on my suitcase and gazed dismally across the railway tracks.

Paragraph for Standard -IX

A small boy, 10 or 12 years old had been watching the plane for some time. The boy emerged from the clump of trees. He had seen the plane shot down by the enemy army and the pilot parachute down from the flaming plane. He came out from the darkness of trees and reached the wounded pilot. "Come with me Sir. The enemy will be here any minute. There is no time to lose. There is an army camp nearby. You will be safe there". The pilot staggered, tried to take 2 or 3 steps but collapsed on the ground. "I can't walk on my legs. My legs are probably fractured. Can you do something for me?" asked the pilot.

NOTE: When it is necessary curriculum based testing for reading may be carried out for all children.

Paragraph for Standard X

Norms were not collected for children studying in X standard. 10 children studying in X standard were randomly selected to read, comprehend and copy 3 paragraphs from their English textbook. All were able to read, comprehend and copy without any mistakes. Curriculum based testing (State/ CBSE/ ICSE) would be needed for assessing the child's learning abilities.

QUESTIONS FOR COMPREHENSION

Instructions for comprehension: "Now, I will ask you some questions about what you have read". (The child can answer orally. The verbatim of answers to be recorded)

Comprehension questions - Standard I

1. What is the big tree?
2. What plant is this?
3. Are the leaves big?
4. What are the colours of the roses?

(If 1 colour, 1 mark; 2 colours 2 marks)

Comprehension questions - Standard II

1. Who lived in the house?
2. One day, who came into the house?
3. Who killed the rats?
4. Who was the leader of the rats?
5. What did the rats want to do to the cat?

Comprehension questions - Standard III

1. Who was Tenali Ramakrishna?
2. Where is Tenali?
3. What was he given?
4. What did the king ask Tenali Rama to do?
5. Where did Tenali Rama tie the horse?

Comprehension questions - Standard IV

1. Where did the monkey live?
2. What animal lived near the river?
3. What was the monkey eating?
4. What did the crocodile ask?
5. What did the monkey suggest to the crocodile?

Comprehension questions - Standard V

1. Who was Ananda?
2. What did he do?
3. Why did he borrow the money?
4. Who was Shrikanth?
5. What was strange penalty fixed by Shrikanth?

Comprehension questions - Standard VI

1. Who was Alibaba?
2. Who came to the forest when he was cutting wood?
3. What did they do?
4. After they went away what did he do?
5. What did he see?

Comprehension questions - Standard VII

1. Who drove through the thick forest?
2. What did he see from a distance?
3. What did he do?
4. What happened to the deer?
5. Why did all the animals run away?

Comprehension questions— Standard -VIII

1. What is the name of the station?
2. What is the name of the train?
3. What did his parents think?
4. What was he doing at the station?
5. What happened when the train arrived?

Comprehension Questions

1. What is the age of the boy?
2. What did the boy see?
3. What did the boy want to do?
4. Why did he take the pilot there?
5. Why the pilot was unable to walk?

SCHONELL GRADED SPELLING TEST

Spelling test procedure:

- 1) Test entire group at one time (paper and pencil are required)
- 2) Use foolscap paper. On the top of the paper should be written
 - a. Child's name
 - b. The date
 - c. A or B test (Individual or group administration)
 - d. The score
- 3) Explain the procedure
 - a. Words given and marked in group of ten
 - b. A boy continues his test until he makes ten mistakes in succession
 - c. Words to be written in a column, not across the page, with space left between groups of 10
 - d. Each word will be given only three times
- 4) Giving the test
 - a. Say the word (i.e. 'meat') The word given in our study was 'mat'
 - b. Use the word in a sentence so that its meaning is clear (i.e. "The meat was too raw to eat) "we sit on the mat".
 - c. Say the word again- meat/mat
 - a. The child is then required to write the word.

5) Explain you cannot tell them where they went wrong as it is a test

6) Scoring (The Indian norms are presented in terms of mean and SD. Spelling age is not calculated). The procedure given by the author is as follows:

- Keep a running total as you mark each group of ten (i.e. 10 20 26 30 32 36)
- Take the raw score (the total number of words spelt correctly) divide by 10 and add 5, anything over a whole year must be converted to twelfths. i.e. $36/10 + 5 \text{ years} = 8.6 = 8.7$ (corrected to twelfths)
- Write correct word besides the wrong word.

Conversion Table

<u>Tenths</u>	<u>Twelfths</u>
1	1
2	2
3	4
4	5
5	6
6	7
7	8
8	10
9	11

SCHONELL GRADED SPELLING TEST

See	Cut	Mat	In	Ran
Bag	Ten	Hat	Dad	Bed
Leg	Dot	Pen	Yet	Hay
Good	Till	Be	With	From
Time	Call	Help	Week	Pie
Boat	Mind	Sooner	Year	Dream
Sight	Mouth	Large	Might	Brought
Mistake	Pair	While	Skate	Stayed
Yoke	Island	Nerve	Join	Fare
Iron	Health	Direct	Calm	Headache
Final	Circus	Increase	Slippery	Lodge
Style	Bargain	Copies	Guest	Policy
View	Library	Cushion	Safety	Patient
Account	Earliest	Institution	Similar	Generous
Orchestra	Equally	Individual	Merely	Enthusiastic
Appreciate	Familiar	Source	Immediate	Breathe
Permanent	Sufficient	Broach	Customary	Especially
Materially	Cemetery	Leisure	Accredited	Fraternally
Subterranean	Apparatus	Portmanteau	Politician	Miscellaneous
Mortgage	Equipped	Exaggerate	Amateur	Committee

COPYING

Instructions: 'I want you to copy this passage. Copy as neatly as possible'. (Ruled sheets to be provided)

(Time limit: 10 minutes)

Paragraph for Copying - Standard- I

There are many kinds of dogs.

There are big dogs and small dogs.

There are tall dogs and short dogs.

(20 words)

Paragraph for Copying - Standard -II

My little brother Bob is three years old. He loves everybody and everything. But what he likes best is the bath tub. Bob doesn't care if the bath tub is full or empty.

(33 words)

Paragraph for Copying - Standard -III

Once in a month Mr. Simple went to see his old mother. He would catch the ten 'o' clock train in the morning. He would spend the day with his mother and come back in the evening. He would come back by the six 'o' clock train.

(47 words)

Paragraph for Copying - Standard-IV

Tiny is a little brown mouse. He lives in the wall of Peter's bedroom. Peter does not know he has a house guest. Tiny comes out only when peter is not there. One day Peter went to buy something from the market.

(42 words)

Paragraph for Copying – Standard – V

Once, not so long ago, two sisters inherited land from their father. The younger sister Amy received land that was hilly and rocky. The elder one Dora was happy to have property with planted fields and orchards. There was even a stream near her orchard so she could water her plants and trees.

(53 words)

Paragraph for Copying – Standard- VI

Little children often want to stay at home. We often see little kids with shining faces going unwillingly to the school. Some of them can be seen crying when they are bundled off in their school rickshaws or vans. It is quite natural because little kids don't want to leave the cosy lap of their mothers. But some children carry this habit even when they grow older.

(67 words)

Paragraph for Copying – Standard -VII

In order to feel fit, you need strength, stamina and suppleness. All these are provided by exercise and to exercise well, you need all three.

Exercise is something that should be started gently, and built up gradually at a pace which you find comfortable.

Exercise has been proven by research to improve cardio vascular health and a human being's immunity; thereby keeping illnesses at bay. Most importantly, it enhances cognitive functions, making a person attentive and sharp. Therefore, exercise is good for both your body and mind.

(87 words) (Only the first paragraph was given while collecting the norms. This should be kept in mind while interpreting from the table).

Paragraph for Copying – Standard - VIII

His name was Alexander Fleming, and he was a poor Scottish farmer. One day, while trying to eke out a living for his family, he heard a cry for help coming from a nearby bog. He dropped his tools and ran to the bog. There, mired to his waist in black muck was a terrified boy, screaming and struggling to free himself. Farmer Fleming saved the lad from what could have been a slow and terrifying death.

The next day, a fancy carriage pulled up at the Scotsman's sparse surroundings. An elegantly dressed noble man stepped out and introduced himself as the father of the boy farmer Fleming have saved. (110 words)

Paragraph for Copying – Standard –IX

My grandmother and I were good friends. My parents left me with her when they went to live in the city and we were constantly together. She used to wake me up in the morning and get me ready for the school. She said her morning prayer in a monotonous sing-song while she bathed and dressed me in the hope that I would listen and get to

know it by heart; I listened because I loved her voice but never bothered to learn it. Then she would fetch my wooden slate which she already washed and plastered with yellow chalk, a tiny earthen ink-pot and a red pen tied them all in a bundle and hand it to me. After a breakfast of thick, stale chapati with a little butter and sugar spread on it, we went to school. She carried several stale chapatis with her for the village dogs. (150 words)

Paragraph for Standard X

Norms were not collected for children studying in X standard. 10 children studying in X standard were randomly selected to read, comprehend and copy 3 paragraphs from their English textbook. All were able to read, comprehend and copy without any mistakes. Curriculum based testing (State/ CBSE/ ICSE) would be needed for assessing the child's learning abilities.

WRITTEN EXPRESSION

Instructions: "I want you to write an essay/ few sentences on the topic 'My School'. Write all that you know". (Time limit: 15 minutes)

ARITHMETIC TEST (SCHONELL'S (I to V Std))

Instructions: " I want you to carry out the following problems". (Time limit: 30-40 minutes).

List of sums are given in the record form. Answer key is provided in page number 107

If the child expresses inability to carry out, the test can be discontinued.

Record the time taken and the procedure used to solve the sums.
Record examples of mistakes made and total score.

MATHEMATICS TEST (6TH STANDARD AND ABOVE)

Instructions: "I want you to carry out the list of statement sums given below".

(Time limit: 60 minutes) The test is provided in the page number 127

If the child expresses inability to carryout, the test can be discontinued.

Record the time taken and the procedure used to solve the sums.
Record examples of mistakes made.

(answer key provided in page number 107)

Other tests of the battery include intelligence test, test of perception and tests of memory. Follow the procedure given in the manuals of these tests. the instructions for auditory memory test is given below.

Instructions : "I will be reading out a list consisting of pairs of words. Afterwards I will read out 1 word of the pair. I want you to tell me other word of the pair". Presentation of the pairs of the words is done once. If the child's answer is wrong, the examiner gives the correct word & proceeds.

INTERPRETATION:

A child who obtains an IQ of 85 and above on an intelligence test, with performance on any of the academic skills is more than two standards below the standard he is studying in currently, indicates presence of the 'learning disability'. If the child's performance is just one or two standards below, it indicates 'leaning difficulty'. The procedure for arriving at severity rating is given below:

Answer Key for diagnostic arithmetic test.

- I Standard- 1) 1, 2) 4, 3) 7, 4) 11, 5) 9, 6) 1, 7) 2, 8) 5, 9) 4, 10) 5
II Standard- 1) 3, 2) 0, 3) 54, 4) 40, 5) 72, 6) 2, 7) 1, 8) 3, 9) 8, 10) 9
III Standard- 1) 11, 2) 7, 3) 42, 4) 6, 5) 17, 6) 97, 7) 28, 8) 118, 9) 125, 10) 95, 11) 131, 12) 69, 13) 313, 14) 109
IV Standard- 1) 88, 2) 10908, 3) 2964, 4) 7632, 5) 4176, 6) 11, 7) 302, 8) 200, 9) 303, 10) 31.2
V Standard- 1) 16/21, 2) 23/3, 3) 3/4, 4) 5/9, 5) 27/11, 6) 75/12, 7) 15/16, 8) 2, 9) 6/35, 10) 32/5, 11) 3/2, 12) 7/5

Answer key to Mathematics tests (VI standard and above)

1. 50, 220
2. Sixty eight thousand eighty seven
3. 7423656 > 7242365
4. 3699014
5. 73.59
6. HCF-12
7. LCM-273
8. 3768
9. 0.33
10. 0.56
11. 1/5
12. 4/10000 or 1/2500
13. 1.1289

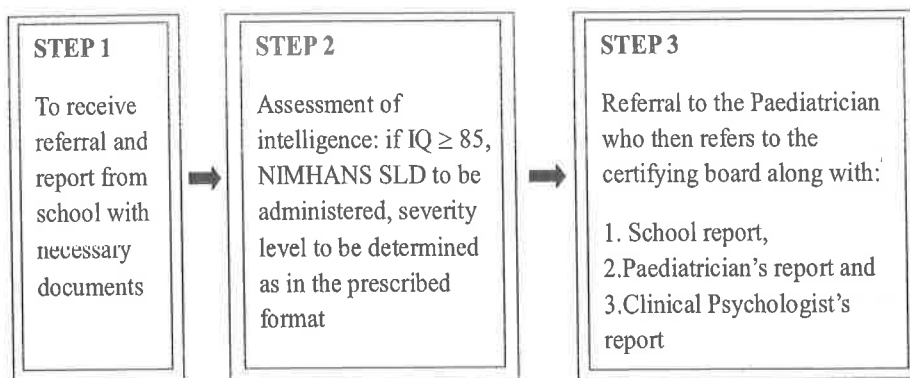
14 $X=1$
15 $X=5$
16 13
17 16
18 14
19 500064005028
20 Twenty seven thousand eight hundred and twenty six
21 Seven Crore Four Lakhs Ninty Six Thousand and Twenty Three.
22 52469
23 3003
24 25
25 2401
26 81
27 1728
28 15625
29 1000
30 XXIX
31 XXXXXV
32 12
33 29
34 409
35 78
36 28505
37 18318802
38 42,000
39 67
40 0.66
41 463.6
42 HCF-8

43 LCM-20
44 4.7 litres
45 1.95
46 20
47 $1/12$ kg or 833.3gms
48 $6/5$
49 0.3
50 0.4
51 .005 :0.5%
52 5%
53 6.33% Loss
54 675
55 0.095
56 4%
57 0.002%
58 27.50
59 $X=6$
60 $DC=1/2$ BC so DC not equal to BC, so triangle ABC is not equal to triangle CDA
61 $3/5 > 3/8$
62 $4/13 < 4/9$
63 $1/7 < 3/7$
64 Count the number of triangles correctly written
65 253.75
Total Score= 65

GUIDELINES FOR CLINICAL PSYCHOLOGISTS

(Prepared by the Core Committee, Department of Clinical Psychology, NIMHANS, 2017)

I. Procedure to be followed by certified Clinical Psychologists:



***Assessment of Intelligence:** Decision on the Intelligence assessment to be taken on a case-by-case basis; the Gazette of India states the use of WISC-III or MISIC for this purpose. In cases when Wechsler scales or MISIC is not possible or applicable, (MISIC not having norms for older children-above 15 years) Wechsler Performance test of Intelligence (WAPIS) may be considered. WISC-IV is recommended to be used only with children and adolescents who are able to converse fluently in English.

II. Determining severity ratings for each domain of learning disability:

Following assessment of intelligence, if IQ is 85 or above the NIMHANS Index for Specific Learning Disabilities has to be administered. Based on the results of the assessment, the severity rating for each domain of disability can be given, as follows:

Domain	Severity Rating
Reading* / Spelling Reading & Listening Comprehension Needs to be supplemented with curriculum-based testing when needed and based on the child's current class level.	• ○ (No disability) • Disability present, Severity level 1 (Functioning at 3 Standard below the current class of the child). • Disability present Severity Level 2 (Functioning at 4 standard or lower than the current class of child).
Writing / Copying Written expression	• ○ (No disability) • Disability present, Severity level 1 (Functioning at 3 Standard below the current class of the child).
Mathematics	• ○ (No disability) • Disability present, Severity level 1 (Functioning at 3 Standards below the current class of the child). • Disability present severity Level 2 (Functioning at 4 standard or lower than the current class of child).

* Severity rating for reading comprehension to be given only when there is no disability in reading

III.: Summary Table of Severity Ratings (SR) in each domain of disability:

Each psychological assessment report from a qualified clinical psychologist should also include a summary of severity ratings for every domain of disability. The range of the scale is 0- 2, 2 being the highest. The following summary table is provided as an example for the same (tick as applicable):

Intelligence Assessment (Specify) : Name of the test and findings:				
NIMHANS Index for Specific Learning Disabilities (Level II)	Domain	Disability absent 0	Disability Present: Severity Level 1	Disability Present: Severity Level 2
	Reading			
	Reading Comprehension*			
	Listening Comprehension			
	Spelling			
	Writing - Copying			Not applicable
	Written expression			Not applicable
	Mathematics			
*Severity of Reading Comprehension to be rated only when there is no disability in reading				

Signature of Clinical Psychologist

DEPARTMENT OF CLINICAL PSYCHOLOGY

NIMHANS, BENGALURU - 29

NIMHANS INDEX FOR SPECIFIC LEARNING DISABILITIES

RECORD FORM - LEVEL - II

Name of the child	:	O.P.No.	:
Age	:	School	:
Sex	:	Syllabus	:
Class / Medium	:		
Language Spoken by the child		Handedness	:
(1)		Parents Education :	
		Father	:
(2)		Mother	:
(3)			
Date of referral/Testing :		Parents Phone no. :	
Presenting Complaints/Brief History :			

Reasons for referral : Language spoken by the parents

Delay in milestones : Yes No

Parents estimation of child's development : In parent's view the child's development is - Appropriate to CA/more than CA/less than CA (by _____ years)

Multiaxial Diagnosis :

I

II

III

IV

V

Date :

Examiner

PROFILE ANALYSIS

NIMHANS INDEX FOR SPECIFIC LEARNING DISABILITY- REVISED (LEVEL-1 and LEVEL-2)

(Hirisave U, Kapur M, John A, Rozario J and Oommen A, 2019)

General Abilities :

Malin's Intelligence Scale for Indian Children (MISIC) is used to assess the general intellectual functioning of the child. WISC-III UK (Indian adaption) WISC IV can also be used.

MISIC PROFILE

Raw Score-TQ											
	SCALES										
	Information	Comprehension	Arithmetic	Similarities	Vocabulary	Picture Completion	Picture Arrangement	Object Assembly	Block Design	Coding	Digit Span

VERBAL IQ =
PERFORMANCE IQ -
FULL SCALE IQ =
* also known as intelligence scale for Indian Children.

Level II (for 8 to 16 years)

I. ATTENTION TEST

1 3 5 4 2 1 3 1 2 4 2 1 3 5 4 3 5 2 1 4 4 2 5 3 1
 3 1 2 5 4 1 5 4 3 2 5 4 1 3 4 4 2 1 5 3 5 1 4 2 3
 5 1 4 3 2 4 2 5 3 3 3 2 3 4 4 2 1 3 5 4 1 3 5 4 2
 2 4 5 1 3 3 1 4 5 2 5 2 3 4 4 5 4 5 2 3 2 4 1 5 3
 4 5 3 2 1 2 4 3 1 5 1 4 2 3 5 3 2 4 1 3 5 2 2 4-1
 5 1 4 3 2 3 1 4 5 2 4 2 1 5 3 4 3 5 3 1 2 1 3 1 5
 4 5 3 2 1 2 4 5 1 3 5 1 4 3 2 5 1 2 5 4 1 3 5 4 2
 2 4 3 1 5 3 1 4 5 2 4 2 5 1 3 1 5 4 3 2 5 3 1 2 4
 1 4 2 5 3 5 2 3 4 1 3 5 2 1 5 2 5 1 3 4 2 1 3 5 4
 4 2 5 2 1 5 1 4 2 3 1 3 5 4 2 2 4 1 5 3 3 5 4 2 1
 4 5 3 2 4 2 4 5 1 3 1 4 2 5 3 5 3 2 4 1 3 5 4 2 1
 2 2 5 1 3 3 1 4 5 2 5 2 3 4 1 1 4 5 2 3 2 4 1 5 3
 3 1 2 5 4 1 5 3 3 2 2 5 1 3 4 4 2 1 5 3 5 1 4 2 3
 1 0 5 4 2 5 3 1 2 4 2 1 3 5 4 3 5 2 1 4 4 2 5 3 1
 5 1 4 3 2 4 4 4 5 1 3 3 5 2 1 4 2 1 3 5 4 1 3 5 4
 2 4 5 1 3 3 1 4 5 2 5 2 3 4 1 1 4 5 2 3 2 4 1 5 3
 3 1 2 5 4 1 5 4 3 2 2 5 1 3 4 4 2 1 5 3 5 1 2 5 3
 1 3 5 4 2 5 3 1 2 4 2 1 3 5 4 2 5 2 1 4 4 2 5 3 1
 5 1 4 3 2 4 2 5 1 3 3 5 2 1 4 2 1 3 5 4 1 3 5 4 2
 4 5 3 2 1 2 4 5 1 3 1 4 2 5 3 5 2 2 4 1 3 2 5 4 1
 3 1 4 5 4 1 6 4 3 2 2 5 1 3 4 4 2 1 5 3 6 1 4 3 3
 1 3 5 4 2 5 3 1 2 4 4 2 3 5 4 2 5 2 1 4 4 2 5 3 1
 5 1 4 3 2 4 2 5 1 3 3 5 2 1 4 2 1 3 5 4 1 3 5 4 2
 4 5 3 2 1 2 5 5 1 3 1 4 2 1 5 3 5 3 2 4 1 3 5 4 2
 2 4 5 1 3 3 1 4 5 2 5 2 3 4 1 1 4 5 2 3 2 4 1 5 3

No. of Correct R Errors (Commission / Omission) Time taken

Single Number cancellation

Double Number cancellation

Remarks :

II LANGUAGE

(A) READING

	1 st Std.	2 nd Std.	3 rd Std.	4 th Std.	5 th Std.	6 th Std.	7 th Std.	8 th Std.	9 th Std.
Reads word by word									
Ignores Punctuation									
Adds words									
Omits words									
Can't use phonetic cues									
Spells out words									
Guesses at words									
Reversals									
Time Taken									

Remarks : Example of errors

Time Taken :

(Adequate means there are not many errors)

Class Level :

Impression :

B. Comprehension : (Note response in verbatim)

(Reading (R) & Listening (L) Comprehension)

Question No.	1st Std.		2nd Std.		3rd Std.		4th Std.		5th Std.		6th Std.		7th Std.		8th Std.		9th Std.	
	R	L	R	L	R	L	R	L	R	L	R	L	R	L	R	L	R	L
1																		
2																		
3																		
4																		
5																		

Remarks : Eg. of Mistakes :

If reading comprehension is inadequate then test listening comprehension.

Class Level :

Impression.

(C) SPELLING (Asked to write the word for dictation)

- 1 _____ 18 _____ 35 _____
 2 _____ 19 _____ 36 _____
 3 _____ 20 _____ 37 _____
 4 _____ 21 _____ 38 _____
 5 _____ 22 _____ 39 _____
 6 _____ 23 _____ 40 _____
 7 _____ 24 _____ 41 _____
 8 _____ 25 _____ 42 _____
 9 _____ 26 _____ 43 _____
 10 _____ 27 _____ 44 _____
 11 _____ 28 _____ 45 _____
 12 _____ 29 _____ 46 _____
 13 _____ 30 _____ 47 _____
 14 _____ 31 _____ 48 _____
 15 _____ 32 _____ 49 _____
 16 _____ 33 _____ 50 _____
 17 _____ 34 _____ 51 _____

E. Written Expression-Write an essay / few sentences on the topic "My school"

Total number of words :

Time Taken :

Remarks :

III. Perceptual Motor Abilities

(1) Bender Gestalt Test Score

Fig. A	Fig. 1	Fig. 2
1a-	4-	7-
1b	5-	8-
2-		
3-	6-	9-
Fig. 3	Fig. 4	Fig. 5
10	13-	15-
11-	14-	16-
12a		17a
12b		17b

Fig. 6	Fig. 7	Fig. 8
18a-	21a	24-
18b-	21b	25-
19-	22-	
20-	23-	

Developmental Score =

(No. of errors) =

Developmental Age =

Chronological Age =

Remarks :

III(2) Developmental Test of Visual - Motor Integration

1	2	3	4	5	6	7	8	9	10	11	12	13
R/W -	-	-	-	-	-	-	-	-	-	-	-	-
14	15	16	17	18	19	20	21	22	23	24	Total	Correct

R/W - - - - - - - - - - - - -

Remarks :

IV. Memory

(i) Visual Memory

Benton Visual Retention Test Score

No.	1	2	3	4	5	6	7	8	9	10	Total Correct
W/R	-	-	-	-	-	-	-	-	-	-	-

Remarks

(ii) Auditory Memory

(a) Familiar pairs

(b) Unfamiliar pairs

Trials					Trials				
1	2	3	4	5	1	2	3	4	5
Dog -	Cat				Cow -	Spoon			
Tree -	Plant				Water -	Paper			
Chair -	Table				Lamp -	Stick			
Man -	Woman				Floor -	People			
Day -	Night				Star -	River			
Black -	White				Boy -	Fruit			

Remarks :

Adequate means for familiar pairs, correct recall by third trial and for unfamiliar pairs, correct recall by fifth trial.

IV (A) GRADED MULTIPLICATION

(1) $22 \times 4 =$

(2) $303 \times 36 =$

(3) $76 \times 39 =$

(4) $106 \times 72 =$

(5) $348 \times 12 =$

(B) GRADED DIVISION

(6) $4 \overline{)44}$

(7) $3 \overline{)906}$

(8) $4 \overline{)800}$

(9) $5 \overline{)1515}$

(10) $5 \overline{)156}$

V (A) ADDITION (FRACTIONS)

(1) $\frac{3}{7} + \frac{1}{3} =$

(2) $7 + \frac{2}{3} =$

(3) $\frac{5}{8} + \frac{1}{8} =$

(B) SUBTRACTION (FRACTIONS)

(4) $\frac{7}{9} - \frac{2}{9} =$

(5) $2\frac{7}{11} - \frac{2}{11} =$

(6) $9\frac{5}{6} - 3\frac{7}{12} =$

(C) MULTIPLICATION (FRACTIONS)

(7) $\frac{3}{16} \times 5 =$

(8) $7 \times \frac{2}{7} =$

(9) $\frac{2}{5} \times \frac{3}{7} =$

(D) DIVISION

(10) $8 \div \frac{5}{4} =$

(11) $\frac{1}{2} \div \frac{1}{3} =$

(12) $5\frac{3}{5} \div 4 =$

Adequate Means

at Ist Standard (1-10)

Correct Answers

I A = 3/5

B = 3/5

at II Standard (1-10)

II A = 3/5

B = 3/5

at III Standard (1-14)

III A = 2/4

B = 3/5

C = 3/5

at IV Standard (1-10)

IV A = 3/5

B = 3/5

at V Standard (1-12)

V A = 2/3

B = 2/3

C = 2/3

D = 2/3

Revised Norms :

V, VI, VII, VIII std Children can solve all problems till 4th std. Refer to table 33 for norms on fractions test. VIII, IX std Children can solve all problems till V Std. Refer to table 34 for norms on mathematics test.

MATHEMATICS TEST (6TH STANDARD AND ABOVE)

Instructions: "I want you to carry out the list of statement sums given below".

(Time limit: 60 minutes)

If the child expresses inability to carryout, the test can be discontinued.

Record the time taken and the procedure used to solve the sums.
Record mistakes made.

(answer key provided in page number 107)

1) Write in numbers

Fifty Thousand two hundred twenty

2) Write in words

68,087

3) Write greater than or lesser than in symbols: > OR <

7242365 7423656

4) Answer:

$3114001 + 232756 + 475346 - 123089$

5) Answer:

$17589 \div 239$

6) HCF

24, 36

7) LCM

39, 91

8) Remove brackets & solve:

$$[632 + (411 - 101)] \times (12 \div 3)$$

9) Write in decimals:

3

4

10) Write in decimals

56

100

11) Write in fraction

0.2

12) Write in fraction

0.0004

13) Solve:

$$1.2576 - 0.1257 - 0.003$$

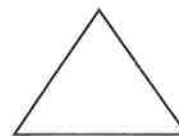
14) Find the value of x

$$2x + 5 = 7$$

15) Find the value of x

$$3x - 6 = 9$$

16, 17, 18) Write the perimeter for the following figures in centimetres



19) Write in Numbers:

Five billion, Six crore, Forty lakh, Four thousand twenty eight

20) Expand:

27826

21) Expand:

70496023

22) Write in simple manner:

$$5 \times 10^4 + 2 \times 10^8 + 4 \times 10^2 + 6 \times 10^1 + 9 \times 10^0$$

23) Write in simple manner:

$$3 \times 10^3 + 3 \times 10^0$$

24) Solve:

$$(-5) \times (-5) = (-5)^2 =$$

25) Solve:

$$(-7) \times (-7) \times (-7) \times (-7) = (-7)^4 =$$

26) Solve:

$$(-3) \times (-3) \times (-3) \times (-3) =$$

27) Find the value:

$$12^3$$

28) Find the value:

$$25^3$$

29) Nagesh sells 10 boxes of biscuits. Each box contains 10 packs, each pack has 10 biscuits. How many biscuits (altogether) has he sold?

30) Write this in Roman numerals:

29

31) Write this in Roman numerals:

55

32) Write this in Roman numerals:

555

33) Write in Hindu Arabic way:

XII

34) Write in Hindu Arabic way:

XXIX

35) Solve:

$$23 + 15 + 40$$

36) Solve:

$$20000 + 7965 + 456 + 81 + 3$$

37) Population of the following states in 1981:

Delhi- 5727730

Bombay: 8202759

Calcutta: 9165650

Madras: 427663

Write the total population

38) Kiran earns Rs. 3,500 per month. What are his earnings for the whole year?

39) Oral sum:

$$67 \div 1$$

- 40) Oral sum:
 $6 \div 9$
- 41) Solve:
 $8810 \div 19$
- 42) Solve:
HCF: 8, 16
- 43) Solve:
LCM: 5, 20
- 44) One bottle has $\frac{3}{4}$ litres of juice, another bottle has $2\frac{7}{9}$ and the third bottle has $\frac{7}{6}$. What is the total quantity of juice? (Write in litres)
- 45) A Room's width is $4\frac{7}{8}$ meters; its length is $6\frac{5}{6}$ meters.
How much is the length of the room more than it's width?
- 46) 'A' Gets 80 marks, 'B' gets $\frac{1}{4}$ of 'A'. How much has 'B' got?
- 47) 4 rice bags contain $\frac{1}{3}$ kg of rice each. If all bags have equal amounts of rice, how much does each bag contain?
- 48) Convert % into fraction: 120%
- 49) If $100\% = 3$; then, $30\% = ?$
- 50) If $100\% = 2$; then $40\% = ?$
- 51) If $\frac{1}{2} : 0.5 : 50\%$: then $\frac{1}{200} : ? : ?$
- 52) 'O' buys 1 kg of butter, he gets only 950 gm. What is the % of loss incurred?
- 53) 'A' buys a cycle for Rs.15, 001. He sells it for Rs.14, 051. What is the percentage of loss or percentage of profit incurred?

- 54) 'A' takes a loan of Rs. 1500 at the rate of 15% interest. What interest does he will have to give after 3 years?

- 55) Convert % into decimal:
9.5%

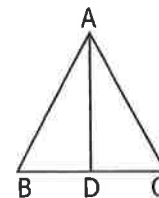
- 56) Convert decimal into %:
0.04

- 57) Convert decimal into %:
0.002

- 58) The length of one wire is 40.25 cms, if we cut 12.75 cms from the wire, what is the length of the remaining wire?

- 59) Solve: $x^2 - 12x + 36 = 0$
Find the value of x

- 60) $AD = AC$
AD bisects BC



Prove that $\triangle ABC$ is not similar to $\triangle CDA$

- 61) Write > or < symbol:
 $\frac{3}{5}$ ___ $\frac{3}{8}$

62) Write > or < symbol:

4/13 ____ 4/9

63) Write > or < symbol:

1/7 ____ 3/7

64) Write your name in English capitals and count the number of angles in it (number of acute, obtuse and right-angles)

65) Bharath buys 2 ½ kg of apples which is Rs. 35 per kg; 1 ½ kg of oranges which is Rs. 28 per kg; and 3 kg of mangoes which is Rs. 40 per kg; ¼ kg of banana which is Rs. 17 per kg. Prepare a receipt based on this as would be given in a shop.

Revised Norms :

V, VI, VII, VIII std Children can solve all problems till 4th std. Refer to table 33 for norms on fractions test. VIII, IX std Children can solve all problems till V Std. Refer to table 34 for norms on mathematics test.

Table - 1

Performance of normal children on Brigance Diagnostic Inventory

a) Characteristics of the Sample (N = 50)

Age (in Months)		Sex		Education			Class Performance	
Mean	SD	Boy	Girls	LKG	UKG	1 st	Adequate	Inadequate
		N(%)	N(%)	N(%)	N(%)	N(%)	N(%)	N(%)
59.2	6.8	26(52)	24(48)	17(34)	25(50)	8(16)	48(96)	2(4)

b) Performance on Basic Skills of Brigance Inventory

	N	%
Adequate (no difficulty)	47	94
Mild Difficulty	3	6

Table - 2

Performance of normal children on NIMHANS SLD Index (Level - I)

a) Characteristics of the Sample (N = 35)

Age (in Months)		Sex		Education
Mean	SD	Boys	Girls	1 st Std.
		N%	N%	N%
78.1	6.1	21(60)	14(40)	35(100)

b) Performance on NIMHANS SLD Index (Level - I)

Subtests

	Visual Motors Skills		Writing of Capital Letters		Writing of Small Letters	
	N	%	N	%	N	%
Adequate (No Errors)	33	94	27	(77)	24	(69)
Few Errors	2	6	8	(23)	11	(23)

	Writing of Alphabet Preceding a specific alphabet		Writing of alphabet Succeeding a specific alphabet	
	N	%	N	%
Adequate (No Errors)	27	(77)	24	(69)
Few Errors	8	(23)	11	(31)

Table (b) Continued.....

	Writing of Numbers (Serially)		Writing of Number Preceding a Specific Number		Writing of No. Succeeding a Specific Number	
	N	%	N	%	N	%
Adequate (No Errors)	35	(100)	30	(85)	31	(88)
Few Errors	0	(0)	5	(15)	4	(12)

Table (c) Performance on NIMHANS SLD Index (Level-II)

ATTENTION : Number Cancellation (N-26)

	Mean	SD	Range
Adequate (Correctly Cancelled)	32	10.5	(21.5 -42.5)
Numbers Wrongly Cancelled	1.0	2.3	(1.3 - 3.3)
Numbers Omitted	5	8.94	(-3.4-13.94)

Table (c) Continued.....

	Reading (I Grade Text) N-17)		Reading Comprehension (I Grade) N=17	
	N	%	N	%
Adequate (No Errors)	8	(47)	3	(18)
Few Errors	7	(41)	11	(64)
Inadequate	2	(12)	3	(18)
Mean Errors (SD)	1.4	(0.6)	Mean Correct (SD)	2.5 (1.1)
Range	(0.7 - 2.1)		(1.3 - 3.6)	

Table (c) Continued.....

	Simple Digit Addition		Simple Digit Subtraction		Simple Digit Multiplication		Simple Digit Division	
	N	%	N	%	N	%	N	%
Adequate (No Errors)	16	(76)	11	(53)	1	(4)	1	(4)
Few Errors	4	(19)	8	(38)	10	(48)	-	-
Inadequate	1	(5)	2	(9)	10	(48)	20	(96)
Mean Correct (SD)	4.0	(1.0)	4.0	(1.0)	1.6	(0.79)	0.14	(0.2)
Range	(3.0 -5.0)		(3.0 - 5.0)		(0.8 - 2.3)		(-0.1 - 0.3)	

**Performance of typically growing children on
NIMHANS SLD Index - Level - II**

Table - 3

a) Characteristics of the Sample (N = 20)

Age (in months)		Sex Boys girls		Education II std.	
Mean	SD	N%	N%	N	N%
90.0	5.1	9 (47)	11(53)	20	(100)

Table b) : Attention (Number cancellation)

	Mean	SD	Range
Adequate (Correctly Cancelled)	39	11.7	(20 - 50.7)
Numbers Wrongly Cancelled	0.7	2.5	(-1 - 3.2)
Numbers Omitted	8.92	10.9	(-1.9 - 19.8)

Table (b) Continued.....

	Reading (II Grade Text) N = 20		Reading Comprehension (II Grade) N = 20	
	N	%	N	%
Adequate (No Errors)	13	(65)	6	(33)
Few Errors	4	(20)	10	(50)
Inadequate	3	(15)	4	(17)
Mean Errors (SD)	1.5	(0.7)	Mean Correct (SD) 2.8	(1.0)
Range	(0.7-2.2)		Range	(1.8 - 3.8)

Table (b) Continued.... Arithmetic Tests (N = 20)

	Graded Addition		Graded Subtraction		Graded Multiplication		Graded Division	
	N	%	N	%	N	%	N	%
Adequate (No Errors)	12	(60)	12	(60)	4	(20)	1	(5)
Few Errors	8	(40)	8	(40)	12	(60)	10	(50)
Inadequate	-	-	-	-	4	(20)	9	(45)
Mean Correct(SD)	4.6	(0.1)	4.5	(0.5)	3.9	(0.8)	2.2	(1.2)
Range	(3.4 - 4.7)		(3.5 - 5.0)		(3.1 - 4.7)		(0.9- 3.4)	

Table (c) Visual Perceptual abilities as assessed on BGT (N = 20)

	Developmental Age (in years)	
	N	%
Adequate (DA Equal to CA)	5	(25)
Above Average (DA is > CA)	15	(75)
Mean DA (SD)	8.9	(1.3)
Range	(7.6 - 10.2)	

STANDARD III

Table - 4

Number Cancellation Test N = 44

	No. Correct	No. Wrong	No. Omitted
Mean	39.77	0.19	4.64
SD	9.85	0.70	12.47

Table - 5
BENDER GESTALT TEST N = 39

Errors	Distortion	Rotation	Integration	Perseveration	Bender score	Age Equivalent (Months)
Mean	0.52	0.63	0.74	0	1.95	132.00
SD	0.98	0.96	0.99	0	1.65	41.30

Table - 6
Simple Arithmetic N = 45

	ADD Co.	ADD WR	ADD OM	SUB CO	SUB WR	SUB OM	MUL. CO.	MUL. WR	MUL. OM	DIV CO.	DIV WR.	DIV OM	MISC CO	MISC WR	MISC OM
Mean	4.76	0.11	0.02	4.60	0.22	0.18	3.96	0.44	0.58	0.11	0.02	4.80	1.96	0.36	1.64
SD	0.80	0.32	0.15	0.89	0.52	0.78	1.02	0.66	1.06	0.75	0.30	0.84	1.15	0.57	1.23

Table - 7
GRADED ARITHMETIC N=45

	G.ADD CO	G.ADD WR	G.ADD OM	G.SUB CO	G.SUB WR	G.SUB OM	G.MUL CO	G.MUL WR	G.MUL OM	G.DIV CO	G.DIV OM	G.DIV WR
Mean	4.42	0.47	0.11	3.22	1.38	0.40	0.38	0.02	4.71	0.20	0.02	4.78
SD	1.01	0.76	0.75	1.59	1.37	1.29	1.21	0.15	1.04	0.94	0.15	1.04

Graded - G Miscellaneous - Misc
Addition - Add Correct - Co
Subtraction - Sub Wrong - Wr
Multiplication - Mul- Omitted - Om
Division - Div

Table - 8
ACADEMIC SKILLS

	No. Adequate	No. Inadequate
Reading N=47	79%	21%
Comprehension N = 47	92%	8%

STANDARD - IV
Table - 9
Number Cancellation Test N = 38

	Number Correct	Number Wrong	Number Omitted
Mean	46.00	0.02	15.53
SD	10.32	0.16	26.47

TABLE - 10
BENDER GESTALT TEST N=41

Errors	Distortion	Rotation	Integration	Perseveration	Bender Score	Age Equivalent (Months)
Mean	1.29	0.29	0.44	0.02	2.02	139.15
SD	1.12	0.56	0.63	0.22	1.56	41.57

Table - 11
SIMPLE ARITHMETIC N = 41

	ADDITION			SUBTRACTION			MULTIPLICATION			DIVISION			MISCELLANEOUS		
	CO	WR	OM	CO	WR	OM	CO	WR	OM	CO	WR	OM	CO	WR	OM
Simple Addition															
Mean	4.78	0.02	0.12	4.88	0.12	0	4.85	0.15	0	4.90	0.02	0.02	3.46	0.02	0.51
SD	0.82	0.30	0.64	0.51	0.51	0	0.36	0.36	0	0.37	0.22	0.31	1.32	0.22	1.33

TABLE - 12

GRADED ARITHMETIC

	GA		GA		GS		GS		GS		GM		GM		GM		GD	
	CO	WR	OM	WR	CO	WR	CO	WR	OM	OM	CO	WR	OM	WR	CO	WR	OM	OM
Mean	4.85	0.15	0	4.63	0.37	0	4.63	0.37	0	2.88	1.66	0.45	3.12	1.02	1.02	1.05		
SD	0.36	0.36	0	0.70	0.70	0	0.70	0.70	0	1.38	1.28	1.11	1.29	0.91	1.55			

GA - Graded Addition CO = Correct
GS - Graded Subtraction WR = Wrong
GM = Multiplication OM = Omission
GD = Graded Division

TABLE - 13
FRACTIONS

	FAD		FAD		FSU		FSU		FSU		FMU		FMU		FDI		FDI	
	CO	WR	OM	OM	CO	WR	WR	OM	OM	CO	WR	OM	CO	WR	CO	WR	OM	OM
Mean	0.20	0.51	2.29	0.12	0.12	0.24	0.24	2.56	0.24	0.24	0.02	2.66	0	0.29	2.71			
SD	0.40	0.87	1.23	0.33	0.33	0.70	0.70	1.03	0.70	0.70	0.37	0.94	0	0.90	0.90			

FAD - Fractions Addition
FSU - Fractions Subtraction
FMU - Fractions Multiplication
FDI - Fractions Division

Table -14 ACADEMIC SKILLS

	Number Adequate %	Number Inadequate %
Reading N = 41	100	0
Comprehension N = 41	100	0

STANDARD - V

TABLE - 15

NUMBER CANCELLATION TEST (SINGLE) N=40

	Correct	Wrong	Omitted
Mean	59.10	0.20	10.48
SD	16.78	0.22	18.70

TABLE - 16

NUMBER CANCELLATION TEST (DOUBLE) N = 40

	Correct	Wrong	Omitted
Mean	60.08	0.68	25.40
SD	12.47	1.19	52.40

Table - 17

BENDER GESTALT TEST N = 38

Errors	Distortion	Rotation	Integration	Perseveration	Bender Score	Age Equivalent (Months)
Mean	1.08	0.24	0.47	0.02	1.87	138.47
SD	1.17	0.59	0.69	0.27	1.63	39.38

**TABLE - 18
SIMPLE ARITHMETIC**

	AD	AD	AD	SU	SU	SU	MU	MU	DI	DI	DI	MI	MI	MI
	CO	WR	OM	CO	WR	OM	CO	WR	CO	WR	OM	CO	WR	OM
Mean	4.90	0.01	0	4.83	0.18	0	4.88	0.13	4.73	0.10	0.18	3.78	0.15	0.02
SD	0.30	0.30	0	0.81	0.81	0	0.46	0.46	0.88	0.38	0.81	0.58	0.43	0.27

**TABLE - 19
GRADED ARITHMETIC**

	GA	GA	GA	GS	GS	GS	GM	GM	GM	GID	GID	GID
	CO	WR	OM	CO	WR	OM	CO	WR	OM	CO	WR	OM
Mean	4.73	0.28	0	4.50	0.50	0	2.88	1.53	0.60	2.85	0.93	1.25
SD	0.55	0.55	0	0.64	0.64	0	1.40	1.24	1.15	1.56	1.07	1.60

**TABLE - 20
FRACTIONS**

	FAD	FAD	FAD	FSU	FSU	FSU	FMU	FMU	FMU	FDI	FDI	FDI
	CO	WR	OM	CO	WR	OM	CO	WR	OM	CO	WR	OM
Mean	1.13	1.43	0.45	0.85	1.33	0.83	1.73	0.53	0.75	0.02	1.25	1.78
SD	0.76	0.90	0.90	0.48	0.89	0.96	1.18	0.82	1.21	0.32	1.28	1.29

Table - 21
ACADEMIC SKILLS

	Number Adequate %	Number Inadequate %
Reading N = 39	49	51
Comprehension N = 39	44	56

Table 22

PERFORMANCE OF CHILDREN ON READING AND READING COMPREHENSION TESTS

Performance of children studying in VI Standard (N =20), VII Standard (N=20), VIII Standard (20) and IX standard (20) on Reading and Reading Comprehension Tests.

Standard	Adequate Reading (2-7 mistakes)	Adequate Comprehension (3-5 correct answers)
VI Standard	N=18	N=18
VII Standard	N=16	N=18
VIII Standard	N=14	N=16
IX Standard	N=16	N=18

(Frequently made errors in reading include Guessing, Addition of a letter, Inability to use a phonetic cue)

SPELLING, WRITTEN EXPRESSION AND COPYING TASKS

Table 23

Performance of 1st standard children (N=36) on spelling, written expression and copying task

	Mean	SD	Range (Mean±2SE)
Schonell's (Total No. of Words=100) Spelling	12.9	3.16	12-14
Written expression (Correct Number of words written)	14.41	6.79	12-17
Errors in written expression			
Spelling	1.11	1.2	0-1
Omission	0.22	0.54	0
Grammar	1.41	0.87	1-2
Capital vs Small	0.1	0.23	0
Punctuation	0.11	0.52	0
Reversal	0.1	0.23	0
Copying (Total No. of Words=20)	19.4	3.4	18-20
Errors in copying			
Spelling	0.6	0.8	0-1
Omission	0.1	0.23	0
Grammar	0	0	0
Capital vs Small	0.02	0.2	0
Punctuation	2.1	1.3	1-3
Reversal	0.2	0.4	0

Table 24

Performance of 2nd standard children (N=29) on spelling, written expression and copying task

	Mean	SD	Range (Mean±2SE)
Schonell's (Total No. of Words=100) Spelling	23.93	10.69	20-28
Written expression (Correct Number of words written)	21.20	10.99	17-26
Errors in written expression			
Spelling	2.07	3.63	1-4
Omission	0.07	0.25	0
Grammar	0.27	0.59	0-1
Capital vs Small	0.24	0.57	0-1
Punctuation	2.44	2.28	2-3
Reversal	0.03	0.18	0
Copying (Total No. of Words=33)	32.96	0.19	32-33
Errors in copying			
Spelling	1.07	1.49	1-2
Omission	0.24	0.63	0-1
Grammar	0.17	0.38	0
Capital vs Small	0	0	0
Punctuation	2.75	1.50	2-3
Reversal	0.10	0.4	0

Table 25

Performance of 3rd standard children (N=28) on spelling, written expression and copying task

	Mean	SD	Range (Mean±2SE)
Schonell's (Total No. of Words=100) Spelling	37.14	8.03	34-40
Written expression (Correct Number of words written)	28.16	3.01	24-33
Errors in written expression			
Spelling	1.58	1.65	1-2
Omission	0.16	0.37	0
Grammar	1.67	1.37	1-2
Capital vs Small	0.58	0.99	0-1
Punctuation	1.87	1.99	1-3
Reversal	0	0	0
Copying (Total No. of Words=47)	44.25	2.65	43-45
Errors in copying			
Spelling	0.32	0.54	0-1
Omission	0.03	0.18	0
Grammar	0	0	0
Capital vs Small	0.17	0.39	0
Punctuation	0.67	1.38	0-1
Reversal	0	0	0

Table 28

Performance of 6th standard children (N=25) on spelling, written expression and copying task

	Mean	SD	Range (Mean±2SE)
Schonell's (Total No. of Words=100) Spelling	57.88	17.52	51-65
Written expression (Correct Number of words written)	69.8	15.9	63-76
Errors in written expression			
Spelling	3.51	2.69	3-4
Omission	0.05	0.32	0
Grammar	3.72	2.40	3-5
Capital vs Small	0.86	1.11	1
Punctuation	3.10	2.59	2-4
Reversal	0.02	0.16	0
Copying (Total No. of Words=67)	66.64	1.9	66-67
Errors in copying			
Spelling	0.36	0.76	0-1
Omission	0	0	0
Grammar	0	0	0
Capital vs Small	0.32	0.56	0-1
Punctuation	0.48	0.71	0-1
Reversal	0	0	0

Table 29

Performance of 7th standard children (N=25) on spelling, written expression and copying task

	Mean	SD	Range (Mean±2SE)
Schonell's (Total No. of Words=100) Spelling	55.14	15.30	50-60
Written expression (Correct Number of words written)	80.76	22.13	72-90
Errors in written expression			
Spelling	1.8	1.80	1-3
Omission	0.08	0.27	0
Grammar	1.4	1.22	1-2
Capital vs Small	0.92	1.29	0-1
Punctuation	2.4	2.12	2-3
Reversal	0	0	0
Copying (Total No. of Words=44)	42.6	3.65	41-44
Errors in copying			
Spelling	0.64	1.11	0-1
Omission	1.4	4.55	0-3
Grammar	0	0	0
Capital vs Small	0.48	0.87	0-1
Punctuation	1.2	1.19	0-2
Reversal	0	0	0

Table 30

Performance of 8th standard children (N=37) on spelling, written expression and copying task

	Mean	SD	Range (Mean±2SE)
Schonell's (Total No. of Words=100) Spelling	62.88	19.23	55-71
Written expression (Correct Number of words written)	97.44	37.42	83-112
Errors in written expression			
Spelling	1.88	2.02	1-3
Omission	0.08	0.28	0
Grammar	2.76	2.35	2-4
Capital vs Small	1.36	1.38	1-2
Punctuation	2.48	1.96	2-3
Reversal	0	0	0
Copying (Total No. of Words=110)	110	0	110
Errors in copying: no errors			

Table 31

Performance of 9th standard children (N=30) on spelling, written expression and copying task

	Mean	SD	Range (Mean±2SE)
Schonell's (Total No. of Words=100) Spelling	67.1	18.1	61-74
Written expression (Correct Number of words written)	99.4	36.8	85-114
Errors in written expression			
Spelling	1	1.3	1-2
Omission	0.2	0.5	0
Grammar	1.3	1.3	1-2
Capital vs Small	0.6	0.8	0-1
Punctuation	1.2	1.6	1-2
Reversal	0	0	0
Copying (Total No. of Words=150)	145	2.5	143-149
Errors in copying	No errors of spelling, omission, grammar, capital vs small, punctuation or reversal		

Table 32

Performance of 10th standard children (N=34) on spelling task

	Mean	SD	Range (Mean $\pm$ 2SE)
Schonell's (Total No. of Words=100) Spelling	75.1	9.7	72-78

*Norms were not collected for 10th standard children on copying and written expression.

These children have to be tested using their curriculum based reading tasks.

Table 33

**PERFORMANCE OF CHILDREN ON FRACTIONS SUB-TEST
(SCHONELL ARITHMETIC TEST) AND ON MATHEMATICS TEST**

Performance of children studying in V Standard (N=15), VI standard (N= 15), VII Standard (N= 18), VIII standard (N=15), on Fractions test (Total=12 sums).

Standard	Mean	SD	Range (M $\pm$ 1SD)
V standard	1.07	0.6	0-1
VI standard	4.90	1.2	4-6
VII standard	6.1	1.2	5-7
VIII standard	11.02	0.94	10-12

Table 34

Performance of children studying in VIII standard (N=37) and IX standard (N=30) on Math Test (Total=65 sums).

Standard	Mean	SD	Range (M $\pm$ 2SE)
VIII standard	10.20	5.92	8-12
IX standard	20.5	9.56	17-24

*Norms were not collected for 10th standard children. These children have to be tested based on their syllabus (Arithmetic, algebra and geometry).

CHAPTER 5

INTERVENTION STRATEGIES FOR LEARNING DISABILITIES

Joe Rozario, Anna Oommen, Uma Hirisave

Each student is unique. One differs from another intellectually, emotionally, socially and physically. They get their regular education in classrooms. This is adequate for most of the students. Some students, however, deviate so greatly from normal limits that the regular class room teaching is inadequate. Special needs of these students must be met with additional inputs from outside the regular class rooms. These students may have difficulties in one or several of the academic areas such as reading, arithmetic, spelling and writing. Some of these students exhibit good skills and excel in many areas other than the problem area. Some others are merely slow in acquiring school related skills. A few, besides academic problems also exhibit emotional or conduct problems.

The causes of scholastic problems are many and complex. Seldom, if ever, is a single cause of difficulty unequivocally identified. Attempts to pinpoint specific biological correlates of individual learning problems have not often been productive (Hallahan, Kauffman, and Lloyd, 1985). However, biopsychosocial factors can contribute to the development of learning problems which may result in academic backwardness. Family disintegration, poverty, social class discriminations and other negative social and cultural factors can give rise to learning problems (Berkowitz and Rothman, 1960). Educational factors such as crowded class rooms, poor study habits, inadequate and inappropriate teaching can also give rise to learning problems (Kauffman, 1985).

Students who have academic problems are a heterogeneous group. Theoretically they may have (i) academic discrepancy; (ii) cognitive problems; (iii) biological correlates and (iv) social emotional problems.

The first area of academic discrepancies includes basic reading skills, reading comprehension, maths calculation, written expression, oral expression and listening comprehension. The second area of cognitive skills includes short attention, perceptual deficits, poor memory and problem solving abilities. The third area, biological correlates, includes neurological signs, chromosomal abnormalities, biochemical factors and laterality of the brain. The fourth area, socio emotional problems, includes motivation, poverty, learned helplessness and other school related problems. A student with learning problem may show a combination of the above mentioned factors. When severity or degree of each problem area and its combinations are considered, the possible outcome of varieties will be enormous (Adelman and Taylor, 1985).

Instructional Framework

Medically oriented theories view learning disabilities as overt symptoms of underlying biological pathology. The inferred pathology has been conceptualized by different theorists as affecting perceptual systems, for example, perceptual motor functioning as described by Mann in 1979. According to these theoretical view points, the content for instruction should consist of skills related to the deficient underlying pathology (e.g. visual discrimination, fine motor skills, eye movements). The major emphasis in remediation is on strengthening the process weaknesses.

Proponents of psycho-educational model of learning disabilities interpret academic deficits as reflecting aberrations in the ability to perceive, integrate and remember auditory and visual information associated with

listening, speaking, reading and writing tasks (Johnson and Myklebust, 1967). Assessment procedures are used to identify specific strengths and weaknesses in auditory and visual processes related to academic functioning and this information is used for clinical teaching. In contrast to the medical model, the psycho-educational approaches advocate teaching academic skills by taking into account the students' modality preferences or areas of processing strength (visual, auditory or kinesthetic), the nature of the content (verbal or non-verbal) and the response requirements (oral or written). The emphasis on remediating the underlying process deficiency is secondary. Unfortunately, it is difficult, if not impossible, to find any evidence beyond testimonials and anecdotal reports that support the assumptions, treatment methods and stated outcomes associated with medical and psycho-educational models (Lyon, 1985). No data exists that indicates that underlying processes can be strengthened and that such improvements generalize to academic skill development (Wong, 1986).

Neuro-psychological models and theories of learning disorders conceptualise learning strengths and weaknesses as manifestations of efficient and inefficient brain regions or systems (Fisk and Rourke, 1983). Within this context, instruction is designed so that an intact hemisphere or neural system will be exploited in bypassing the areas of dysfunction. For example, a number of researchers have extended theoretical concepts relevant to cerebral asymmetry and hemispheric specialization to explain both the information processing and the response characteristics of learning disabled children (Kirby and Robinson, 1987). In this context, neuropsychological aptitudes such as simultaneous and sequential processing and linguistic (Phonological, semantic, syntactic) and visual-spatial reasoning capacities are considered when direct instruction of

academic skill is undertaken. Evidence to support the clinical efficacy and scientific validity of these neuropsychological approaches to assessment and instruction of learning disabled students remain sparse. In significant contrast to medical, psycho-educational and neuropsychological models and theories, behavioural approaches conceptualize learning disabilities as resulting from characteristics of academic tasks (Treber and Lehey, 1983). There is no assumption of underlying pathology or process deficiency. Assessment and instructional activities are directed towards identifying academic skills deficits and modifying them with techniques derived from learning theory. The major assumption guiding this approach to instruction is that academic content consists of skill hierarchies and that complex academic behaviours such as reading, writing and mathematics are dependent on adequate development of pre-requisite academic enabling behaviours. Recent reviews of the effectiveness of behavioural interventions with learning disabled individuals have generally indicated favourable results with respect to increasing attentional and academic skills (Gadow, Torgeson and Dahlem, 1985).

Another theoretical context within which instructions are planned and presented is exemplified by studies that have characterized many learning disabled children as deficient in a variety of language functions. In contrast to other broader instructional models, these models are primarily used to conceptualize the nature of linguistic deficits associated with reading, writing and spelling disorders. More specifically deficiencies in phonological coding and short term memory for linguistic material have been linked to poor reading, spelling and word retrieval problems (Mann and Liberman, 1984).

Cognitive psychologists have provided a view of learning disabilities suggesting that instructions should be directed toward enabling students

to exercise self-conscious, deliberate, and strategically applied efforts and when learning academic content (Brown and Compione, 1986). Instructional emphasis is placed on (a) increasing the learners' awareness of task demands (b) teaching the students to use appropriate strategies to facilitate task completion and (c) teaching the student to monitor the success of the strategy. Support for the application of cognitive principles to assessment and instruction has come from studies that have characterized the learning disabled as one who fails to enlist efficient, task appropriate strategies and orchestrate their use.

For many years psychologists and educationists have tried to come up with techniques that would be of value to teachers in dealing with children with various kinds of scholastic problems. A number of programmes and techniques have been developed but have been too elaborate, time consuming and requiring great skills for carrying out remedial work. It has also been demonstrated that suitable simple strategies could be used by teachers to help students with scholastic problems in the school setting.

Reading

Learning to read presents a considerable obstacle for 4-10 percent of the children in a normal class room (Mann and Brady, 1988). Mercer and Mercer (1985) report that about 10 to 15 percent of the general school population experience difficulty in reading. Two studies (Sheldon and Carrillo, 1952 & Carrillo, 1976) on the relation between environmental differences and reading problems concluded that :

- i) home environment is important in the genesis of reading ability.

- ii) retarded readers seem to show a background of slow development in verbal skills, speech defects and motor skills.
- iii) The emotional histories of poor readers revealed a lack of adjustment to change, friends and independence. Wedge and Prosser (1973) found that there were more poor readers in the disadvantaged group than in the control groups. Mann and Brady (1988) uncovered an association between early difficulties in learning to read and impairment in one or more aspects of language processing. A longitudinal study (Lesgold and Resnick, 1981) found that the lack of word processing efficiency contributes to reading disability.

The study done by the author (Rozario, 1991) indicated that 25 children studying in fourth standard English Medium schools, identified by the teachers as poor readers were poor in recognizing simple common words and they were unable to exploit the system of phonics. Therefore a target was set for remedial education - (1) improving their basic sight words (2) improving their phonic skills.

To teach the basic sight words the list of basic sight words given in their text book was used. These words were written on flash cards and everyday the students were made to be familiar with 4 to 5 words. They pronounced the words loudly and wrote these words on the black board without looking at the card. They were encouraged to use those words in a sentence and made to use these words atleast once during the remedial education class. At the end of the week specific attention was paid to revise all the words during the week.

To teach the phonic skills Rosner's (Rosner, 1991) decoding units were used. This list contains 108 decoding units and each unit has a set of words. Each set is organized into four levels of increasing difficulty. The students were requested first to start with level A from set 1, then proceed to level A from set 2 and to the further set. Then he was asked to proceed to level B as before and then to level C. The level at which a student can read adequately all the words was his level of reading. That level was kept as the starting point.

First the decoding unit was taught as a - n means / an / when the student can pronounce the word, the first word from the unit of that level was taken. The student was told then / b.an/ means/ban/. Then the student repeated it. Then the next word in the unit was taught. In addition to teaching them sight words and training in phonic skills they were also given reading practice every day for 10 to 15 minutes. The common type of errors they made were repeating words, spelling out words, omitting words, difficulty in pronouncing certain syllables, ignoring punctuation and mumbling. The mistakes were noted down and were informed at the end of the reading. They were made to correct them on the following day before reading. They were reminded of the mistakes and advised to avoid such mistakes. If the mistakes were corrected or the errors reduced, they were given verbal reinforcements by the group. At the end of 25 sessions, the reading performance of the group showed a significant improvement. The overall errors made in reading came down from 59 to 13 after the therapy sessions.

Spelling

Spelling is the forming of words through traditional arrangements of letters. The ability to spell is essential because it allows one to read

correctly what is written. Carpenter and Miller (1982) found that children who had trouble recognizing words in reading had poor spelling skills. Ekwall (1985) noted that phonetic spellers mispronounced phonetically irregular words. He suggests that a child in order to spell must be able to read the word, possess knowledge and skill in certain relationships of phonics and structural analysis, apply phonic generalizations, visualize the word and use motor ability to write the word. According to him, spelling problems may stem from problems in visual memory, auditory memory, auditory and visual discrimination.

Lovitt, Guppy and Blattner (1969) used the study-test technique with fourth grade children and found significant improvement in the spelling skills. Bryand, Drabin and Gettinger (1981) used fixed and flow word lists wherein spelling words are presented and taught in fixed word lists. They reported improvement in spelling performance of children using this method. Kauffman, Hallahan, Haes, Brame and Boreu (1978) tested the effectiveness of the imitation method in teaching spelling skills. In this method, the teacher provides an oral and written model of the spelling word and the child is required to imitate the model by spelling the model aloud and writing it. The child receives immediate feedback and praise for correct responses. Incorrect reponses are followed by retraining. They found this method was particularly useful for words that did not follow regular phonetic rules words for which the child should use visual memory.

The author (Rozario, 1991) taught the students letter sound combinations through phonics. The mismatches, that is, the words with irregular spellings were taught using rote memory. The student was encouraged to use some of the other senses than just eyes and ears. The word to be learnt was shown to the student. The student pronounced it loudly,

pronounced every letter. He wrote the letter down in his note book. Then he wrote the word five times with his eyes closed. Every day they were given drill exercises using the above mentioned techniques. They were given five words at the beginning of the session and asked to look at them carefully. Then they wrote the spelling of each word without looking at the list. If they made mistakes they were asked to look at it again, use the cues given as before and write the correct spelling. The post therapy assessment indicated significant improvement. The average spelling score of the students went up to 8.76 from 6.00.

Writing

Writing is a tool for communication and it is both a skill and means of self expression. The complex process of writing integrates visual, motor and conceptual abilities and it is a major means through which students demonstrate their knowledge in academic subjects : Classroom instructions in handwriting usually begins in kindergarten. After the third grade, emphasis is placed on writing as a form of meaningful self expression. Mercer and Mercer (1985) list the following common errors among the learning disabled children - slowness, incorrect directionality of letters, too much or too little slant, spacing difficulty, messiness, inability to stay on a horizontal line, illegible letters, too much or too little pencil pressure or mirror writing.

Weiderholt, Hemmill and Brown (1983) suggest that teachers in the primary grades should devote at least 10 minutes each day to teaching handwriting. Teachers should demonstrate the correct way to form letters and supervise students' handwriting efforts carefully. Also, the teacher should help the student develop a positive attitude towards handwriting. Hofmeister (1981) lists six instructional errors to avoid, they are

unsupervised handwriting practice, lack of immediate feedback to errors, lack of emphasis on student analysis of errors, failure to provide close range models of correct letter formation, repeated drill of both correct and incorrect letter production and misplaced emphasis on activities of limited value.

The author (Rosario, 1991) found the following type of errors in his study of 25 learning disabled children with writing disabilities; improper holding of pen, wrong positioning of the writing paper, illegibility of some letters, incorrect spacing, errors in punctuation, omission of words, adding words and overwriting. It was also found that many students were slow in writing and the handwriting became illegible under pressure of speed.

The target set was to improve the speed of writing and to reduce the number of errors in writing. To improve the speed of writing, behavioural charts were maintained. Every day they were given a passage of 100 words to copy. The time taken to copy the passage, number of errors and the type of errors were recorded on a chart. At the end of the week, a curve was drawn on the board to show the progress made by the student. Depending upon the progress made, the student concerned was given verbal reinforcements by others in the group. Simultaneously, the student was also made aware of his errors in writing and advised to look at the previous days' writing and take a note of the time taken, total number of mistakes and the type of errors.

There was an overall improvement in the speed of writing. The average time taken by the group came down to 18.32 minutes from 20.20 minutes. Similarly, the average number of errors made by the group came down to 7.16 from: 10.48 after 25 sessions of therapy.

Arithmetic

Students with learning disabilities often have difficulty mastering arithmetic skills and concepts. Arithmetic problems are common at all age levels. During the pre-school and primary years they have difficulty in sorting objects by size, matching objects, understanding the language of arithmetic or grasping the concept of rationale of counting or one to one relationship. During elementary school level, they have trouble with computational skills. In the middle and upper grades they have problems with fractions, decimals, and measurements. Also many secondary students face problems in place values and basic facts like addition, subtraction, multiplication and division.

Ashlock (1982) found that the error patterns that were developed by children were the result of incomplete concept formation. Roberts (1968) identified four error categories with third grade students: (1) wrong operation, the student subtracted when he should add (2) obvious computational errors, student applied the correct operation but made errors in recalling a basic number fact (3) defective algorithm, the student added without regard for place value (4) random response, there was no relationship between the problem solving process and the problem.

Otto and Smith (1980) offer the following principles for remedial math teaching-write specific objectives to deal with specific problems, arrange regular practice sessions, provide immediate and positive feedback during practice sessions, provide for concrete learning experiences, keep an accurate record of error levels and response rates, encourage children to set goals, ask questions, match the task activities to the levels of competence of the child and diagnose errors regularly.

The author (Rozario, 1991) taught the 25 learning disabled students over 25 sessions the basic operations of addition, subtraction, multiplication and division by first using concrete operations then semi-concrete operations and then abstract operations. At the concrete level, to teach the concepts, sticks and beads were used. At the semi-concrete level, beads and sticks were removed, instead lines were drawn on the black board or in the note book and the same procedure was followed. At the abstract level the students were encouraged to do the calculations mentally without using any aids. When they were learning multiplication, they were asked to learn multiplication tables everyday. The average test score of the students went up to 24.8 from 22.4 and statistically this was a significant improvement.

Behavioural problems and learning disabilities

It is generally believed that the prevalence rate of behavioural and emotional problems is higher among the poor performers than the normals. This view is supported by various studies (Critchley 1968, Scott 1981, Sturge 1982, Krouse and Krouse, 1981). Krouse and Krouse (1981) reported that there was greater maladjustment, immaturity and inadequacy among underachieving children than among students performing at their expected levels.

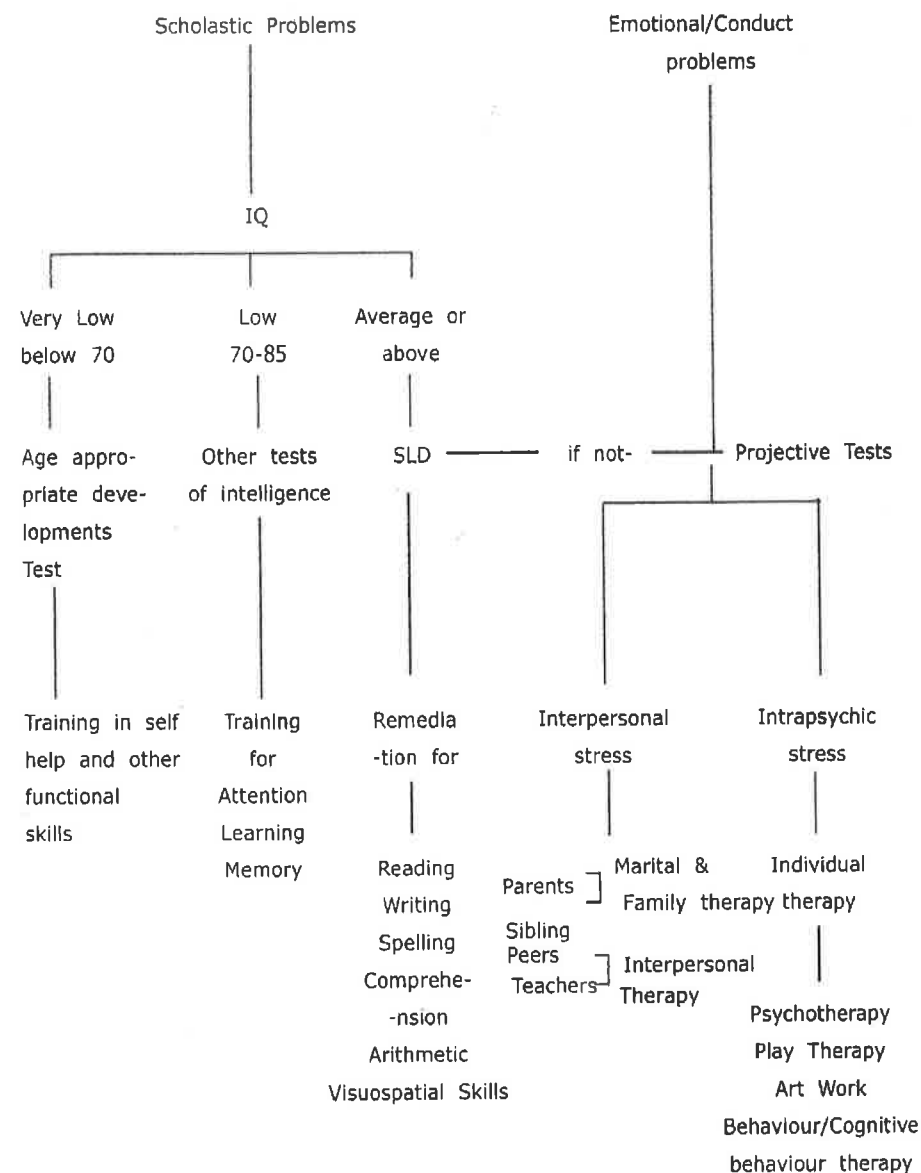
In the study with 25 learning disabled students, the author (Rozario, 1991) found two students were disturbed as indicated by Rutter Proforma - B. The first student was found to exhibit aggressive behaviour and picked up fights with others even with little provocation. His father was an alcohol dependent and a lorry driver by profession. Mother was an illiterate housewife. The house atmosphere was not congenial as there were frequent quarrels. The boy was seen over ten therapy sessions. During the initial sessions he accused others of picking on him. He was

made to role play some of the incidents, which helped him realize his responsibility in those episodes and the bad outcome of his aggressive behaviour. He was made to reflect what other alternate behaviour he could have adopted and the possible consequences of such alternate behaviours. Parents were counselled regarding the need to foster a congenial home atmosphere for the good of the child, over three sessions. The teachers reported 60 percent improvement at the end of the second month.

The second student was a girl who was sensitive, shy and withdrawn. It was found that she had lot of negative feelings about herself. She was given ten sessions of rational emotive therapy which focused on building positive thinking. The mother and teachers were counselled regarding the need of positive reinforcements to build her self image. At the end of ten sessions both the teacher and the mother reported significant improvement.

Remedial education need not be carried on through long drawn sessions. Many students can show significant improvement even with as many as 25 sessions. This remedial education can be carried out within the school set up by teachers. Training teachers for remedial education has to include (1) assessing the educational needs of the students (2) the ability to design and implement suitable intervention programmes (3) developing the ability to work effectively and harmoniously with class teachers (4) and finally the remedial teacher has to be taught to help the students with behavioural problems and counsel the parents and teachers as and when necessary. Learning disabled students need remedial education which has varied instructional objectives depending on the individual's needs, varied entry points into curriculum, varied pacing, active participation of the learner, use of varieties of instructional techniques and finally criterion-referenced evaluation of the learner.

Flow Chart for Assessment and Intervention



Some Exercises in Reading

Learning to read is an important skill for a child. However, millions of children experience difficulty in reading. Reading difficulties are not isolated problems but can affect other curricular areas as well as personal, social and behavioural adjustment.

The skills involved in learning to read are many and varied. Generally, these skills are classified as either word recognition or comprehension skills. Word recognition skills are needed to decode printed letters and word with sounds. Comprehension skills are needed to understand the meaning of what is read. Children with reading handicap typically experience difficulty with both types of skills. It is important to be aware of specific reading deficits in order to plan for appropriate methods and materials to alleviate the reading problems.

Here are some of the commonly used methods for developing reading skills.

1. High interest /low vocabulary materials

Students with reading problems are often frustrated with reading materials because books geared to their interest level are beyond their reading ability. Therefore books that are designed to be of high interest yet at an easier level are to be used.

2. Multisensory methods

Reading approaches that are labeled multi sensory usually attempt to teach reading skills through kinesthetic and tactile stimulation along with the visual and auditory modalities.

Two methods of teaching that emphasize a multisensory approach are : Fernald method, known as whole word approach. It involves four stages in stage one, the student selects a word which is written in large letters in a flash card. The student traces the word with his finger and is instructed to say it aloud as it is traced. The student repeats this process as often as necessary until the word can be written as a unit from memory. Each new word that the student learns is filed alphabetically and these words are frequently utilised in stories.

Stage two is identical to stage one except that tracing is eliminated.

In stage three, words are no longer written on cards.

The fourth stage encourages the students to learn new words from generalising from words already known, to survey a reading passage and pick out new words and learn.

Gillingham approach

This method is known as phonic method. Phonetic drill cards are used.

1. A small card with one letter 'printed on it is exposed to the student and the name of the letter is spoken by the teacher. The name of the letter is then repeated by the student.

2. As soon as the name of the letter is mastered, its sound is made by the teacher and repeated by the student. The original card is then exposed and the teacher asks "What does this letter say?" The student is expected to give the sound.

3. Without the card exposed, the teacher makes the sound represented by the letter and says, "Tell me the name of the letter that has this sound" The student is expected to give the name of the letter.

4. The letter is then written by the teacher and its form is explained to the student. That letter is then traced by the student, copied, written from memory and then written again looking at it.

5. Finally, the teacher makes the sound and instructs the student to write the letter that has this sound.

Programmed Reading

Most programmed reading materials are designed to teach reading skills through a concisely organised and sequential approach. In reading most programmed instructions take the form of work books. Short learning units require the student to be actively involved in the reading process by responding to each frame and immediately checking the correctness of the response. Students are usually encouraged to complete the material at their own pace.

Remedial Reading Drills

The programme consists of lists of words emphasizing specific sounds and combination of letters. The student is instructed to use various lists to learn to blend various sounds. Drill exercises are used to reinforce the sound symbol relationship.

Neurological Impress Method

This approach uses a system of union reading, whereby the student and the teacher read aloud simultaneously at a rapid rate. The disabled reader is placed slightly to the front of the teacher with the student and the teacher holding the book jointly. As the student and the teacher read the material in union, the teacher's voice is directed into the student's ear at closed range.

New Alphabets - Some approaches to beginning reading instruction have modified the traditional 26 letter alphabet in order to assure a regular correspondence between sound and symbol. The initial teaching alphabet includes 44 character alphabets, composed of 24 letters from the conventional Roman alphabet (q and x eliminated) and 20 additional letters.

Transition to traditional orthography is usually recommended at the end of first grade or at the beginning of second grade.

Reading Mastery : Distar (Englemann and Brunen, 1984)

Distar Reading programme is a highly structured, fast-paced and intensive programme of reading instruction that emphasizes decoding skills. Students are taught sequencing skills through symbolic game. Blending exercises are designed to teach the synthetic relationship reinforcement throughout the programme. Their mastery of various skills is appraised through criterion referenced tests.

None of these methods are tailor-made to suit the individual needs. The best approach is to go in for an eclectic approach depending upon the type of the individual child's needs based on a complete and detailed assessment.

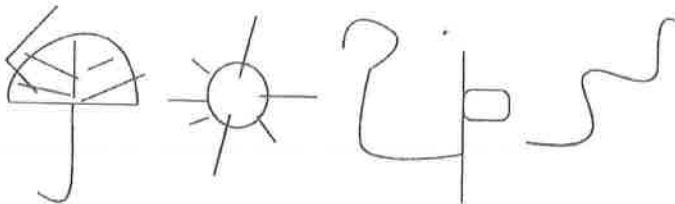
Some exercises in writing activities

Readiness Activities

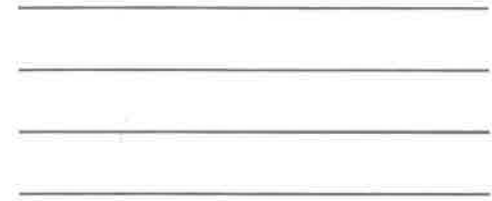
- * Body exercises to practise movements
- * Scissor work.
- * Colouring
- * Practice drawing circles, balloons, coins.
- * Finger painting and clay modeling
- * Nuts & bolts, buttoning, bottle caps
- * Black board activities
- * Stencils

Find hidden letters

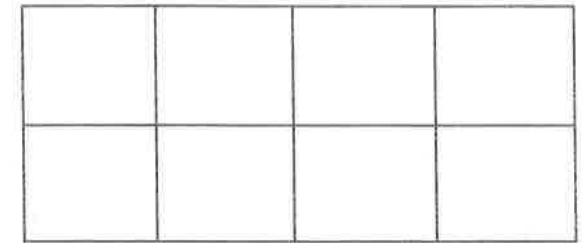
* Pictures



* Draw between the lines



* Square note book



Writing skills include competence in writing, spelling, punctuation, capitalization, making sound - letter correspondence knowing the alphabet, and distinguishing one letter from another.

HANDWRITING PRACTICE

STRAIGHT FAMILY

i l j t k u y

BUMP FAMILY

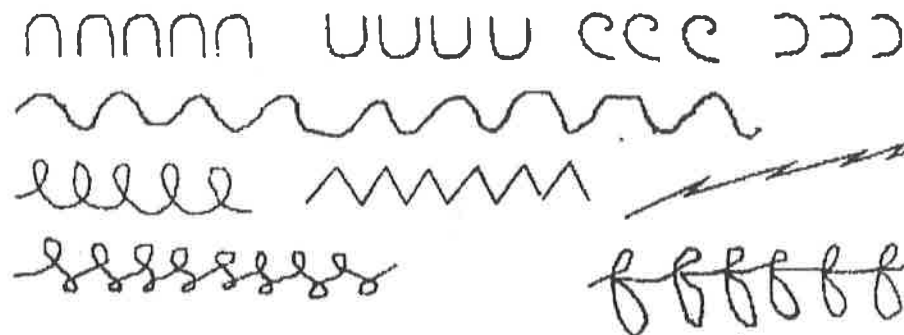
w m n p b h

ONE - 'O' - CLOCK FAMILY

a a c g d g s

SLANT FAMILY

V X W Y N



MANUSCRIPT WRITING

a b c d e f g

CURSIVE WRITING

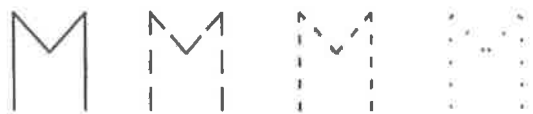
a b c d e f g

MULTISENSORY APPROACH TO TEACHING LETTER FORMS

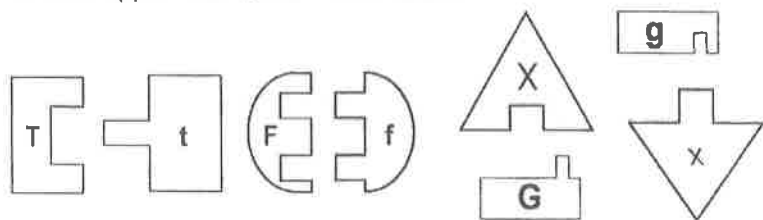
Steps -

1. The T shows the child the letter to be written
2. The T says aloud the letter name and stroke directions
3. The child traces the model with his finger and may report his movements aloud as he traces
4. The child traces the letter model with his pencil
5. The child copies the letter on paper while looking at the model

Fading model



* match upper and lower case letters



* for children with reversals, make associations



(small b is the same direction as big B)



(P is like a flag)

- * for children with difficulty in spacing between words
- Square book
- 2 - leaded pencil
- finger between words
- * put dots / arrows to show direction



SPELLING ACTIVITIES

SENSORY MODALITY PREFERENCE (WESTERMAN, 1971)

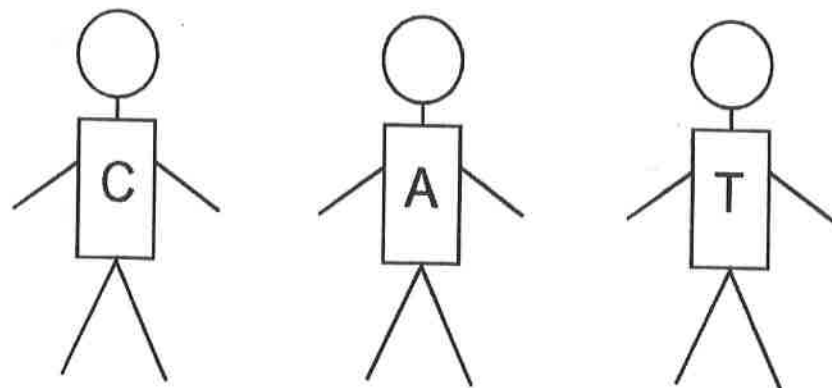
1. Auditory - Vocal
 - T spells the word aloud
 - child orally spells words
2. Auditory - Motor
 - T spells the word aloud
 - Child writes word on paper
3. Visual- Vocal
 - T shows word on flash card
 - child orally spells the word
4. Visual - Motor
 - T shows word on flash card
 - child writes word on paper
5. Multisensory combination channel
 - T shows word on flash card and spells it aloud
 - child orally spells word and writes it on paper

COVER & WRITE METHOD

- I look at the word and say it
- II look at it and write twice
- III cover word and write again
- IV check by looking

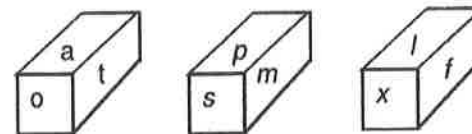
Some Exercises in Spelling

SPELLING GAMES

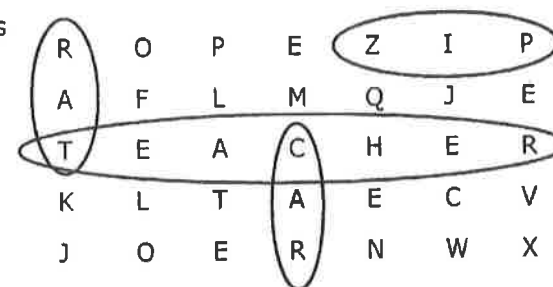


T
O
P I N
E S
S I T

- Fill in the blanks
- spill and spell



- scrabble
- boggle
- find the hidden words



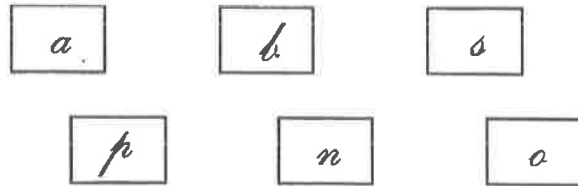
- spelling bee
jump forward if correct
- anagrams
smile resist
- word building
- crossword puzzles
- word cards game
- dictionary use
- practise on typewriter
- develop memory devices to help remember spelling. eg-stationery
is writing paper

t is a wall for no reason at all
match watch catch

e follows i except after c
receive field

THE GILLINGHAM METHOD

(Gillingham & Stillman, 1968)

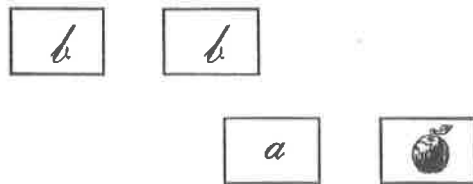


- Association of the visual symbol with the name of the letter
- Association of the visual symbol with the sound of the letter
- Association of the feel of the child's speech organs in producing the name or sound of the letter as he hears himself say it

1. Card is exposed
2. T says name of letter
3. Child repeats
4. Therapist says sound of letter
5. Child repeats



One key word for each letter



T says sound without showing cards

Child says letter

T says sound

Child says letter

Child writes letter

Fernald's Multisensory Approach (VAKT) (1943)

Important areas in learning spelling

- a) Clear preception of word form
- b) Development of a distinct visual image of the words
- c) Habit formation through repetition of writing until the motor pattern is automatic

STEPS:

1. Write and say a word while the student watches and listens.
2. The student traces the words while simultaneously saying the word. Then on careful pronunciations, with each syllable of the word dragged out slowly as it is traced or written.
3. Next, the word is written from memory. If it is incorrect, step 2 is repeated. If the word is correct, it is put in a file box. Later the words in a file box are used in stories.
4. At later stages
 - no need for tracing.
 - student learns the word by observing the teacher write and say it and then he writes and says it himself.
 - he learns the word by looking at it in print and writing it and finally merely by looking at it.

Kaufman et al (1978)

- Show child's wrong spelling written down
- then show correct spelling

"tot"	"taught"
"cam"	"calm"
"wach"	"watch"
- especially for words that do not follow regular phonetic rules - words for which the child must use visual memory.

Some Exercises in Written Expression

WRITTEN EXPRESSION

fluency vocabulary structure content

- * Encourage spontaneous writing
- * Functional writing
 - conveying information in a structured form
 - business letters
 - invitations
 - reports
 - minutes of meeting
- * Creative writing
 - personal expression of thoughts and experiences
 - poetry
 - story writing
- * Unfinished stories
- * Wordless picture books

Fluency

- * From jumbled - up words, form a sentence
- * Complete sentences
- * Connect 2 simple sentences to make 1 compound sentence
- * Highlight noun/verb/adjective in a sentence

Vocabulary Development

- * Reading books, mag. / newspaper
- * Discussions
- * Word file with word and meaning

- * Story building
- * In a written passage substitute words/phrases
- * Crossword puzzles

Structure Development

- * insert punctuations
- * wrong punctuations - correct
- * fill in the blanks - verbs
- tenses
- * COPS

Content Development

- * Complete story
- * Give key words to build up story
- * Encourage letter writing
- * Story with the wrong sequence-rewrite
- * Cartoons / pictures without captions
- * Maintain diary / journal
- * Class magazine

ERROR MONITORING EXERCISES

Error Monitoring Strategy

(Schumaker et al. 1981)

COPS

C - Capital

O - Overall appearance

P - Punctuation

S - Spelling

THEME WRITING

T - Think

O - Order ideas

W - Write

E - Edit

R - Rewrite

Number work excercises

Readiness for number instruction

(Piaget & Inheldor 1969)

- a) CLASSIFICATION - grouping of objects according to some common distinguishing characteristics
size, shape, colour
- b) ORDERING - important for sequencing
counting objects in order
- SERiation - Ordering objects on the basis of their properties
- length 
 - size 

- c) ONE - TO ONE CORRESPONDENCE
- basis for counting - involves understanding that one object in a set is the same number as one object in a different set, whether or not characteristics are similar



- distributing pencils among children
- matching school bags to children

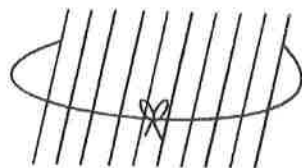
- d) CONSERVATION - fundamental to numerical reasoning
- the quality of an object or the number of objects in a set remains constant regardless of spatial arrangement



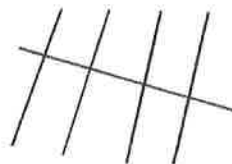
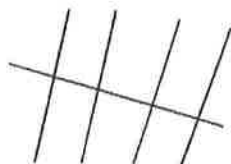
TEACHING ARITHMETIC SKILLS

CONCRETE LEVEL

- block, beads, straws



SEMICONCRETE LEVEL

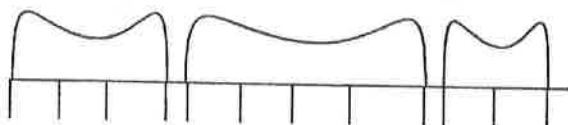


ABSTRACT LEVEL

H	T	U
5	3	1

H	T	U
5	9	6

Multiplication addition etc. with beads



number lines

Math Concentration

$$7 \times 7$$

$$5 \times 6$$

$$20$$

$$30$$

$$4 \times 2$$

$$9 \times 3$$

$$49$$

$$8$$

$$6 \times 3$$

$$4 \times 5$$

$$\square$$

$$18$$

Fraction Games

Weave math into daily life

- Rhymes & Songs
- Stories

Note : For additional, information refer the book 'Seven ways to help your child with math's by Geere B.

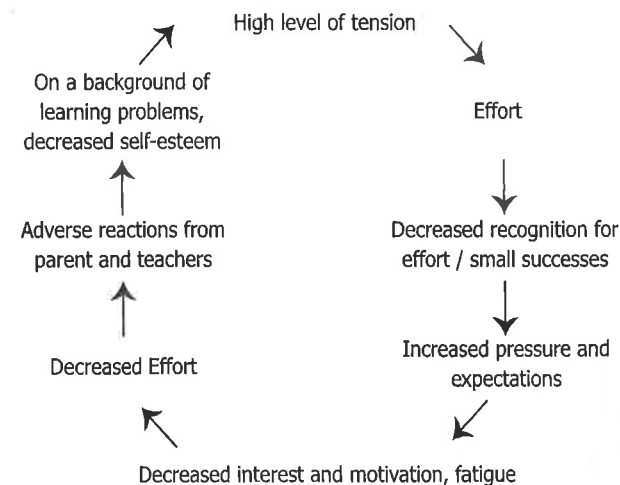
Intervention for Children with SLD - other considerations

In addition to remediation for specific learning disabilities, it is essential to identify and treat the socio-emotional disturbances often found in children with these disabilities.

Based on the findings of a series studies in this area Hirisave (2003) has formulated a model (figure) from Clinical Psychological perspective. When learning disability begins in early school years, children though temperamentally and socially well adjusted suffer from feelings of low self esteem, which leads to high levels of tension and apprehension in situations involving academic evaluations. When they experience increased academic pressure, they do attempt to put in some efforts to improve their performance, but they fail to get reinforcement for their efforts. When there is increased parental expectation they suffer from feelings of poor motivation, which may result in decreased efforts. This further leads to adverse reactions from parents and teachers and the vicious cycle goes on.

Figure :

Model for understanding psychosocial characteristics of children with learning disabilities



Broadly speaking, work in this direction can be divided into three parts : (a) supportive work with the child; (b) counselling of parents; (c) counselling of teachers.

a. Supportive work with the child

Let the child know that the treating professional is interested in him as a person with interests and abilities. The child with learning disabilities may have associated emotional and / or conduct problems. It is necessary to help the child to understand the nature of these problems. He needs to be taught to express feelings with parents, peers and teachers. He may also need to develop better frustration tolerance and appropriate ways of handling criticism. In some children, it is essential to offer suggestions for coping with feeling of sadness, anger and guilt. Training in relaxation, anxiety, tension, apprehension, worry, especially in the testing situations demanding his performance. Reassurance, encouragement of other activities of his interest may enhance the self-esteem of the child. Ultimately the child should be helped to realise his potential to the maximum extent possible.

b. Counselling of parents

The parents should learn not to blame or criticise the child or make unfavourable comparisons. They need to decrease their academic expectations from the child and understand the importance of appreciating whatever little the child does in the area of academics. They should be willing to put in efforts for many days and months.

If the child's disability is severe then attempts should be made to reduce the home work, which is actually a source of strain both for the child and his parents.

The parents can help the child in organising his daily routine structuring the time for studies and other activities of his interest.

The parents are counselled to use simple sentences in a clear manner especially if the child has poor reading comprehension. The child needs guidance from parents in taking appropriate small steps in mastering pre-academic skills.

The parents should be helped in interacting with the school personnel. They could be having difficulties in communicating what the mental health professionals have to say about the child. If necessary school visits should be planned wherein the psychologist can meet the teachers along with the parents.

The parents should help the adolescent learning disabled child in taking decisions about vocational training and placement. Finally the parents can be encouraged to interact with other parents who have children with learning disabilities.

c. Work with the teachers

The teachers play a major role in making the child feel secure and comfortable in the school. It is essential for the teachers to accept the child with his strengths and weaknesses and to have a non-judgemental attitude. Both the teachers and peers should be made aware of the fact that the child has weakness in the area of academics but does not have general dullness.

The teachers need to establish rapport with the child so that the child trusts her and enjoys the experience of working with her. It is important to have a positive dialogue with the child, always appreciating the small gains made by

the child. Punishing the child for not doing well can affect the child permanently.

The teacher can help the child to plan and organise the activities in the classroom. The activities planned should include non-academic activities, and should provide ample opportunities for the child to succeed. The teachers/should make the learning easy, simple and fun. Gradually the complexity of the tasks can be increased and include academics. It should be made clear to the child that he can ask questions without fear of ridicule. In the process of learning he needs to be helped in handling his own emotional reactions and frustrations. He can be taught relaxation in situations calling for his performance. He can be encouraged to self correct the answers.

The homework given should be appropriate to the ability of the child. While evaluating the work done it is important to give feedback regarding accuracy, neatness and legibility. It is also necessary to write the correct answers in the margin.

The teacher can plan group sessions for the children with learning disorders in the school. This helps the child to share his experiences, problems with other children having similar difficulties. The children also require constant encouragement to form healthy peer relations, and to participate in sports, music, and other extracurricular activities.

The teachers should understand the need to have regular contact with professionals and parents.

Recently a study by Moulya (2018) examined the usefulness of group intervention for children with learning disabilities in an outpatient setting and found phonological intervention useful in remediating reading and spelling difficulties. Similar group work can be conducted by teacher s in the school setting.

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CHAPTER 6
PHONOLOGICAL MODEL OF DYSLLEXIA AND
ITS RELEVANCE FOR INTERVENTION OF READING DISABILITY

Priya Kayastha and Uma Hirisave

Among the various causal explanations proposed for dyslexia, the phonological theory has received strong support in last two decades. Current theories on development of reading skills in English accentuate that phonological processing is the most significant underlying cognitive process. The significant function of phonological processing is the association of sounds with letters and or combination of letters. This function is referred as *grapheme-phoneme conversion rules*. This skill is the basis of decoding print and therefore, phonological route is critical in the early development of reading skills.

The theory focuses on the relationship between spoken and written language and explains the phonological basis of reading difficulty. According to this theory, letters and letter string (i.e. the orthography) are represented by the sounds of spoken language. In order to develop reading skill the child has to develop an understanding that spoken words can be broken down into individual sounds (phonemes) and that each of these sounds are represented by the letter in the word. For e.g. the word 'bat' has three phonemes b/ .../aaa/.../t/.

Spoken language is segmental in nature (Liberman, Cooper, Shankweiler, Studdert, Kennedy, 1967). Children with dyslexia are unable to appreciate this characteristic of spoken language. This poses challenge to them in establishing a systematic relationship between the phonemes in the spoken language and letters in print. Children, who are skilled readers,

are able to form association between spoken words and corresponding phoneme and thus are able to use this knowledge to identify words. Generally, children use many strategies to identify unknown words. However if they do not skillfully use letter-sound correspondence early in development, they invariably remain poor readers (Foorman, Francis and Shaywitz, 1997).

An Indian study (Ralte, 2012) revealed that children with reading and spelling disorders had more pronounced inadequacy in verbal intelligence on WISC III, verbal working memory and overall deficits in phonological skills as compared to children without reading and spelling problems.

Studies indicate deficit in phonology represents the most robust and specific correlate of reading disability (Morris, Stuebing, & Fletcher, 1998). For this reason most of the contemporary approaches for identifying reading disability are based on phonological model to understand how children gain access to words. Based on the result of longitudinal studies evidence based interventions are designed to improve phonemic awareness skill in children (Gillon, 2004). Phonemic awareness refers to the ability to hear and manipulate the individual sounds within words.

Schatschneider and Torgesen (2004) have pointed out that phonemic awareness supports growth in accurate and fluent reading skill in children in following three ways:

- It helps them to understand the alphabetic principle.
- Helps children notice the regular ways in which letters represent sounds in words. E.g. the four letters in the word 'clap' correspond to four different sounds.
- Facilitates generation of possible words in context that are only sounded out partially. E.g. consider the child is given the following sentence to read. The baby boy is.

If the child has to guess the correct word in context and differentiate from words in their mental dictionary (e.g. playing, running, crawling), his/ her knowledge of phonemic awareness of first two letters will help him to guess the correct word. Therefore, if they know 'pl' makes /pl/ sound, they can complete the sentence with playing as the correct answer.

Phonological approach for intervention

Phonological awareness is a multi-level skill of breaking down words into smaller units. It can be described in terms of syllable awareness, onset-rime awareness and phoneme awareness (Gillon, 2004). The terms with definition and examples are given in table 1.

Levels	Definition	Eg. of Tasks
Syllable awareness	Refers to awareness that word can be divided into syllables.	- Syllable segmentation- "how many syllables in the word <i>pillow</i>?" - Syllable completion-show the picture of a rabbit and say "I will say the first part and u have to complete the word. This is a ra_____" - Syllable Identity-"which part of <i>boy</i> and <i>ball</i> sound the same?" - Syllable deletion - "Say <i>football</i>. Now say it again without the <i>foot</i>"
Onset-rhyme awareness	Awareness of intra-syllabic level	- Spoken rhyme recognition "do <i>coat</i> and <i>goat</i> rhyme" - Rhyme oddity task- "tell me the odd word - <i>cat</i>, <i>bat</i>, <i>cot</i>, <i>sat</i>, <i>hat</i>" - Rhyme generation- "tell me words that rhyme with <i>lot</i>"

Phoneme awareness

Refers to the ability to hear and manipulate the individual sounds within words.

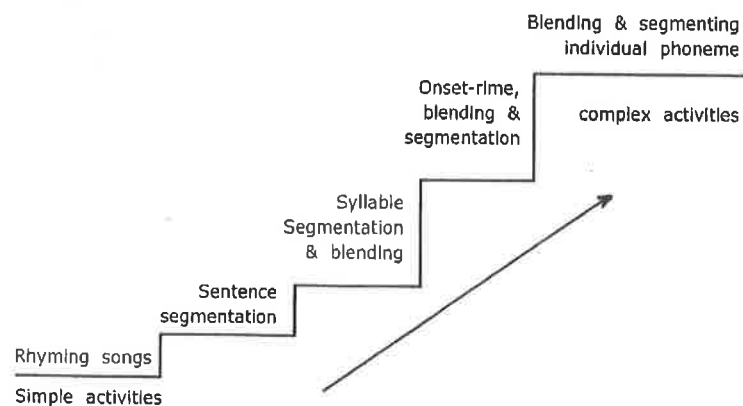
- Phoneme detection- "Which word begins with a different first sound: *cat*, *call*, *pet*, *come*."
- Phoneme matching-"which word begins with the same sound as *hat*: *pig*, *hut*, *car*"
- Phoneme isolation-tell me the beginning sound of the word "*leg*"
- Phoneme elision: say *mat*. Now say it again without /m/
- Phoneme blending: "what word do these sounds make: *ch...al....r*"
- Phoneme segmentation: how many sounds can you hear in the word *worm*
- Phoneme manipulation: say *fan*. Now instead of /a/ say /i/.... (*fin*)
- Spoonerisms- *felt made* becomes *melt fade*

Intervention:

The interventions with young children who are at-risk of reading difficulties mainly focus on strengthening their phonological skills. Though the interest in teaching phonological awareness skills to young school-age children started in early 1970's, it was Bradley and Bryant (1983) who conducted a longitudinal and training study and found a causal relationship between phonological awareness and reading development. This prompted interest in numerous researches in the area of phonological awareness training

leading to strong scientific basis. Studies have demonstrated the usefulness of phonological intervention among first standard school going children. (Kayastha, 2011) and among clinic children aged 5-10 years (Moulya, 2018). Both studies have used Gillon's phonological awareness program. While Kayastha used individual format, Moulya adopted 15 group session program for children. The program led to a significant improvement in the phonological skills that reflected in improved reading and spelling skills.

Before we familiarize ourselves with effective remedial methods, it is important to know the levels of phonological awareness as it contributes in better designing and implementation of an intervention programme. The development of phonological awareness takes place at various levels. It begins when the child start recognizing that speech is composed of words (Pikulski, 1989) and later extends to recognizing and manipulating sounds or phonemes in the word. The various levels of phonological awareness are represented in the following figure.



There is ample evidence to suggest that phonological awareness is a crucial skill in developing reading skills. Research indicates that relationship between phonological awareness and reading is not unidirectional but reciprocal in nature (Stanovich, 1986). On one hand, it is understood that training in phonological awareness skills support in building reading

abilities in children. On other hand, it is seen that children who engage themselves in reading have some understanding of internal structure of words and providing additional instructions in phonological awareness will promote reading. Studies indicate that children as young as four years old can benefit from training (Bradley & Bryant, 1983) and phonological awareness skill in kindergarten is a strong predictor of later reading success. (Ehri & Wilce, 1980). Remediation focusing on strengthening phonological processing ameliorates decoding and word-reading weaknesses in dyslexia (Hatcher, Hulme & Ellis. 1994). For effective intervention children who are at risk for reading difficulty need to be identified before the age of nine years for intervention to show encouraging results (Lyon,1996).

Language is generative in nature. Though, there are just 26 letters in English alphabets, its different combinations can produce 44 phonemes in the language creating tens of thousands of words. The interventions done with children helps them to explore this aspect in detail and facilitate smooth process in learning phonological awareness skills.

Various activities related to phonological awareness training are as follows:

Rhyming games:

- Teach nursery rhymes.
- Simple games with rhyming words.
- Sentences with rhyming words e.g "boys run in the sun to have fun".
- Read stories that contain rhyming words.

Letter-sound correspondence:

- Teach sounds of each alphabet. (Children pick up learning sounds of consonants first and then vowels!)
- Show pictures which start with same sound. Eg. "Which pictures starts with /c/ sound?".

- Identify letter and the picture with corresponding phoneme (Refer Figure 1 and 2 on Page No. 172).

Phoneme identity:

- "I spy" game-the child is asked to recognize the given phoneme in spoken word. Instructions- I spy something that starts with /m/ sound in.... rose, mango, book.
- Create a Sound box containing objects or pictures that begin with same phoneme. E.g. /h/ sound is represented in hat, horse and house.
- Sort out picture cards based on phoneme selected for grouping.

To make the task more challenging the instructions can be changed to Recognizing initial, middle and ending sounds in the word.

Elision

- Omitting syllable / phoneme in the word. Eg. Say toothpaste without the toothpaste.
- Say fat without /f/at.
- Say sling without /l/.....sing.
- Use both blocks and pictures to demonstrate teaching.
- At concrete level, use blocks to practice deleting phoneme in the words (Figure 3, Page No. 173).

Blending & segmentation skills:

- Speak out individual phonemes of the word and ask the child to guess the word (Figure 4, Page No. 173).
E.g. f...oo....d= food, t...or...ch=torch, b...ir...d=bird.
- "Tapping or clapping to indicate individual phoneme in the word. E.g. SMART /s/ /m/ /ar /t/ (4 times), LEANED /l/ /ea/ /n/ /d/ (4 times), WHALE /w/ /ai/ /l/ (3 times).

Remember, the segmentation skill is based on the number of phoneme in the word and not on the actual number of letters in the word.

Phoneme substitution:

- To facilitate learning at concrete level, bring into play alphabet blocks to make a word and then replace a phoneme (represented in a block) with another phoneme to make different word.
- Teach to substitute a phoneme with another phoneme.
- E.g. CAT substitute /c/ with /h/..... it becomes HAT (initial sound) SIT substitute /i/ with /a/..... it becomes SAT (middle sound) MAP substitute /p/ with /n/..... it becomes MAN (ending sound)
- Riddle format. E.g What rhymes with box and starts with /f/fox
What goes with hand and starts with /s/sand

Blends (use flash cards):

B: BL, CL, FL, GL, PL, SL (E.g blow, clap, flag, glass, play, sleep)
R: BR, CR, DR, FR, GR, PR (E.g break, crow, drop, frog, grape, pray)
S: SK, SM, SN, SP, SPR, ST, STR, SW, TR, TW (E.g sky, smoke, snake, spring, strong, swing, try, twin)

Consonant Digraphs:

CH, SH, TH, WH, GH, PH (E.g. chair, shop, throw, where, through, phenomena)

Exercise:

Choose the correct blends to complete the following sentences: BL, SK, DR, SM, BR, SW, GR, SL, PR, CR, FR.

The _____ y is full of _____ oke. (bl, sk, dr sm,).

My little _____ other went to _____ eep.

- The _____ apes are _____ eet in taste.
- Children will _____ ay in the school.
- Mother said "choose your _____ ock _____ om the cupboard".

Using phonemic knowledge in group sessions:

- All children whose name start with /p/ sound to sit on left side and the ones with /s/ sound sit on right side.
- Find a partner whose name start with / n/ sound.
- Alphabet game sound- find animals, place, things that begin with e.g. / b/ sound.
- Snail talk-display pictures and then say the word by stretching out the phonemes. E.g mmmmaaaannnn. Children have to look the picture and guess the word.

Role of parents:

In addition to providing training in phonological awareness skills, research indicates that parent's early participation in reading stories and books to children significantly contributes in developing reading skill (Snow & Tabors, 1996). It is well known that child's exposure to early literacy starts with family members at home. Behaviors such as holding book, turning pages, identifying pictures, verbal interactions about story characters are early experiences of literacy at home. Broadly covered under **home literacy environment**, such experiences influences learning in school and should be encouraged before starting formal schooling (Hart & Risley, 1995, Bhola, 1995).

While providing training in phonological awareness skills, it will be helpful to use words from familiar storybooks. This can be advantageous in two ways. Firstly, the child will be able to relate meaningfully to phonological awareness activities and connect phonemes and letters /words with print. Secondly, using familiar storybooks will motivate children in reading thus resulting in more practice to read.

Published literature shows that interactive storybook reading (Karweit & Wasik, 1996) and daily exposure to being read by parents as a predictor for child's language and cognitive abilities at three years (Raikes, Luze,

& Brooks-Gunn, 2006), particularly for children belonging to under-privileged background or those with history of language delay. In India, a study examining the role of Indian bilingual parent's book reading practices on developing literacy skills in English (as second language) among preschoolers indicated that exposure to book reading in English was associated with bilingual's oral language, narrative and phonological awareness skills in second language (Kalia,2007). These promising findings recommend parents involvement with children in reading activities.

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Figure 1: Pictures which start with same sound.....

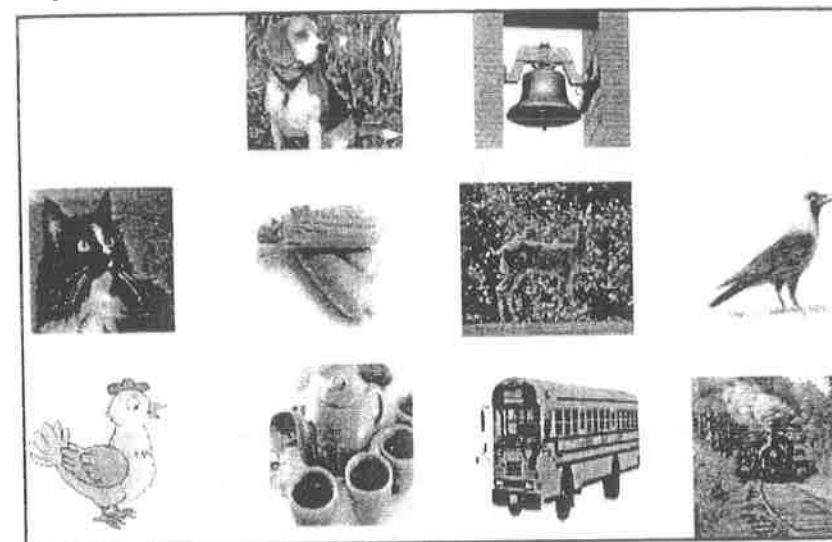


Figure 2: Identify letter & find picture with corresponding sound

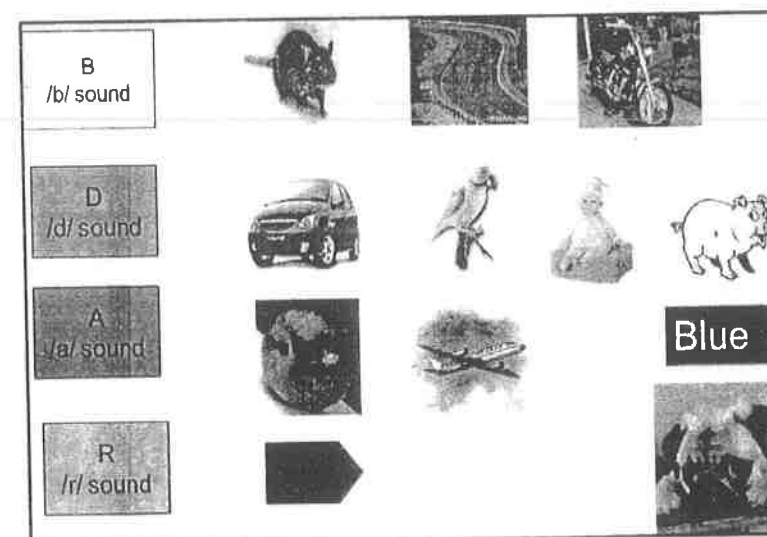


Figure 3: To analyze & manipulate sounds in words.

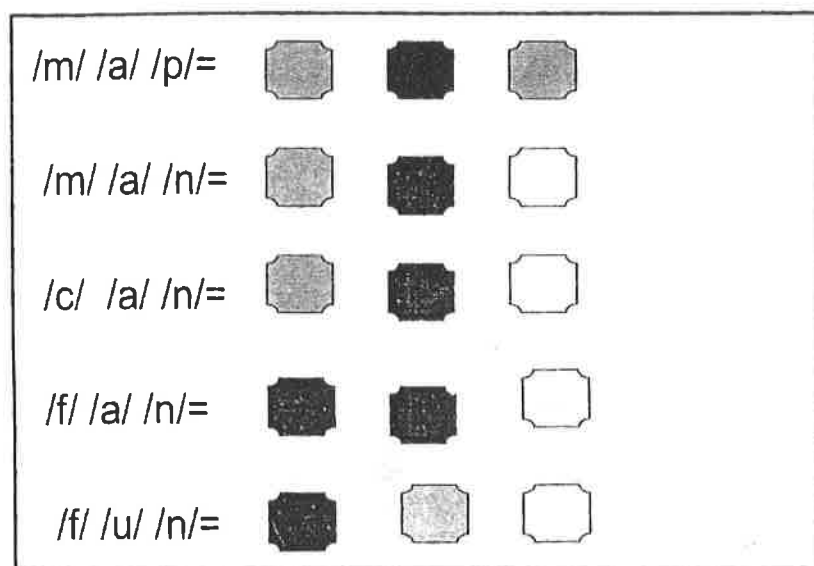


Figure 4: Phoneme blending

- *H...e...n* = *hen*
- *F...oo....d* = *food*
- *b...i...r...d* = *bird*
- *s...w...i...m* = *swim*
- *t...o...r...ch* = *torch*

CHAPTER 7

RELIABILITY AND VALIDITY OF THE NIMHANS INDEX FOR SPECIFIC LEARNING DISABILITIES

Uma Hirisave, Anuja S. Panicker and Nalini N.R.

INTRODUCTION

The term learning disabilities emerged from a need to identify and help children who repeatedly show poor scholastic performance, yet are not mentally retarded. According to ICD-10, Specific Developmental Disorders of Scholastic Skills (SDDSS) is a condition associated with clinically significant degree of impairment in scholastic skills, not solely explained by mental retardation, developmental in onset and not explainable by any external factors. Also known as Specific Learning Disabilities (SLD), it is a condition in which children are unable to perform adequately - as compared to children with average/ above average intelligence, studying in the same grade. In the West, this condition is seen in about 3-6% of school children. In India, recent research has indicated that 31% of primary school children have been diagnosed as having learning disabilities. Considering the high prevalence of SLD in primary school children, it is important that an effective tool be used for assessment of SLD.

The NIMHANS Index for Specific Learning Disabilities was initially developed by Kapur, John, Rozario and Oommen in 1991 in the Department of Clinical Psychology, NIMHANS, Bangalore. The profile of the performance of normal children (Std. I-V) has been published. This tool has been used in a number of studies to confirm the diagnosis of SLD and differentiate children with SLD and those without. The SLD index has been

mainly used to confirm the diagnosis of SDDSS made by the clinician as per the ICD-10 criteria to enable to children to get benefits offered by the state Government.

Unlike Western achievement tests, where it is possible to arrive at a percentile point as well as derive age appropriate grades, the NIMHANS SLD index is a curriculum-based assessment.

The usefulness of a tool cannot be commented upon unless its statistical properties have been established. The aim of the current study was to establish the reliability and validity of the NIMHANS Index for Specific Learning Disabilities for a sample of children with SLD.

METHODS

Sample

The sample consisted of 40 children (boys and girls) recruited the out patient Department of the Child and Adolescent Mental Health Unit, NIMHANS, currently named Department of Child and Adolescent Psychiatry.

Inclusion Criteria

- Age between 6-16 years at the time of first contact
- Diagnosis of average to above average intelligence on Axis III of Rutter's multi-axial classification
- Children diagnosed as have Specific Learning Disabilities or difficulties on the Axis II of Rutter's multi-axial classification.

Exclusion Criteria

- Children with comorbid Attention Deficit Hyperactivity Disorder, Emotional
- Disorder, Conduct Disorder, Dissociative Disorder or Eating Disorder on Axis I

Tools

1. NIMHANS Index for Specific Learning Disabilities

The NIMHANS Index for Specific Learning Disabilities was developed in the Department of Clinical Psychology, NIMHANS, Bengaluru. It was initially developed by Kapur, John, Rozario and Oommen in the year 1991 to screen children with SLD. The NIMHANS Index for Specific Learning Disabilities consists of tests of reading, writing, spelling and arithmetic abilities, to identify children with disabilities in these areas. It consists of two levels:

Level I: Consisting of tests of Attention, Visual Discrimination, Visual Memory, Auditory Discrimination, Auditory Memory, Speech and Language, Visuo - Motor skills and Writing Skills.

Level II: Consisting of tests of Attention, Language (Reading, Writing, Comprehension), Spelling, Perceptual Motor Abilities, Memory and Arithmetic. (More details of these tests are given in Chapter 4)

The norms of this battery for children in Std. I-V have been published. This test has been used for confirming an initial diagnosis SLD as well as for certification of these children to enable them to avail benefits. This tool is used in several studies (Mukherjee et al. 1995, Lall et al. 1997, Shanti et al. 1999, Bhola et al. 2000) to compare the psychosocial characteristics of children having SLD and normal children. It has also been used for assessment of improvement after remediation. (Rozario 1991, Uma et al. 2004).

For the purpose of the present study, children whose performance was below (less than two standards) their expected grade were considered as having mild SLD. Those whose performance was much below (less than four standards) their expected grade were considered as having severe SLD.

2. Developmental Psychopathology Checklist for Children (DPCL)

The Developmental Psychopathology Checklist has been standardised on 221 children below 16 years of age. The DPCL consists of 124 items divided into six subsections: (1) Developmental History, (2) Developmental Problems, (3) Psychopathology, (4) Psychosocial Factors, (5) Temperamental Dimensions and (6) Helpful Factors. The psychopathology subsection consists of items pertaining to Attention Deficit Hyperactivity Disorder, Conduct Disorder, Learning Difficulties, Emotional Disorders, Obsessive Compulsive Disorder, Somatic Disorders and Psychosis. The scores obtained on the Learning Difficulties subscale were used for the present study. The reliability of the scale has been found to be 0.96. Criterion validity on comparison with the Child Behaviour Checklist (CBCL) has been established. (Details of this tool are given in Chapter 2)

3. Malin's Intelligence Scale for Indian Children (MISIC)

Malin's Intelligence Scale for Indian Children (MISIC) is the Indian adaptation of the Wechsler Intelligence Scale for Children (Wechsler, 1949) Malin (1969) standardised the MISIC on 1,200 children. It consists of 11 subtests, classified into Verbal and Performance subtests. The test - retest reliability is 0.91. Concurrent as well as congruent validity have also been established. This tool has been widely used in the Indian context for assessing intellectual abilities in children. (Uma et al. 1990, Mukherjee et al. 1995, Lall et al. 1997, Shanti et al. 1999, Uma et al. 2004).

4. Checklist for School Report

This Checklist for School Report is routinely used in the CAMHU to get the teacher's evaluation of the child's academic performance, consisting of items related to: (1) Overall performance in class, (2) Regularity in classwork, (3) Difficulty in paying attention in class, (4) Presence of

specific difficulties in reading, writing and arithmetic, (5) Behavioural problems, (6) Social interaction with other classmates and (7) Any other comments regarding the child. For the purpose of the current study, this tool was used for comparing the teacher's evaluation of specific learning difficulty with the child's reading, writing and arithmetic abilities. Thus, item 4 on this checklist was used to obtain the teacher's evaluation regarding the presence or absence of Specific Learning Disability in reading, writing and arithmetic.

Statistical Analysis

Test-retest reliability of the NIMHANS Index for Specific Learning Disabilities was determined by comparing assessment and post assessment scores using Intraclass Correlation Coefficient (ICC). Criterion validity was computed by comparing the Teacher's rating and Clinical rating with the child's performance on NIMHANS Index for Specific Learning Disabilities using Pearson's correlation coefficient. Chi-square was also used to determine the association between Parental rating on DPCL and severity of SLD as assessed using the NIMHANS Index for Specific Learning Disabilities.

Procedure

The sample was taken from the Child and Adolescent Mental Health Unit, (CAMHU), NIMHANS. The children were individually evaluated by the Post Graduate trainees using the Child and Adolescent Mental Health Proforma. The diagnosis was confirmed by the Child Psychiatry consultants (who have more than 20 years of experience in child and adolescent mental health).

Forty children were selected as per the inclusion and exclusion criteria. They were called for assessment during the same week of their registration

in the CAMHU. They were also asked to bring the Checklist for School Report completed by the class teacher.

Each child was individually assessed using MISIC and NIMHANS SLD index. The parents were also seen separately and administered the DPCL. All assessments were carried out by a Clinical Psychologist (III Author) with adequate training in child assessment.

After assessment all forty children and their parents were given feedback about the nature of difficulties. The parents and teachers were counselled regarding possible causes for the child's difficulties in scholastic performance and on different methods of remediation in a supportive framework. They were given exemption certificate to seek exemption from second and third languages in accordance with the Karnataka Government Order. The parents were offered further sessions of counseling as well as sessions for remediation for the child.

Out of the 40 children tested, 35 children were confirmed as having Specific Learning Disability and five did not have SLD as per the NIMHANS Index for SLD. Of these 35 children, five children underwent remediation sessions.

Thirty children who did not avail any type of remediation either in NIMHANS or outside were contacted for post assessment after a gap of 6 months for test-retest reliability. Some of the reasons cited for not availing remedial services were: lack of remedial center either at their school or near their residence and financial burden. Even though parents were ready to take away children from regular school to give special inputs for reading and writing, the school authorities were not ready to guarantee re-admission. Afterwards thirty children reported for post assessment after a gap of 6 months.

RESULTS

The sample characteristics of the children are indicated in Table I. Table I indicates that the children in the present sample range from 6-16 years of age. Majority of children were studying in English medium, state syllabus schools.

Table I: Sample Characteristics

	Variable	Frequency	Percentage
Sex	Male	29	73
	Female	11	27
Age (years)	6-8	5	12
	9-11	15	38
	12-14	16	40
	15-16	4	10
Medium of Instruction	English	38	95
	Kannada	2	5
Class	Primary	8	20
	Middle	17	43
	High	15	37
Place	Bangalore	34	85
	Outside Bangalore	6	15
Mother Tongue	Kannada	18	45
	Telugu	10	25
	Tamil	8	20
	Urdu	3	7
	Tamil	1	3
Syllabus	State	33	83
	ICSE	4	10
	CBSE	3	7

The Assessment findings are indicated in Table II. It was found that using the NIMHANS SLD battery, 88% (N=35) of children were diagnosed as having Specific Learning Disability (SLD). Based on teacher's rating, 92% (N=37) were diagnosed as having SLD. Table II shows the similarity in findings when NIMHANS Index for Specific Learning Disabilities, Teacher and Clinical ratings were compared.

Table II: Assessment Findings (N = 40)

Variable	SLD Present		SLD absent	
	N	%	N	%
NIMANS SLD battery	35	88	5	12
Teacher rating	37	92	3	8
Clinical Rating	38	95	2	5

Significant correlation was present when Teachers rating and Clinical Rating were compared with the Presence of SLD. This indicates the high criterion validity of the NIMHANS Index for Specific Learning Disabilities with Teacher and Clinical Rating.

Table III: Presence of SLD vs. Teacher rating and Clinical Rating (N = 40)

	Teacher rating		Clinical rating	
	r	sig.	r	sig.
Presence of SLD	0.75**	.01	0.61**	.01

** p < .01

Table IV indicates moderate association between severity of SLD as assessed on the NIMHANS Index for Specific Learning Disabilities with the Parental rating on the Learning Difficulties subscale of the DPCL.

Table IV: Parental rating compared with severity of SLD (N = 40)

SLD Severity (NIMHANS SLD Index)	Parental rating (DPCL)		TOTAL	X ² (sig.)
	< cutoff	> cutoff		
Nil	2	3	5	0.35 (.06)
Mild	1	18	19	
Moderate/ Severe	1	15	16	
TOTAL	4	36	40	

Table V indicates high test-retest reliability for the NIMHANS Index for Specific Learning Disabilities with respect to severity of SLD.

Table V : Test-retest reliability (N = 30)

variable	r	sig.
SLD severity	0.53***	0.001

*** p < .001

DISCUSSION

The present study, having a longitudinal design, aims to establish the reliability and validity of the NIMHANS Index for Specific Learning Disabilities for a small sample of children with SLD.

In a review on evidence-based assessment of cognitive, neuropsychological or intellectual skills to identify children with Learning Disability, a hybrid model was proposed. (Fletcher et al. 2005) This model incorporated two models, viz., achievement model and response to intervention model, which were found to be more reliable and valid.

In the Indian educational system, there is a lot of heterogeneity in the inputs given. Often children are promoted to higher classes even when they are not grade appropriate. Because of these inherent problems, it

is difficult to devise a test that can reliably and validly classify children with SLD, due to the vast differences in the inputs they get. Lack of early intervention in majority of the schools further complicated their education.

Keeping these difficulties in mind, the NIMHANS Index for SLD was developed. It has a modest goal of screening children with SLD and it should not be used for subtyping of SLD or for identification of those who would respond for intervention.

Assessment of the test-retest reliability of the test indicated that there was high significant correlation (0.53) between test findings after a gap of 6 months, indicating high temporal consistency of the test.

Criterion validity was established by comparing the NIMHANS Index for Specific Learning Disabilities findings with Teacher and Clinical ratings. Teacher rating was based on a letter from the teacher in which he/ she rated the child's academic performance. Clinical Rating was based on the diagnosis with respect to Learning Disability given by the CAMH Unit prior to assessment. The correlations of these ratings with findings on the NIMHANS Index for Specific Learning Disabilities were found to be 0.75 and 0.61 respectively. Both correlations were found to be highly significant.

On comparison of parental rating of the child's learning difficulties with the severity of SLD as assessed by the NIMHANS Index, only moderate association was present. This can be because the DPCL is a tool for screening psychopathology and not learning disability in detail. Thus comparison with a different standardised test specifically used for assessment of learning disability could have been used for a more reliable measure of concurrent validity.

Overall, it can be concluded that the NIMHANS Index for Specific Learning Disabilities has high test-retest reliability and criterion validity.

This test is thus a valid and reliable tool for assessment of Specific Learning Disabilities in children.

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Chapter - 8

Adaptation of Grotberg's tool for assessing resilience in children and adolescents

Uttarachari, Anupama V, Ratnesh Chopra, and Uma Hirisave

There are several studies examining prosocial behaviors in typically growing children and adolescents (Akoijam 2003, Vishwanatha 2009, Chari 2013, Bhattacharya 2016). The interesting finding is that both parents and teachers report high prosocial behavior in children and adolescents. There are not many studies examining child's self report of strengths. Resilience is considered as one of the strengths of children and adolescents.

Resilience is the capacity to bounce back from, or go forth despite adversity. Resilience is promoted by factors that lie within the individual and in the outside environment. The International Resilience Research Project (IRRP) was among the pioneering few to examine resilience promoting factors among children across countries. The IRRP sought to answer the question – *“What actions do parents or other caregivers themselves take that seem to promote resilience in the children upto age 12?”*

IRRP was carried out across the globe, with various cultures and countries, comprising collaborations with national and international agencies (Grotberg, 2001). Based on Erikson's developmental framework, resilience promoting factors were elicited from responses given by participants (parents/caregivers and children) on questions following a given scenario of adversity. Adversities were considered as those occurring in everyday life situations such as abandonment, bullying, aggression, etc. A total of 15 scenarios were constructed. All scenarios were administered on parents/caregivers, and six of the 15 were administered on children. Questions that followed these scenarios were:

Parents/Caregivers	Children
1. What did the adult do?	1. What did the adult do?
2. What did the adult feel?	2. What did the child do when the adult did that?
3. How did the child feel when the adult did that?	3. How did the child feel?
4. How did the child feel?	4. How did things come out or how are things now?
5. How did things come out or how are things now?	

Scoring for resilience promoting factors was done based on a 3 point system. A score of 3 was given to a response which fully endorsed resilience promoting factors. A score of 2 was given to a response displaying both resilience and non-resilience promoting factors, and a score of 1 was given to a response that was counter to promoting resilience.

Resilience promoting factors were categorized under 'I have' (external supports), 'I am' (inner strengths), and 'I can' (interpersonal and problem solving skills) factors. The factors, guided by prior research, were verbalized as under (Grotberg, 2001, p. 76-77):

I have

1. People around me I trust and who love me, no matter what.
2. People who set limits for me so I know when to stop before there is danger or trouble.
3. People who show me how to do things right by the way they do things.
4. People who want me to learn to do things on the own.
5. People who help me when I am sick, in danger or need to learn.

I am

7. A person people can like and love
8. Glad to do nice things for others and show my concern
9. Respectful of myself and others

10. Willing to be responsible for what I do
11. Sure things will be all right

I can

11. Talk to others about things that frighten me or bother me
12. Find ways to solve problems that I face
13. Control myself when I feel like doing something not right or dangerous
14. Figure out when it is a good time to talk to someone or to take action
15. Find someone to help me when I need

Psychometric properties were established on inter-rater reliability of coding the responses and convergent validity with tools such as Social Skills Rating System, Nowicki-Strickland Locus Control Test, and Parental Bonding Inventory. Findings on the IRRP revealed variations in use of resilience promoting factors across age, gender, culture, and socioeconomic status.

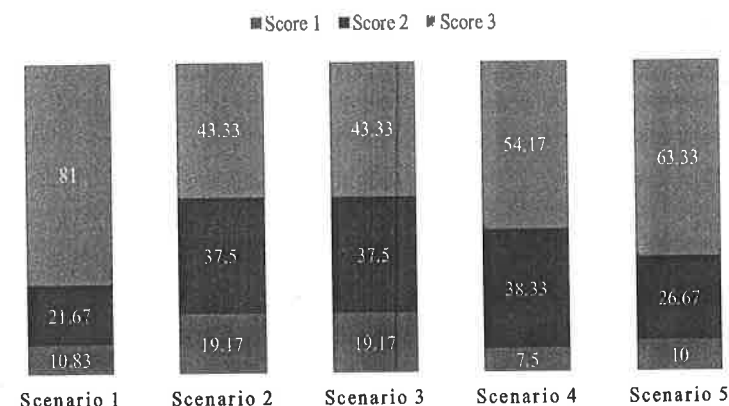
Studies in India

The method of utilizing Structured Scenarios with Questions (SSQ) to examine resilience was attempted in India by Chari and Hirisave (2017), towards examining resilience among children and adolescents (60 children & 60 adolescents) from the lower socioeconomic strata of the country. The study was carried out in the city of Bengaluru, and participants were drawn from government-run/aided school. The scenarios of the SSQ were adapted to make them contextually relevant for the study sample. The scenarios were adapted based on data from another study on a similar population (Vishwanatha, 2009). There were a total of 10 scenarios that were used, 5 each with children (8-12 years) and adolescents (13-16 years), which covered contexts of family, intra and interpersonal domains, and academic performance. Following each scenario, questions were asked regarding the actions and feelings of the characters and the expected outcome. The four questions were the same as those asked on the IRRP (Grotberg 2001). Only child questions were asked from the participants, as parents/caregivers

were not included as participants. The scenarios are given at the end of this chapter. Inter-rater agreement on scoring of responses was established.

Majority of participants' responses demonstrated complete use of resilience promoting factors (score of 3). Figure 1 displays the percentages of participants whose narratives were scored as 1, 2, and 3 on each scenario.

Figure 1: Percentage of scores of 1, 2, and 3 on scenarios on SSQ



N=120 (60 adolescents and 60 children)

Common 'I have' factors were structure and rules at home, parental encouragement of autonomy, stable home environment, and trusting relationships. 'I am' factors frequently endorsed were being autonomous, achievement oriented, of appealing temperament, with internal locus of control. 'I can' factors noted were of effective communication, problem solving, controlling impulses, and seeking trusting relationships. These were reflected as:

I have

1. People around me I trust and who love me, no matter what
2. People who set limits for me so I know when to stop before there is danger or trouble

3. People who want me to learn to do things on the own
4. People who help me when I am sick, in danger or need to learn

I am

5. A person people can like and love
6. Respectful of myself and others
7. Willing to be responsible for what I do
8. Sure things will be all right

I can

9. Talk to others about things that frighten me or bother me
10. Find ways to solve problems that I face
11. Control myself when I feel like doing something not right or dangerous
12. Figure out when it is a good time to talk to someone or to take action
13. Find someone to help me when I need

The SSQ-adapted version was subsequently used to assess resilience in 30 typically growing children between the ages of 5-7 years (Anupama,2018). The results are depicted in the table below

Table 1 - Mean resilience score of typically growing children as measured by the Structured Scenarios with Questions

Mean+SD	Range	
2.35 ± 0.36	Minimum	Maximum
	1	3

Range of Score - 1-3; High Score indicate greater resilience

The results indicated that these children have a moderate level of resilience. The responses largely had more of an 'I have' component reflective of support from parents, teachers and others.

Further, The SSQ was also utilized to explore the resilience of children and adolescents who had lost a parent in a road traffic accident (Anupama, 2018). 10 scenarios that were utilized to measure resilience previously in the Indian context (Chari, U & Hirisave, U, 2017) were utilized in this study. The study was carried out on 23 children and adolescents in the age group of 5-16 years. The results are depicted in the table below:

Table - 2 - Mean resilience scores of parentally bereaved children as measured by the Structured Scenarios with Questions – (N = 23 – Age range – 5 – 16 years)

Mean±SD	Median	Obtained range of scores
2.18 ± 0.44	3.00	1-3

Range of score= 1 – 3; Higher score indicates greater resilience.

On the whole, the group displayed a moderate level of resilience. The children gave responses that were broadly reflective of having emotional support both within and outside the family, a stable school environment, some structure and rules in the home setting.

Chopra (2017) explored resiliencies in children and adolescents with psychiatric disorders. The sample comprised of 63 participants with psychiatric disorders which was divided into children (participants between the ages of 8 years through 12 years 11 months) and adolescents (participants between the ages of 13 years through 16 years.). There were 37 adolescents and 26 children in the sample. The mean age of the children was 10.76 years and the mean age of the adolescents was 15.08 years. There were 21 girls and 42 boys in the sample. Children across both the groups had adequate intelligence. The sample was pooled from the Child and Adolescent Psychiatry Department of NIMHANS. They were all diagnosed cases of behavioural or emotional disorders and were stabilized on

medication. Most of them belonged to middle socio-economic status and came from urban backgrounds. Greater percentages of children were from nuclear families.

Resilience was assessed using Grotberg's Structures Scenarios with Questionnaire- adapted by Chari (2017).

Table - 3 - The mean resilient score of the sample (N=63)

Score	Mean	SD
Total Resilience score	2.26	0.46

It was found that the mean scores for all the participants (n=63) were above 2 indicating presence of mixed resilient and non-resilient promoting factors. There was no significant difference in resilience scores between children and adolescents and also between boys and girls.

Analysis of resilient responses from children indicated 'I have' factor in terms of emotionally supportive relationships at home and school and 'I am' factor in terms of seeing oneself being loved, valued and understood. Most of the children reported of relationships wherein adults come to their aid and provide support and soothe them in times of distress making them feel loved and valued. However, it appears that there is still scope for optimizing their response since although there is presence of support, it is not a complete episode promoting resilience as most often it ends up compromising the child's mood and functioning. It was interesting to note that very often, there is use of soothing behaviour accompanied with criticality or making the child settle for less or delaying effective action- which although settles the child for time being but does not completely pacify him/her. This indicates presence of both resilient and non-resilient promoting factors in children and also a scope for further refinement of response to promote complete resilience.

Adolescents reported 'I have' factor in terms of having supportive relationships within the family and 'I can' factor in terms of seeing oneself capable of solving problems. There is presence of tendency towards sharing of difficulties, using effective communication and seeking timely help. In most of the scenarios, children received support from their parents in times of distress and were helped to solve the problem jointly. However, it was noteworthy that although children received support from their parents but very often it also entailed use of coercion or curtailing the child's autonomy or displaying lack of trust and restraining the child or promising and providing hope but not fulfilling it or reassuring the child but not taking effective action to overcome the stress. Such support and aid from the parents, though rescued the child from the initial crisis, but does not promote complete resilience as it is not optimal and impacts the child's mood, performance and coping behaviour in the long run. Thus, it indicates presence of both resilient and non-resilient promoting factors in adolescents and a scope for further improvement in response.

The above findings indicate that moderate level of resilience factors in terms of 'I have, I can and I am' were found in all the participants. It suggests the need for intervention to foster resilience in children and adolescents with psychiatric disorders. There is a need to standardise this tool on large number of children and adolescents. Strength based interventions focussing on promoting protective factors could examine the utility of this tool in fostering resilience.

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Age group 8 - 12 years

Scenario I: During the day, a girl goes to school while her mother works as a maid. On holidays, the girl goes with her mother to the house where she works. One such day, this girl starts playing with the dolls and toys of the house owner's daughter. The mother sees this and snatches the toys away from the girl to put them back. The girl starts screaming and crying in protest.

Scenario II: An important guest comes home while the mother and son are at home. The mother goes to speak with the guests. The son feels hungry and demands for food. The mother does not pay attention to the son's demand as she is busy talking to the guest. The boy starts crying and makes a lot of noise which disturbs the mother and the guest. The mother asks him to stop making so much noise, following which he starts screaming in protest.

Scenario III : Few students in the class are repeating the year because they had failed. They look bigger than the rest of the children because they are older. The other younger children in the class frequently tease these children. One day the older slow learning children decide to teach a lesson to the younger children who tease them. The teacher sees these older children chase after these younger children.

Scenario IV: A boy studies hard for his examination and wants to do well. His father is very strict and the boy's brother is very good in studies. However, the boy does not get good marks and his father has to come to school to collect and sign the report card. The boy is scared what his father will say and do on seeing the marks.

Scenario V: Both parents of a girl work as they are poor and need money. The girl has to stay at home when the parents go to work. The girl is scared of being alone at home. She tells her parents about this and asks whether she can go with them.

Age group 13 - 16 years

Scenario I: A girl is studying in Class X. Both her parents work and she is responsible for the care of her younger brothers. Every day, this girl has to get herself and her brothers ready for school, assist her mother with household work and also help her brothers in their studies. She gets very tired and is not able to spend enough time for her studies. Her parents observe that the girl is worried about studying for the Board examination.

Scenario II: A girl and a boy were talking to each other after school regarding class notes. Some of their classmates saw them and informed their parents. The boy and girl were slightly late in returning home. They noticed that their parents looked angry.

Scenario III: Few boys are playing cricket after school. They had finished a very difficult test and had two more tests left. They were very excited and went to play instead of studying. They had planned to start studying for the next test after the game. Their fathers came back from work and noticed that the boys were not at home. Their mothers informed them that they had gone to play.

Scenario IV: A boy goes to a friend's house. There he saw his friend playing on his mobile phone. He asked his friend where he got it from as he always dreamed to buy one. So far, his family could not afford it. Anyway he went back home and asked his parents again.

Scenario V: Two girls are studying in the same class. One girl's parents are close friends of the teacher. Both girls study hard for the exam and do equally well. However, the girl known to the teacher is given more marks for the same answer. The other girl notices this and tells her mother when she returns home.

CHAPTER 9

MY EXPERIENCES SCALE (MES)

Snigdha Bhattacharyya, Mariamma Philip and Uma Hirisave

Studying experiences is of immense value as the ability to organize and understand our experiences are a basic part of how we make sense of our world. From a very early age, children are actively constructing how things happen, and forming memories of the events of their lives. (Fivush, 1998) The word 'experience' is perhaps the most commonly used word in behavioural research with children and adolescents. Ironically, the term 'experience' as well as 'various types of experiences' of children and adolescents are largely conceptualised by adults. Furthermore, 'experience' is most often defined either from the perspective of philosophy or a rudimentary dictionary version.

Children's early experiences and relationships build a foundation for their subsequent learning and social interactions. A child is most likely to reach his/her full potential if he/she experiences consistent, predictable, enriched, and stimulating interactions and positive experiences in context of attentive and nurturing relationships (Jackson & Cartmel, 2010). However, beyond an ideal world, children are also exposed to negative experiences. There are a wide variety of negative experiences which includes accidents, painful medical procedures, or life changing emotional events and abuse such as physical or sexual abuse or serious neglect and trauma. (Ziegler, 2002)

Traumatic experiences can have a devastating impact on the child, altering their physical, emotional, cognitive and social development. Traumatic events in childhood increase risk for a host of social (e.g., teenage pregnancy, adolescent drug abuse, school failure, victimization, anti-social behaviour), neuropsychiatric (e.g., post-traumatic stress disorder, dissociative disorders, conduct disorders) and other medical problems (e.g., heart disease, asthma). (Perry, 2000)

In order to understand different experiences of children and adolescents and how they work through their experiences, it would be important to evaluate the techniques/ methods used to access the experiences that children go through in clinical as well as community settings. The various methods that researchers and clinicians adopt to gain understanding of experiences among children and adolescents are many and can be classified as qualitative methods and quantitative methods. Quantitative methods are primarily self-report techniques which includes surveys and questionnaires. These techniques are objective in nature, time saving and less labour intensive. Quantitative techniques are useful in capturing the diversity of childhood and of children's daily life experience. (Qvortrup, 2008) On the other hand qualitative methods provide rich information, are open ended and holistic. Ideally for a more detailed viewpoint and better access to a child's world of experiences both qualitative and quantitative methods are used in adjunct with each other.

In the Indian context, a careful evaluation of various researches conducted as reported in the review of 50 years of research in NIMHANS (Shah & Rao, 2007) highlights that experiences in children and adolescents are explored primarily through interviews. Apart from this the only available standardised tool is Life Events Scale for Indian Children. (Malhotra, Kaur & Nehra, 1993) This is a parent report scale, adapted from the British Life events scale (Manoghan and Dodge, 1979) and the clinical validity of the scale is promising. Apart from this, very little work has been done to understand about the prevalence and occurrence of various experiences both traumatic and non-traumatic among children and adolescents. The dearth of studies in India to understand experiences in children may be attributed to challenges at different levels including administrative limitations and lack of clear guidelines in working with children especially special populations like those from difficult circumstances. However, the challenge of handling a diverse Indian population without efficient and standardized measures makes

research in the area a steep climb. Therefore, it was felt that there is a need to develop more standardized techniques for the assessment of experiences that children and adolescents go through.

Procedure of Scale Construction

Scale Construction involves systematic and step wise process of data collection and data analysis. The ethical considerations were reviewed and approved by the institution before the initiation of the study.

The sample of the study comprised of 1,897 participants from Bangalore city of urban India. Sampling was purposive and the participants were children and adolescence studying in 6 schools from different socio-economic backgrounds. The participants were between the age ranges of 8 to 16 years. Efforts were taken to ensure that the sample had equal representation of both genders, age ranges and socioeconomic background.

Two measures were used in the study for the purpose of establishing validity of the experiences scale. The Strengths and Difficulties Questionnaire (Goodman, 2005), the self-report version of SDQ was used. Self-Perception Profile for Children (Harter, 1982), The Global Self –Worth subscale was used which consists of 6 items

Written Informed consent was taken from the head of the schools and parents at every stage of the study. Only after receiving the consent forms, written assent was taken from the child for participation in the study. They were explained that participation in the study was voluntary. Participants were assured of confidentiality, especially from school authorities and were debriefed at the end of the interaction.

The process of developing an experiences scale and establishing its validity was carried out in 4 phases.

Phase 1 – Process of Item generation

The objective of this phase was to generate items for the scale through focus group discussions and semi structured interviews, so that the scale is truly representative of the voices of the participants. Twelve Focus Group Discussions and Semi Structured Interviews were carried out with 110 children and adolescents. The age ranges (8-10yrs, 11-13yrs and 14-16yrs) ensured homogeneity within the groups. Since boys and girls were in separate groups, it ensured that there was lesser inhibition and more comfort in the discussion.

The 12-audio taped Focus Group Discussions and Semi Structured Interviews were transcribed, emerging themes were coded and content analyzed for the purpose of item generation and understanding the construct of 'experience'. The Focus Group Discussions once transcribed were analyzed and reported using the Kreuger's manual on 'Analyzing and Reporting Focus Group Results' (Kreuger & Casey 2014). Similarly, the semi structured interviews once transcribed were analyzed following the Interpretative Phenomenological Analysis (Smith, Flowers & Larkin, 2009).

The themes and sub themes under positive and negative experiences was then categorized. The maximum themes of in the domain of positive experiences reported by participants were in the area of family relations (42%), followed by experiences in school (26%). Peer relationships (16%) related positive experiences themes and personal positive experiences themes (16%) were reported in equal percentage by participants.

Similarly, under the domain of negative experiences maximum negative experiences reported by participants were in the domain of family relationships (32%). This is closely followed by personal experiences (17%), school related experiences (15%) and abusive experiences (15%). Apart from this, negative experiences related to peer groups (12%) and experiences of societal pressure (9%) was also reported.

There were many sub themes under the broad themes in each domain of positive and negative experiences. The researcher and mentor discussed the findings and reached a consensus regarding dominant themes. An exhaustive list of initial pool of 75 items was sifted from the verbatims of the participants in the domains of positive and negative experiences.

Phase 2 – Scale Construction

The objectives of this phase were construction of the experiences scale and establishing the content validity of the scale. There were 5 steps involved in the process of Scale Construction

Step 1: - To carry out item reduction based on repetitiveness and relevance. The initial pool of 75 items were reduced to 60 items based on repetitiveness and relevance. The study aimed to develop common measure for both genders, a wide age range of 10 to 16 years and across different socio-economic background; it was decided that **only significant experiences that have been repeated across age ranges, socio economic background and both genders would be retained.**

Step 2: - To decide the Question Format of the newly developed scale. Various formats of item presentation in self-report scales with children and adolescents were looked into. It was decided to frame questions with four

criteria that are direct, clear, precise and relatively short. Keeping these criteria as a point of reference it was decided to frame questions as 'Have you experienced.....?'. This would make the questions straightforward and less confusing especially considering that the scale would be applicable across a heterogeneous sample of boys and girls from different age groups and different socio-economic backgrounds.

The questions were framed in a way that it has a closest resemblance to the verbatim provided by the participants in the focus group discussions and interviews.

Step 3: -To finalize Response Format of the newly developed scale.

Based on the extensive review of scales used to measure experiences, it was decided to have a four-point likert type frequency measure indicating how often the experience has occurred was finalized for the scale: 'Never'; 'Once'; 'A few times'; 'Many times. This would be a simple format and address issues pertaining to central tendency.

Step 4: -To develop the scoring and instructions for the newly developed scale.

The scoring of the items was finalized as 0(never), 1(once), 2(A few times) and 3(Many times). There was a total of 40 items measuring negative experiences. Therefore, maximum score on negative experiences is $3 \times 40 = 120$. Higher the score would be indicative of higher the number of negative experiences that a child has been through. On the other hand, positive experiences were reverse scored. There was a total of 20 items measuring positive experiences. Therefore, maximum score on positive experiences is $3 \times 20 = 60$. Higher the score would be indicative of lesser number of positive experiences that a child has been through. Total Score is the sum of the positive and negative experiences score. Higher the total

experience score would indicate more negative experiences and lesser positive experiences. Apart from the total score, the subscales of positive and negative experiences need to be looked into separately too. The scale was named 'My Experiences Scale'.

Step 5: - To establish the Content Validity of the scale.

The content validity of 'My Experiences Scale' was established after being reviewed by three experts (2 Females and 1 Male). The experts were mental health professionals with over 10 years of experience. Each expert was given, the newly developed scale, instructions, scoring and verbatim from which the items have been extracted. Apart from this the experts were also given a feedback form with the 4 criteria for evaluating the questions: wording, simplicity, clarity and repetition. The experts had consensus on most of the items and provided their feedback with regard to wording of certain items to simpler language. which was incorporated.

Phase 3 – Try Outs and Factor Analysis

The main objective of this phase was to understand the scale structure. For this purpose, repeated administration of the scale was done on large number of participants and the responses were subjected to factor analysis.

First Administration: Try out 1 of MES was on a sample of 705 freshly recruited participants. A total of 897 parental consent forms were distributed of which 705 students consented to participate in the study. Therefore, the participation percentage was 78.5%. The scale was administered in groups of 25 -30 participants. The completed 705 questionnaires were number coded and entered in Microsoft excel. The data set was then transferred to SPSS for factor analysis to be carried out. The number of missing data was found to be for a total of 50 items, across the 705 questionnaires, which was found to be 0.118%, which is a negligible percent. Following this, the data was subjected to exploratory factor analysis using Principal Component Analysis and orthogonal rotation. Based on the Kaiser Criterion a total of 16

factors emerged, explaining 56.27% of variance. Nine items were discarded in the first factor analysis. A new set of scoring key was made and the 51 items were rearranged randomly forming the modified version of 'My Experiences Scale'. The 16 factors that emerged were named and the factor structure was understood.

Second Administration: The modified version of 'My Experiences Scale' along with 'Strengths and Difficulties Questionnaire' was administered on a fresh sample of 445 participants. Efforts were made to match the background variable like type of school, background of the participants and even area of the school to a large extent. This was done keeping in mind that the factor structure would perhaps get affected if there is a lot of difference in the type of sample in Try out 2. The data was subjected to exploratory factor analysis using Principal Component Analysis and orthogonal rotation. Based on the Kaiser Criterion a total of 17 factors emerged, explaining 60.49% of variance. The 17 factors that emerged were named and the factor structure was understood.

Table 1: Means and Standard Deviations on My Experiences Scale

Experiences Score	Mean	Standard Deviation
Total Experiences Score	46.097	16.140
Positive Experiences Score	13.092	5.921
Negative Experiences Score	33.004	14.148

Possible Range of score on 51 items 'My Experiences Scale' is 0-153. Higher the score is indicative of higher number of negative experiences and lesser number of positive experiences.

Table 2: Factor Structure, items in each factor and total variance for My Experiences Scale

Factor Number	Factor Name	Number of Items	Total Variance
1	Experiences of Difficulties in Peer Relations	5	13.970%
2	Experiences of Loneliness and Sadness	4	6.648%
3	Experiences of Achievement & Recognition	5	4.471%
4	Experiences of Difficulties in School	5	3.583%
5	Experiences of Friendship	2	3.338%
6	Experiences of Heterosexual Relationships	4	3.106%
7	Experiences of Being Blamed& Compared	3	2.918%
8	Experiences of Stressful Home Environment	3	2.788%
9	Experiences of Togetherness in Family	4	2.555%
10	Experiences of Financial Difficulties	2	2.331%
11	Experiences of Worries	4	2.305%
12	Experiences of Feeling Safe	2	2.221%
13	Experiences of Health and Well Being Problems	2	2.141%
14	Experiences of Loss & Grief	1	2.087%
15	Experiences of Sexual Abuse	1	2.019%
16	Experiences of Hobbies and Activities	1	1.997%
17	Experiences of Opportunities in Hobbies	2	1.962%
TOTAL % Variance			60.49%

A total of 11 factors were found to be common across the factor analysis, thereby increasing the credibility of the scale. Factor 16 and 17 were merged and one item was renumbered. Following which the final version of 'My Experiences Scale' was formatted.

Phase 4 – Reliability and Validity of the Scale

The objective of this phase was to establish the reliability and validity of My Experiences Scale.

For the purpose of establishing reliability, re-administration of the scale was done on 70 participants after a gap of 15 days. At this point of time The Global Self Worth Scale (Harter, 1982) consisting of 6 items was also given to the participants.

Table 3: Reliability Correlation Scores of My Experiences Scale

Experiences Score	Internal Consistency Reliability: Cronbach's Alpha	Test - Retest Reliability	Guttman's Split Half Reliability
Total Experiences Score	.852	.994	.815
Positive Experiences Score	.745	.989	-
Negative Experiences Score	.886	.995	-

The internal consistency reliability of the scale was calculated using Cronbach's Coefficient Alpha was found to be good. Test Retest Reliability was calculated by re administering the test on 70 participants after a gap of 15 days. The test retest reliability was found to be excellent by all standards. Guttman's Split Half reliability is calculated by dividing the items in a test into two halves such that the covariance between scores on the two halves is as high as possible. The correlation coefficient was found to be good.

Table 4: Correlation coefficient of scores on My Experiences Scale with Strengths and Difficulties Questionnaire

Experiences Score	Total Difficulties	Emotional Symptoms	Conduct problems	Hyperactivity/inattention	Peer problems	Prosocial
Total Experiences Score	.635**	.510**	.362**	.459**	.398**	-.145**
	.001**	.001**	.001**	.001**	.001**	.002**
Positive Experiences Score	.266**	.188**	.054	.195**	.300**	-.273**
	.001**	.001**	.259	.001**	.001**	.001**
Negative Experiences Score	.615**	.506**	.385**	.446**	.336**	-.043
	.001**	.001**	.001**	.001**	.001**	.368

**p < 0.01

It was found that the total experiences score was found to be significantly ($p < 0.01$) positively correlated with total difficulties, emotional problems, conduct problems, hyperactivity and peer problems. This would imply that as the total experiences score increases (higher negative and lesser positive experiences); there would also be an increase in total difficulties, emotional problems, conduct problems, hyperactivity and peer problems.

It was also found that the total experiences score was significantly ($p < 0.01$) negatively correlated with prosocial behaviour. This would imply that as the total experiences score increases (higher negative and lesser positive experiences); there would also be a decrease in prosocial behaviour.

Table 5: Correlation coefficient of scores on My Experiences Scale with Global Self Worth

Experiences Score	Global Self Worth
	-.391
Total Experiences Score	.001**
	-.297
Positive Experiences Score	.013**
	-.315
Negative Experiences Score	.008**

**** $p < 0.01$**

The comparison of the findings of Experiences score with Global self-worth was found to be significantly ($p < 0.01$) negatively correlated. This implies that as the experiences scores increases (more negative experiences and

lesser positive experiences) the self-worth reported by the participant was found to decrease.

These findings in turn establish the concurrent validity of the scale.

Therefore, 'My Experiences Scale' is developed and standardised on Indian children and adolescents. The scale has sound psychometric properties and has a promising scope for its application as well.

Application of My Experiences Scale (MES)

- The scale is a comprehensive measure of experience and goes beyond symptoms, thereby providing an experiential viewpoint about the pathology rather than a symptomatic viewpoint as often seen in the clinical setting.
- The findings on MES can provide direction in therapy and will aid in identification of areas of work with children and adolescents.
- The findings on MES taps on experiences of subclinical populations where the manifestation of symptoms is often subtle and hence neglected.
- MES can be a useful screening tool for children and adolescents in schools, institutions, orphanages, street children and children in conflict with law.
- The scale can also aid in the process of forensic interviewing with abused children.
- MES would provide children and adolescents from different backgrounds and circumstances an opportunity to communicate about their experiences.

- MES touches on significant positive and negative experiences across different domains. It would provide the researcher /clinician a comprehensive understanding about the child's life in 45mins.
- MES is easy to administer, score and interpret. Therefore, making it a useful tool for school counsellors.
- MES is developed and standardized in India. Therefore, making it a highly reliable tool for research in India especially in the area of school mental health.

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MY EXPERIENCES SCALE - By Children; For Children
(*Bhattacharyya S and Hirisave U*)

Instructions for administration of the scale-

'Given below are various experiences that boys and girls of your age report. I would like to know about the experiences that you have had. Are you willing to share your experiences with me? (Wait for the boy/girl to respond as Yes, then proceed)....Thank you for agreeing to share your experiences, whatever you share will not be revealed to anyone.'

Experiences are what we personally go through and learn from. Experiences are made up of our memories & feelings and are often beyond our control'. – This is how boys and girls of your age understand experience.

Here is a list of 50 experiences given below. Read it carefully, look back and think if this experience has happened to you. There are four responses given below the question.

They are – 1. Never 2.Once 3. A few times 4.Many times

Based on how often you have had the experience, you can tick on the appropriate box. Try to respond to the questionnaire as truthfully as you can.

**Children will be instructed that if they do not understand the item, they can ask for clarification.

SI No	Experience Item	Never	Once	A Few Times	Many Times
1	Have you experienced having a close friend?				
2	Have you experienced doing well in hobbies and extracurricular activities? (Eg – Sports/Art – Drawing, Painting/ Dance/Music etc)				
3	Have you experienced having difficulties in studies?				
4	Have you experienced not being included by your friends and pushed away by friends?				
5	Have you experienced getting enough chances for participating in extracurricular activities?				
6	Have you experienced any physical discomfort? (Eg – Headaches/ stomach aches)				
7	Have you experienced your parent/s blaming you for the mistakes done by your brother/sister?				

SI No	Experience Item	Never	Once	A Few Times	Many Times
8	Have you experienced having close friend/s with whom you do activities that you enjoy?				
9	Have you experienced feeling very moody? (Irritable, sad for no reason...)				
10	Have you experienced watching your parents get into serious fights/arguments with each other?				
11	Have you experienced doing well in your studies?				
12	Have you experienced having a nice time doing activities or playing with your brother/sister or cousins?				
13	Have you experienced being gossiped/ commented badly about by relatives or neighbours?				
14	Have you experienced your classmates or friends talk not so nice things (spread rumours/ gossip) about you behind your back?				
15	Have you experienced being shouted and scolded at with bad words by a parent?				
16	Have you experienced your teachers appreciating you?				
17	Have you experienced enjoying yourself in various school events? (Eg – Sports day, Teachers day, Childrens' day...etc)				
18	Have you experienced having a friend who helps you in times of need?				
19	Have you experienced your seniors / friends / classmates irritating and troubling you when you are alone?				
20	Have you experienced being teased, stared at and commented on when you walk on the street or neighbourhood?				
21	Have you experienced being hit/pushed/ slapped/kicked/beaten by a parent?				
22	Have you experienced feeling fearful about accidents that happen on the roads?				
23	Have you experienced your parent/s being partial to your brother/sister?				

Si No	Experience Item	Never	Once	A Few Times	Many Times
24	Have you experienced being blamed by your teachers for things that you have not done?				
25	Have you experienced loss (death) of a family member/ someone you were close to?				
26	Have you experienced your teachers encouraging your talents?				
27	Have you experienced doing some activity with your family where everyone has a nice time? (Eg – Shopping, Watching TV together) Any other activity, please mention.....				
28	Have you experienced being hit / pushed / slapped / kicked / beaten by a teacher?				
29	Have you experienced being worried about health problems in any family member?				
30	Have you experienced feeling attracted towards friends of opposite gender?				
31	Have you experienced your birthday being celebrated?				
32	Have you experienced your parent/s misunderstanding your interactions with friends of the opposite gender?				
33	Have you experienced some developments in your body that makes you uncomfortable?				
34	Have you experienced being scared or worried about your parents getting divorced or separated?				
35	Have you experienced your teachers scolding you for your poor performance in studies? (Eg – test performance, completion of HW/CW, poor handwriting)				
36	Have you experienced difficulty in making friends and felt alone?				
37	Have you experienced wanting to end your life?				
38	Have you experienced feeling guilty about feeling attracted towards friends of opposite gender?				
39	Have you experienced being involved in celebrations at your home?				

Si No	Experience Item	Never	Once	A Few Times	Many Times
	(Eg - Festivals, Weddings, other functions/ gatherings)				
40	Have you experienced feeling very upset and wanting to leave everything and go away?				
41	Have you experienced your teacher/teachers giving you responsibilities? (Eg- Board Leader, Teach dance to others ...etc)				
42	Have you experienced your parent/s trusting you even when someone has complained about you?				
43	Have you experienced anyone touch you in a way that made you feel uncomfortable? (Eg – touching your private parts, bad touch)				
44	Have you experienced being disappointed in love/friendship with the opposite gender?				
45	Have you experienced difficulties while doing all your tasks by yourself without any help from a parent? (eg- Making your breakfast, Ironing your clothes, Studying alone)				
46	Have you experienced that nobody is able to understand you?				
47	Have you experienced your parent/s refusing to buy you things as they don't have money? (school fees, school uniform, food at school etc)				
48	Have you experienced being constantly compared with other children (brother/sister/ cousin/ friend)?				
49	Have you experienced being worried about bad habits/behaviours in any family member?(Eg – Drinking, Gambling, Smoking...)Any other, please mention.....				
50	Have you experienced your parent/s worry about money/financial difficulties? (Eg –debts/loans)				

ANY OTHER EXPERIENCES YOU WANT TO MENTION ABOUT

DO YOU WANT TO DISCUSS YOUR EXPERIENCES WITH THE PSYCHOLOGIST/ MENTAL HEALTH PROFESSIONAL?

Kindly Note : The Author's permission need to be taken if the scale is used for research purposes (Email ID: snigdhab1@yahoo.com)

SCORING of 'My Experiences Scale'

- There are 3 scores, total positive experiences score, total negative experiences score and total experience score.
- Total Positive Experience Score is sum of positive experiences.
- Total Negative Experience Score is sum of negative experiences.
- Total Score is the sum of the positive and negative experiences score.
- Score Range is 0 to 150.
- Higher the total experience score would indicate more negative experiences and lesser positive experiences.
- Apart from this, there are 16 factors that group the items and the results can be explained based on these factors and the responses on them.

SCORING KEY

EXPERIENCE	ITEMS		SCORING
Positive Experiences	x 1	x 18	Reverse Scoring Never =3 Once =2 A Few Times =1 Many Times =0
	x 2	x 26	
	x 5	x 27	
	x 8	x 30	
	x 11	x 31	
	x 12	x 39	
	x 16	x 41	
Negative Experiences	x 17	x 42	Regular Scoring Never =0 Once = 1 A few times =2 Many Times =3
	x 3	x 29	
	x 4	x 32	
	x 6	x 33	
	x 7	x 34	
	x 9	x 35	
	x 10	x 36	
	x 13	x 37	
	x 14	x 38	
	x 15	x 40	
	x 19	x 43	
	x 20	x 44	
	x 21	x 45	
	x 22	x 46	
	x 23	x 47	
	x 24	x 48	
	x 25	x 49	
	x 28	x 50	

Chapter 10

NIMHANS SENTENCE COMPLETION TEST FOR CHILDREN AND ADOLESCENTS (NSCA)

Uma Hirisave and L.N. Suman

Introduction

Projective techniques are used for inquiry and diagnosis, using vague or meaningless stimuli to elicit responses that are likely to reveal hidden personality strata in a subject through the projection of inner content onto the external stimulus. The various stimuli used in projective techniques are intentionally vague and open to different interpretations, in the expectation that the subject will give meaning to the stimuli, meaning that emerges from internal personality processes, and thus enable observation of these processes. Various experts believe that the greater freedom subjects have to choose their responses, the more their responses will be charged with meaning for them, since they can "supplement" what the stimulus lacks and accord it meaning solely by means of their own internal content. Thus, there are no "right" or "wrong" responses to the stimuli, but whether the responses are normative or non-normative is of great importance. Projective techniques possess several clear advantages: they do not require subjects to have high reading ability or a particularly high level of articulation (compared to other personality tests), their results are very difficult to falsify, and they enable a wide variety of assumptions and a broad and comprehensive view of the subject's personality (Levin-Rozalis, 2006). Projective techniques have been developed specifically for use with adults as well as children.

Sentence completion tests are a class of semi-structured projective techniques. Sentence completion tests typically provide respondents with beginnings of sentences, referred to as "stems," and respondents then complete the sentences in ways that are meaningful to them. The responses are believed to provide indications of attitudes, beliefs, motivations, or other mental states. A sentence completion test form may be relatively short, such as those used to assess responses to advertisements, or much longer, such as those used to assess personality. The tests are usually administered in booklet form where respondents complete the stems by writing words on paper. The structures of sentence completion tests vary according to the length and relative generality and wording of the sentence stems. Structured tests have longer stems that lead respondents to more specific types of responses; less structured tests provide shorter stems, which produce a wider variety of responses. The uses of sentence completion tests include personality analysis, clinical applications, attitude assessment, achievement motivation, and measurement of other constructs. The data collected from sentence completion tests can usually be analyzed either quantitatively or qualitatively.

In a review of literature on the Sentence Completion Tests (SCT), Holaday, Smith and Sherry (2000) found that they were among the most commonly used personality assessment instruments. They were ranked second by Japanese clinicians, third by clinical psychologists, fifth by clinicians working with adolescents, fourth by school psychologists, fifth by representatives of mental health service providers, and third by members of the Society for Personality Assessment in response to the question: "With what 5 projective tests should the professional practitioner be competent?"

The Sentence Completion Test developed by Sacks and Levy (1950) is used with adult patients in NIMHANS. It was developed to explore specific clusters of attitudes or significant areas of an individual's life.

Recently, Holaday, et al (2000) carried out a survey to examine which Sentence Completion Test (SCT) was most used, why it was preferred and the scoring method adopted. The sample consisted of 60 members of the Society for Personality Assessment, U.S.A. They found that out of 15 Sentence Completion Tests available to the clinicians, Rotter's (1951) Incomplete Sentences Blank was the most popular with 58% of the sample reporting its use. 11% used it with children, 19% with adolescents and 28% with adults. 8 of the tests were not used at all and the rest were used by 4% or less. 80% of the respondents who used SCT with children preferred to read stems aloud and record the answers for their clients. The responses were not scored quantitatively but were interpreted by the examiners based on their own clinical skills. The authors also note that although Rotter's SCT is very popular, it does not have specific items that might be relevant for younger children or children with diverse cultural backgrounds.

Dyken, Maxwell Myatt and Schwenk, (2007) used a Sentence Completion Test, Three Wishes Task and the Child Behavior Checklist to assess self-perceptions, thoughts, hopes and inner lives of 128 persons with intellectual disabilities (ID), including children. They found that the semi-projective test provides persons with ID with just enough structure or cues to convey their self-perceptions in a spontaneous and unbiased manner. Results indicated frequent themes of activities, positive affect, desired objects and relationships with family and pets. The responses were significantly related to either syndrome status or CBCL maladaptive behavior. The study thus indicates the usefulness of SCT in the clinical setting with children.

According to Holaday, et al (2000), SCTs will remain popular with practitioners because the tests provide significant information that is valuable for diagnoses and treatment planning. They suggest that practitioners may be better served by writing some of their own stems, reading them aloud to their clients, and recording the responses themselves. They also recommend that SCTs should (a) be tailored to provide appropriate and accurate information about clients' personalities, (b) be based on the theoretical orientations of the examiner, (c) reflect the needs and expectations of clients' cultural and religious backgrounds, and (d) be appropriate for clients' academic and mental developmental levels. However, there are no well established Sentence Completion Tests developed specifically for children in the Indian setting. In order to overcome this lacuna, Uma and Suman (2002) developed a SCT for use with children and adolescents.

Development of the SCT for Children and Adolescents

Step 1

The aim of developing the test was to assess adjustment of children in different domains. The objective was to assess the child's perceptions, attitudes and interactions to arrive at a comprehensive understanding of the child's adjustment, conflicts, relationship problems and self-concept. Items for the test were generated by examining literature on psychological needs, relationships and aspirations of children. Clinical experience with children attending the Child and Adolescent Mental Health (CAMH) Unit at NIMHANS was also used to generate some of the items. Two psychologists with extensive experience in working with children and their families generated sentence stems. Care was taken to keep the stems brief and the language of the items simple. The total number of items generated was 125. Socio-cultural factors were kept in mind while generating the

items. For example, since Indian families are fairly large and since Indian culture is inter-dependent, items related to siblings and grand-parents were included. Similarly, parental attitude towards the child was considered important in view of neglect of the girl child in the Indian setting.

Step 2

The 125 items, written in English, were categorized into five broad areas: Home; Friends; School; Self-concept; Neutral Items. The Home area has six sub-areas which are: Relationship with Father; Relationship with Mother; Relationship with Siblings; Father-Mother Relationship; Parental Attitude towards Child; Family Unit. The area of friends refers to peers at school as well as friends outside school. The School area has two sub-areas which are: Adjustment to School; Attitude to Teachers. The area of Self-concept has five sub-areas which are: Attitude towards Own Abilities; Emotional Functioning; Coping with Emotions; Guilt Feelings; Goals for the Future. The Neutral items included likes and dislikes as well as relationship with grandparents. Thus there are a total of 15 sub-areas under the five broad categories of adjustment. Items were written to include both positively worded stems and negatively worded stems. The sentence stems representing different sub-areas were ordered in such a way that items in any specific area were not clustered together. The scoring system suggested by Sacks and Levy (1950) was adopted for the test. Each area is rated on a three point scale, i.e., 0, 1 and 2. 0 indicates good adjustment; 1 indicates mild problems in adjustment; 2 indicates significant problems in adjustment.

Step 3

The 125 item test was given to a clinical psychologist with more than 10 years experience in working with children and adolescents for final

selection of the items. Items were examined for face validity, repetition, language, usefulness and appropriateness. Inter-rater agreement was also carried out to arrive at a consensus for including or excluding items in specific areas. The inter-rater agreement was found to be good. That is, the inter-rater agreement was at the 0.001 level. Following these procedures, 35 items were rejected, leading to a final item pool of 90 items. These 90 items are spread over the 15 sub-areas in such a way that each sub-area has 6 items. The instructions were kept simple and an example was given. The instructions, for administration of the test by the examiner, are as follows: *"This is a test to find out how you generally relate to people, your feelings, thoughts, likes, dislikes and hopes. I will be reading out sentences which are incomplete. Your task is to complete the sentence as quickly as possible with the first thought that comes to your mind. Do not spend too much time thinking about your answer. There are no right or wrong answers. You have to complete all the items in the test. Since you are familiar with school tasks such as 'Fill in the Blanks', this task will not be difficult for you to carry out. If you have any doubts about any of the sentence stems, feel free to clarify your doubts before responding. For example, complete the following sentence: 'The color of a crow is _____. Now, I will be reading the other items. Are you ready?'"* For the Self-report method, the sentence *'I will be reading out sentences which are incomplete'*, was replaced by the sentence *"Read carefully the incomplete sentences given below"*.

Step 4

A Pilot Study was carried out to examine the ease with which the test could be administered and to check the ease with which children could comprehend the sentence stems. The sample consisted of 40 children in the age range of 8 to 12 years who were comfortable with the English

language. 20 children were clients attending the CAMH Unit at NIMHANS and 20 were normal children selected from a school. Oral informed consent was taken from all the children. It was found that the children could easily understand the sentence stems and respond appropriately. The time taken to administer the test ranged from one hour to one and a half hour. None of the children found the test too long and they showed keen interest in the task. Some of the children attending the CAMH clinic cried when they came across items that had a direct bearing on their problem. Both quantitative and qualitative analyses of the responses were carried out. Findings revealed that the test was useful in getting a profile of adjustment in children, indicating good clinical utility. Based on the Pilot Study, it was decided that oral administration was a better approach as it took less time, responses could be clarified immediately and it was appropriate for children with reading/writing/spelling difficulties. In the self-report format, some items were left out by the children. They were gently coaxed to complete items that they had left out.

Step 5

This step aimed at obtaining information about adjustment of normal children. The management of various English medium schools were contacted to obtain permission to administer the test on normal school going children. The aim of the study was explained to them along with the ethical principles to which the investigators would adhere. After obtaining the required permission, oral informed consent was sought from the children. None of the children refused consent and the test was administered to children studying in four English medium schools. The total sample size was 162 and the age range was 8 to 14 years. The sample consisted of 93 boys and 69 girls and they showed

tremendous interest in the test. Responses indicated that the test was able to correctly identify children with adjustment difficulties, relationship conflicts and emotional problems. Typical responses of well adjusted children were also noted for purposes of comparison.

Application of the Test in a Research Study

Dhanya (2009) carried out a study to examine the adjustment and social problem solving skills in children with emotional and behavioral problems. The sample consisted of 30 children with a psychiatric diagnosis of Behavioral and Emotional Disorder (F 91-98); Depressive Disorder (F32 and F 33); or Dissociative Disorder ((F 44) on Axis I of the Multi-axial Classification System of ICD-10 (WHO, 2002). Care was taken to include only new cases registered in the CAMH Unit of NIMHANS. The sample also consisted of 30 normal children selected from English medium schools. They were screened on the Children's Behavior Questionnaire (CBCL; Rutter, 1967) to rule out children with psychological problems. The age range of the sample was 8 to 12 years and both boys and girls were included in both the groups. The profile of emotional and behavioral problems was obtained using the Developmental Psychopathology Checklist (DPCL; Kapur, et al, 1994). Adjustment was assessed using the Sentence Completion test for Children and Adolescents (SCT-C) Uma and Suman, 2002) and Social Problem Solving was assessed using Group Social Problem Solving Assessment (GPSA; King, 1986). Written informed consent was taken from parents and teachers and assent for the study was taken from the children.

Results of the study as given in Table 1 indicate that there was a significant difference between the two groups on overall adjustment as measured by the SCT-C.

Table 1

Group		N	Mean	SD	t		
SCT	Clinical	30	41.63	11.89	t value	df	p value
Total Score#	Normal	30	31.77	10.44	3.80	54	.001*

In the study; each item has been rated on the three point scale ranging from 0 to 2; thus the maximum possible score is 168 (84 items multiplied by two;neutral items are not scored) * significant at .001 level

Analysis of the responses in terms of sub-areas revealed significant differences between the two groups in five areas. These were: Adjustment to School; Emotional Functioning; Coping with Emotions; Parental Attitude towards the Child; and Guilt Feelings. There were no significant differences between the two groups in the other ten sub-areas. Details of significant differences are given in Table 2.

Table 2

Area	Group	N	Mean	SD	t
Adjustment to School	Clinical	30	0.73	3.7	.001*
	Normal	30	0.52		
Emotional Functioning	Clinical	30	0.52	2.8	.007*
	Normal	30	0.50		
Coping with Emotions	Clinical	30	0.69	2.3	.022**
	Normal	30	0.50		
Parental Attitude towards Child	Clinical	30	0.52	3.5	.001*
	Normal	30	0.50		
Guilt Feelings	Clinical	30	0.80	3.9	.000*
	Normal	30	0.47		

* significant at .05 level; ** significant at .001 level

In the School area, responses of clinic children revealed more difficulties in school such as fear of school itself and fear of examinations. They also indicated that most of their friends were from their home networks rather than schoolmates. On comparing the responses of the two groups on the domain of Emotional Functioning, it was found that children in the clinical group reported more negative emotions such as anxiety and fears. The two groups also differed in the way in which they coped with emotions. Children in the clinical group had poorer coping skills compared to normal children. Further, children in the clinical group perceived their parents as having less favorable attitude towards them, which in turn contributed to their adjustment difficulties. Children in the clinical group also had significantly high Guilt Feelings. Their guilt was mainly related to being a burden on their parents due to their illness/difficulties.

Children in the normal group were more spontaneous in their responses and needed fewer explanations from the examiner during assessment. They often described their father as a 'friendly person'; mother as one of their 'dearest friends'; sibling as 'cute/helpful' and their families as 'best'. They perceived their parents in a positive light and used more subtle expressions to show their emotions. For example, 'I am disappointed with my father at times' as compared to superlative expressions by children in the clinical group who used phrases such as, 'I hate my father' or 'My father is my enemy'. They also felt that their parents considered them as 'talented or special'. Children in the normal group had positive attitudes towards school and teachers and a good self-concept.

higher the stress, poorer the adjustment as measured by the SCT. Thus, the results obtained in the study revealed that the test could significantly discriminate between the two groups. This indicates good clinical validity and usefulness. Further, qualitative analysis of responses provided information about the nature and extent of the difficulties reported. The responses can be used as a spring board to carry out in-depth interviews to plan psychological interventions. The test can also be used as a pre and post measure of adjustment in intervention studies or in routine clinical practice.

Based on the experience of using SCT with young children, the authors decided to reduce the number of items from 90 to 75. Two items were replaced and 5 items were slightly modified. The items omitted were - 28, 29, 38, 48, 60, 64, 66, 75, 76, 80, 81, 83, 86, 88 and 90. The instructions also were modified. The final version is called NIMHANS Sentence Completion Test for Children and Adolescents (NSCA), given in Page 294.

In a recent study by Anupama (2018) norms have been established for normal children between the ages of 5-7 years for this test. The data was collected on a sample of 30 children in the age group 5-7 years studying in government school. The children were well adjusted with respect to all areas of NSCA Home, School, peers, self-concept. The author calculated total score for each domain instead of sub domains. The mean and SD in different areas were as follows. 0.31 ± 0.14 (Home), 0.08 ± 0.04 (peers), 0.35 ± 0.16 (School) and 0.21 ± 0.13 (self concept).

The results revealed that as a group the children were well adjusted with respect to these areas. Similar results were obtained by the same researcher on a sample of 23 children, age range 5-17 years who had experienced parental loss due to road traffic accident. The children were found to be well-adjusted in all the domains.

Salian (2013) has used a brief version of NSCA for assessing attachment of normal children using items from the domain of 'home' from NSCA. Findings of the study revealed that children reported more attachment with their mothers.

**NIMHANS Sentence Completion Test for
Children & Adolescents (NSCA) Initial Version
(Final version on Page No.287)
Department of Clinical Psychology, NIMHANS, Bengaluru**

INSTRUCTIONS: You will find below a number of incomplete sentences. For example: "The colour of the leaf is, Write/tell quickly in the blank space the first idea that comes to your mind to complete each sentence. Quickness is very important, please answer rapidly.

1. My father is
2. My friends and I
3. My teachers are
4. My mother and I
5. At school, I feel
6. I have the ability to
7. My brother is
8. I am afraid of
9. My mother and my father
10. I like to
11. When I am happy, I
12. My parents think I am
13. My ambition is
14. In my family

15. My biggest mistake was
16. I like my father because
17. Most of my friends
18. My teachers and I
19. I like my mother because
20. I like going to school, but
21. When things go wrong I
22. My sister is
23. I feel sad when
24. My father treats my mother.....
25. My favourite eatable is
26. When I am anxious, I
27. My parents feel happy when I.....
28. My greatest wish is
29. My parents always
30. I wish I had not
31. My father and I
32. A real friend is
33. I feel my teachers

34. My mother
35. I find class tests
36. In order to achieve something, I can.....
37. My brother and I.....
38. I wish I could lose the fear of
39. I wish my father and mother
40. My grand father.....
41. When I am afraid, I
42. I am scolded when
43. I want to become
44. My family treats me
45. I feel guilty about
46. I wish my father
47. I like my teachers but
48. Most mothers are
49. The subject I like most
50. When problem arise, I
51. My sister and I
52. I feel angry when

53. My mother treats my father
54. I am interested in
55. When I am sad, I
56. My parents get angry when I
57. I will be happy if
58. The person I like most in my family.....
59. I wish I could' forget
60. Most fathers are
61. My best friends
62. Most teachers are
63. I dislike my mother because
64. My school is
65. I like my brother but sometimes
66. My fears force me to
67. My favourite game is
68. When I am angry, I
69. I am beaten when
70. I always 'wanted to be
71. I wish my family

72. The worst thing that I ever did was.....
73. I dislike my father because
74. I expect my friends to
75. I wish my mother
76. I wish schools were
77. I like my sister but sometimes
78. I am happy when
79. My grandmother
80. My parents see me as
81. My secret desire is
82. Most families I know are
83. I feel bad about
84. I wish my brother
85. The subject I dislike most
86. I wish my sister
87. I feel fearful when
88. My brother always
89. I feel anxious when
90. My sister always

Examples of responses of some children in the
Normal group on
NIMHANS Sentence Completion Test - Initial Version

FATHER:

Boys:

8-10 Years	11-14 Years
1. My father is an auto driver.	1. My father is Srikant
16. I like my father because he is good.	16. I like my father because he brings me whatever I want and he is happy with me.
31. My father and I play.	31. My father and I make my mother happy.
46. I wish my father happiness.	46. I wish my father get me a car after I grow up
60. Most fathers are very good.	60. Most fathers are unforgettable.
73. I dislike my father because no reason to dislike.	73. I dislike my father because but I never dislike my father.
*R=0	R=0

Girls:

8-10 Years
1. My father is a carpenter.
16. I like my father because he takes me to other places.
31. My father and I joke
46. I wish my father to get more money
60. Most fathers are working hard
73. I dislike my father because he does not give me any toys
R=1

* Rating.

FRIENDS

BOYS:

8-10 Years	11-14 Years
2. My friends and I play	2. My friends and I enjoyed the picnic
17. Most of my friends are good	17. Most of my friends tease me but are friendly
32. A real friend is Vinay and Arun	32. A real friend is always helpful
61. My best friends is Vinay	61. My best friend is a very helpful friend
74. I expect my friends to be my family	74. I expect my friends to be friendly and good
R=0	R=0

TEACHER:

BOYS

8-10 Years	11-14 Years
3. My teachers are teaching lessons	3. My teachers are very good
18. My teachers and I take books	18. My teachers and I help each other
33. I feel my teachers are good	33. I feel my teachers help me in studies
47. I like my teachers but they will beat	47. I like my teachers but I do not feel anything
62. Most teachers give notes	62. Most teachers are very friendly
R=0	R=0

GIRLS

8-10 Years	11-14 Years
3. My teachers are very good	3. My teachers are like my mother.
18. My teachers and I are happy	18. My teachers and I talk in a friendly way.
33. I feel my teachers very good	33. I feel my teachers have the greatest job-teaching.
47. I like my teachers but they beat.	47. I like my teachers but some times I dislike
62. Most teachers are good.	62. Most teachers teach very well.
R=0	R=0

MOTHER:**BOYS:**

8-10 Years	11-14 Years
4. My mother and I help each other	4. My mother and I walk everyday
19. I like my mother because I love her	19. I like my mother because she has helping nature. She is my sweetest mother.
34. My mother likes my teacher	34. My mother forever unforgettable mother.
48. Most mothers are good	48. Most mothers are greatful mother.
63. I dislike my mother because nothing to say.	63. I dislike my mother because I never dislike.
75. I wish my mother	75. I wish my mother happliness
R=0	R=0

GIRLS:

8-10 Years	11-14 Years
4. My mother and I go to park.	4. My mother and I like friends.
19. I like my mother because she teaches me.	19. I like my mother because she is kind and helpful to me.
34. My mother does kitchen work and is busy	34. My mother is very helpful and kind.
48. Most mothers are good.	48. Most mothers are helpful and good.
63. I dislike my mother because of her hitting.	63. I dislike my mother because no I won't dislike my mother.
75. I wish my mother good	75. I wish my mother nothing.
R=0	R=0

SCHOOL:**BOYS:**

8-10 Years	11-14 Years
5. At school, I feel good	5. At school, I am happy with my friends
20. I like going to school, but my teacher will be waiting.	20. I like going to school, but I don't feel like going after 5 to 6 days of holidays.
35. I find class tests good.	35. I find class tests happen very quickly.
49. The subject I like most English, GK, Social science.	49. The subject I like most is maths.
64. My school is a good school.	64. My school is most unforgettable school.
76. I wish schools were having good building.	76. I wish schools were always good.
R=0	R=0

GIRLS:

8-10 years
5. At school, I feel nothing.
20. I like going to school, but my friend don't beat
35. I find class tests good
49. The subject I like most social, GK, Kannada.
64. My school is good school.
76. I wish schools were good.
R=0

OWN ABILITIES:

BOYS:

8-10 Years	11-14 Years
6. I have the ability to play.	6. I have the ability to get first rank.
21. When things go wrong I will be bad.	21. When things go wrong I try to clear it.
36. In order to achieve something, I can do.	36. In order to achieve something, I can show my will power.
50. When problem arise I feel bad.	50. When problem arise I try to solve it.
R=1	R=0

GIRLS:

8-10 Years	11-14 Years
6. I have the ability to read.	6. I have the ability to be successful in life.
21. When things go wrong my mother will beat	21. When things go wrong I feel shamed.
36. In order to achieve something, I can do	36. In order to achieve something, I can work hard.
50. When problem arise	50. When problem arise I help myself.
R=1	R=0

SIBLINGS:

BOYS:

8-10 Years	11-14 Years
7. My brother is studying in V standard.	7. My brother is one sometimes teases but a nice person
22. My sister is student in VI standard.	22. My sister is good
37. My brother and I play.	37. My brother and I will always be joking
51. My sister and I feel good.	51. My sister and I go out to eat without asking parents.
65. I like my brother but sometimes he beats.	65. I like my brother but sometimes not
84. I wish my brother good	84. I wish my brother always be good
86. I wish my sister good.	86. I wish my sister to be good.
88. My brother always beats	88. My brother always help me in everything
90. My sister always good	90. My sister always helpful to me.
R=1	R=0

GIRLS:

8-10 Years	11-14 Years
7. My brother is no brothers.	7. My brother is friendly person.
22. My sister is beats me	22. My sister is my best friend.
37. My brother and I no brother	37. My brother and I never fight.
51. My sister and I read.	51. My sister and I never fight.
65. I like my brother but sometimes no brother.	65. I like my brother but sometimes he is naughty.
84. I wish my brother no brother.	84. I wish my brother explain many good things
86. I wish my sister beats me less.	86. I wish my sister becomes less impatient
88. My brother always no brother	88. My brother always cracks jokes
90. My sister always read.	90. My sister always try to explain what I don't know.
R=1	R=0

EMOTIONS

BOYS

8-10 Years	11-14 Years
8. I am afraid of a sound.	8. I am afraid of devil stories
23. I feel sad when I feel bad.	23. I feel sad when I see/hear that someone has passed away.
38. I wish I could lose the fear of feeling bad.	38. I wish I could lose the fear of some bad things.
52. I feel angry when I feel bad.	52. I feel angry when I see someone hitting.
66. My fear force me to become bad	66. My fear forces me to run away where there are people.
78. I am happy when I am good.	78. I am happy when I get good marks.
87. I feel fearful when nothing.	87. I feel fearful when I see bad things.
89. I feel anxious when nothing.	89. I feel anxious when I see a particular question in test.
R=0	R=1

GIRLS:

8-10 Years	11-14 Years
8. I am afraid of night.	8. I am afraid of fever.
23. I feel sad when nothing to say.	23. I feel sad when a close relationship between two persons is cut off.
38. I wish I could lose the fear of mother's beatings.	38. I wish I could lose the fear of fever.
52. I feel angry when my sister beats.	52. I feel angry when I do wrong.
66. My fears force me to nothing to say.	66. My fear force me to go against it.
78. I am happy when I pass all subjects.	78. I am happy when I am happy.
87. I feel fearful when it is night time.	87. I feel fearful when I am in a dark forest
89. I feel anxious when nothing to say.	89. I feel anxious when I want to know something.
R=0	R=0

Mother - Father relationships:

BOYS:

8-10 Years	11-14 Years
9. My mother and my father go to picture.	9. My mother and my father help me and get me whatever I want.
24. My father treats my mother good.	24. My father treats my mother nicely.
39. I wish my father and mother good.	39. I wish my father and mother will help me.
53. My mother treats my father like god.	53. My mother treats my father nicely.
R=0	R=0

GIRLS:

8-10 Years	11-14 Years
9. My mother and my father do different types of work.	9. My mother and my father are my best friends.
24. My father treats my mother good.	24. My father treats my mother as a wife.
39. I wish my father and mother praise me for my studies.	39. I wish my father and mother would never lose patience.
53. My mother treats my father good.	53. My mother treats my father as a husband.
R=0	R=0

COPING WITH EMOTIONS:

BOYS:

8-10 Years	11-14 Years
11. When I am happy I will play	11. When I am happy I go mad
26. When I am anxious I say bad things.	26. When I am anxious I start to think stupidly
41. When I am afraid I feel bad	41. When I am afraid I start messing up things
55. When I am sad I feel bad	55. When I am sad I cry but then I console myself
68. When I am angry I will behave badly	68. When I am angry I lose patience
R=0	R=1

GIRLS:

8-10 Years	11-14 Years
11. When I am happy I will be happy	11. When I am happy I share it with parents and friends
26. When I am anxious I switch on light	26. When I am anxious I will walk from one side to another
41. When I am afraid I go and sleep with mother	41. When I am afraid I go to parents/ grandparents
55. When I am sad I cry	55. When I am sad I share it with friends
68. When I am angry I beat my sister who always beats me	68. When I am angry I just relax for 5 minutes
R=0	R=0

PARENTAL ATTITUDES:

BOYS:

8-10 Years	11-14 Years
12. My parents think I am good	12. My parents think I am good at sports
27. My parents feel happy when I good	27. My parents feel happy when I got 2nd place in running race
42. I am scolded when I don't do the homework	42. I am scolded when I made mistake
56. My parents get angry when I tell a lie	56. My parents get angry when I do something wrong
69. I am beaten when I am lazy to write/notes are incomplete	69. I am beaten when I do something wrong
R=0	R=0

GIRLS:

8-10 Years	11-14 Years
12. My parents think I am reading well	12. My parents think I am a very determined girl
27. My parents feel happy when I get 1st rank	27. My parents feel happy when I study well/do something good
42. I am scolded when something wrong happens	42. I am scolded when I do mischievous things
56. My parents get angry when I do not read well	56. My parents get angry when I do something very big mistake
69. I am beaten when I do wrong	69. I am beaten when but I don't be beaten so I think so
R=0	R=0

GOAL:

BOYS:

8-10 Years	11-14 Years
13. My ambition is to become a police	13. My ambition is to become a software Engineer
28. My greatest wish is to read	28. My greatest wish is to be always good
43. I want to become police	43. I want to become a software Engineer
57. I will be happy if I am good	57. I will be happy if I am good.
70. I always wanted to be a police	70. I always wanted to not tell anything
R=0	R=0

GIRLS:

8-10 Years	11-14 Years
13. My ambition is to become a Teacher	13. My ambition is IAS officer
28. My greatest wish is to read	28. My greatest wish is to do what I think
43. I want to become a Doctor	43. I want to become courageous and intelligent girl
57. I will be happy if	57. I will be happy if others are helpful
70. I always wanted to be a teacher	70. I always wanted to hit my brother when he is sleeping
R=0	R=1

FAMILY

BOYS:

8-10 Years	11-14 Years
14. In my family I like everything	14. In my family I feel happy
29. My parents always are busy	29. My parents always help me
44. My family treats me good	44. My family treats me well
58. The person I like most in my family my sister and brother	58. The person I like most in my family my parents
71. I wish my family good	71. I wish my family will be happy
82. Most families I know are good	82. Most families I know are my neighbour's family
R=0	R=0

GIRLS:

8-10 Years	11-14 Years
14. In my family	14. In my family I am very friendly
29. My parents always talking	29. My parents always help me
44. My family treats me well	44. My family treats me very lovely
58. The person I like most in my family, father, mother and sister	58. The person I like most in my family I like everyone equally
71. I wish my family well	71. I wish my family very friendly and good
82. Most families I know are I don't know about other families	82. Most families I know are good
R=0	R=0

GUILT FEELINGS:

BOYS:

8-10 Years	11-14 Years
15. My biggest mistake was telling lie	15. My biggest mistake was when I went down in studies
30. I wish I had not done bad	30. I wish I had not made a bad mistake
45. I feel guilty about tell lie	45. I feel guilty about my singing
59. I wish I could forget the time I told lies	59. I wish I could forget bad things
72. The worst thing that I ever did was to tell lies	72. The worst thing that I ever did was getting less marks
83. I feel bad about becoming busy with something	83. I feel bad about seeing bad things
R=1	R=1

GIRLS:

8-10 Years	11-14 Years
15. My biggest mistake was nothing	15. My biggest mistake was but I have not yet made a big mistake
30. I wish I had not played, I got beating	30. I wish I had not harmed anyone
45. I feel guilty about nothing	45. I feel guilty about my mistakes
59. I wish I could forget	59. I wish I could forget all my fears and sadness
72. The worst thing that I ever did was I played all the day	72. The worst thing that I ever did was my mistakes
83. I feel bad about my father beating my mother	83. I feel bad about no one
R=1	R=0

NIMHANS Sentence Completion Test for

Children and Adolescents (NSCA) Final Version

Department of Clinical Psychology, NIMHANS, Bengaluru

INSTRUCTIONS: You will find below a number of incomplete sentences. For example: "The colour of the leaf is _____. Write/tell quickly in the blank space the first idea that comes to your mind to complete each sentence. Quickness is very important, please answer rapidly.

1. My father is
2. My friends and I
3. My teachers are
4. My mother and I
5. At school, I feel
6. I have the ability to
7. My brother is
8. I am afraid of
9. My mother and my father
10. I like to
11. When I am happy, I
12. My parents think I am
13. My ambition is
14. In my family
15. My biggest mistake was

16. I like my father because
17. Most of my friends
18. My teachers and I
19. I like my mother because
20. I like going to school, but
21. When things go wrong I
22. My sister is
23. I feel happy when
24. My father treats my mother.....
25. My favourite eatable is
26. When I am happy, I
27. My parents feel happy when I.....
28. I want to become
29. My grand father
30. I wish I had not
31. My father and I
32. A real friend is
33. I feel my teachers
34. My mother
35. I find class tests

36. In order to achieve something, I can.....
37. My brother and I
38. I feel anxious when
39. I wish my father and mother
40. On Sundays I like to
41. When I am anxious, I
42. My parents scold me when
43. I will be happy if I can
44. My family treats me
45. I feel guilty about
46. I wish my father
47. My best friend
48. I like my teachers but
49. I dislike my mother because.....
50. The subject I dislike most
51. Others say I am capable of
52. My sister and I
53. I feel angry when
54. My mother treats my father
55. I am interested in

56. When I am angry, I
57. My parents get angry when I
58. I always wanted to be
59. The person I like most in my family.....
60. I wish I could forget
61. I dislike my father because
62. I expect my friends to
63. Most teachers are
64. I wish my mother
65. I like my brother but sometimes
66. I feel sad when
67. My grandmother
68. When I am sad, I
69. My parents beat me when
70. I wish my family
71. The worse thing I ever did was
72. Most families I know are
73. I like my sister but sometimes
74. I like my family but sometimes
75. My favourite game is

NIMHANS Sentence Completion Test for Children & Adolescent (NSCA) Final version
SCORING SHEET

I-Father	II-Mother	III-Sibling	IV-Mother-Father relations
1	4	7	9
16	19	22	24
31	34	37	39
46	49	52	54
61	64	65	-
		73	
Rating =	Rating =	Rating =	Rating =
V-Parental attitudes	VI-Family	VII-School	XII-Teachers
12	14	5	3
27	29	20	18
42	44	35	33
57	59	50	48
69	67		63
	70		
	72		
	74		
Rating =	Rating =	Rating =	Rating =
IX-Friends	X-Own ability	XI-Goal	XII-Emotions
2	6	13	8
17	21	28	23
32	36	43	38
47	51	58	53
62	-	-	66
Rating =	Rating =	Rating =	Rating =
XIII-Coping with emotions	XIV-Guilt	XV-Neutral	
11	15	10	
26	30	25	
41	45	40	
56	60	55	
68	71	75	
Rating =	Rating =	Rating =	

*0 = No problem in adjustment; 1 = Mild maladjustment; 2 = Moderate to severe maladjustment;

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CHAPTER 11

TESTS OF MEMORY FOR CHILDREN

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Memory is a process by which information stored is brought into awareness when required. But, it does not refer to a single process or structure or an isolated intellectual skill; rather it refers to the processing - storage - retrieval function of the mind that must be seen as embedded in a context. In the broader sense, memory is the complex higher order process which includes the collection, storage and retrieval of information (Erickson and Scott, 1977). Similarly, it is an assortment of cognitive processes, elaborations, and schemas (Kail & Hagen, 1973); and serves as the closeness of fit between an output and an input on some specific dimensions (Russell, 1975). Memory is intimately involved in many of the individual's intellectual and social endeavours, and is reflected in the diversity of his behaviour.

Nature of Memory Process

The general process of memory has been outlined by several workers (Miller, 1956, Broadbent, 1958, Kesner, 1973) as passing through various stages;

(i) The process of memory begins with an input of sensations into the organism according to its focus of attention and this, in turn, is immediately placed into a very brief sensory register called the sensory memory (Craik, 1979).

(ii) The next stage in the memory process is that of Short Term Memory (STM), a temporary storage that acts as the working memory for

cognitive processing. STM has been called primary memory, immediate memory and span memory (Kelley, 1964).

(iii) The third step in the process appears to be encoding and consolidation of items into Long Term Memory (LTM). LTM has been referred to as long term store, secondary memory, and delayed memory.

(iv) The consolidation of memories involves two different sub-processes - the transfer of material into LTM storage and the consolidation of memory in the storage. Transfer is the movement of material either from STM to LTM storage or perhaps directly into LTM storage from the sensory input, guided by STM (McGough, 1966).

(v) Storage is only one of the several memory processes. Long Term Storage consists of a relatively permanent storage of material that has been consolidated by forming memory traces, a permanent physical change in the brain (Hebb, 1949, 1961; Zangwill, 1969).

(vi) Retrieval is the final process related to LTM and refers to the process of locating and removing material from the long term store and bringing it to STM.

Several models of memory have also been put forward to explain this process. Guilford's structure of Intellect Model says that memory is one of the five operations of the individual's intellectual abilities and different types of memory can be included in different factors. Atleast 18 to 24 such factors have been identified. The Multistore Model of Atkinson and Shiffrin (1968) has proposed a three part model of memory involving (a) a sensory register that decays rapidly; (b) a short-term store with limited capacity which can be enhanced by rehearsal and (c) a long term store with unlimited capacity. According to this model, memory

depends on the efficiencies of strategies used for transferring memory from STM to LTM (encoding) and for retrieving it from the LTM. Piaget's (1971) constructive model of memory says that, in accordance with pre-existing knowledge, the individual assimilates new information into existing knowledge constructing the meaning of what is to be stored and reconstructing what needs to be recalled. The level of Processing Model by Craik & Lockharts (1972) modifies the multistore model and says that memory depends on the level of processing, the deeper the processing, the more durable and deeper is the memory. In the Developmental Model, Brown(1975) proposed three developmental dichotomies to understand the development of memory namely, (a) strategy versus no strategy; (b) mediation deficiency versus production deficiency, and (c) episodic memory versus semantic memory.

Development of Memory

Memory develops as children grow older, i.e they remember more effectively with their growing age. There are a number of different factors that contribute to produce these developmental changes in memory. Cognitive abilities, use of different mnemonic strategies, social maturity etc, are some of those factors. In the course of development cognitive changes are seen in children, they become more abstract and conceptual, which often results in memory development. They use their growing knowledge to establish, elaborate, meaningful relations in the information to be remembered and, as a consequence, remember more accurately. Developmental changes are also seen in the uses of mnemonic strategies by children. In the older children thus, the memory is better. Age related changes in the use of mnemonic strategies like rehearsal, outlining, grouping etc, have been reported by various researchers (Flavell, 1967).

For example, children below 7 years of age hardly use rehearsal with some regularity (Curvo, 1975).

The framework of "good information processing" is an elaboration of the concept of the "good strategy user". The good information processing specifies how basic capacities, knowledge, strategies and meta memory interact and articulate to determine children's memory. There is work on implicit memory, fuzzy trace theory, neural network model of representation, cognitive triage & utilization deficiencies. There is a lot of work that has been done on infant memory & autobiographical memory. (Schneider and Pressley, 2013)

Assessment of Memory

One important implication of research on the development of memory is that we can gain valuable insights into general changes in children's memory functioning. An understanding of deficits in memory functioning is essential for preventive and rehabilitative purposes. Thus, assessment becomes an important task to be performed while studying memory functioning and dysfunctioning of children. Assessment of memory is important because it serves certain purposes, as;

- (i) It provides an estimate of a child's memory ability which can be compared with other children's memory abilities in the same group.
- (ii) It gives an idea about the general strategic abilities of children at different age levels in their performances.
- (iii) Memory impairment is frequently associated with various clinical conditions such as head injury, cerebral tumours, progressive degenerative diseases, nutritional disorders, attention deficit hyperactivity disorder, learning disorder, epilepsy etc. In such conditions evaluation of memory is often sought for diagnostic, therapeutic and prognostic purposes.

(iv) Evaluation or assessment of memory is also required for the evaluation of the outcome of a particular treatment approach for certain clinical conditions in a child mental health service centre.

(v) An empirical study in the area of memory needs absolute demonstration which is possible through assessment.

THE NEED FOR AN ASSESSMENT TOOL AND THE DEVELOPMENT OF THE TEST OF MEMORY FOR CHILDREN

Though a great deal of attention has been given to the study of learning and memory, relatively little has been paid to the construction of tests for assessing memory functions in the clinical settings. The most vital need in the area of memory testing is the development of tests especially for children. Batteries of memory tests which have been used with children so far, by and large, have inadequate norms for children as they have been standardized on adult subjects.

The adult version of Rivermead Behavioural Memory Test (Wilson, Cockburn and Hiorns, 1989) has been used successfully to screen who cannot cope with memory tasks of every day life. There have also been attempts to assess various aspects of learning and memory using simple tasks; however, not many attempts have been made to evolve or construct tests from a developmental perspective to assess various aspects of memory in a comprehensive manner.

There is, thus, a need to develop a battery of memory tests and standardize it on a normal sample of children, with extensive coverage of the various aspects of memory and also to validate it on different clinical populations which are known to produce memory deficits. Being triggered by this vital need, authors have tried to develop the present test with the objectives being -

- (1) To construct a scale to assess various aspects of memory in children.
- (2) To standardize the scale on normal children in the age range of 7 to 11 years.
- (3) To establish reliability.
- (4) To establish validity.
- (5) To provide age norms and a memory profile chart for the individual child.
- (6) To establish clinical validity.

Sample

The samples used at various phases of the scale construction procedure comprised two kinds of groups i.e. Normal Group and Clinical Group.

Inclusion Criteria

1. Age 7-11 years.
2. Studying in an English medium School between 2nd and 6th standards.
3. Residing in the urban area.

Exclusion Criteria

1. History of any psychiatric illness.
2. History of any physical or mental handicap.
3. Specific learning difficulty.
4. History of any associated organic condition excluding epilepsy.

Tools : The tools used for screening were :

1. Children's Behaviour Questionnaire (CBQ) (Rutter, 1967)
2. Coloured Progressive Matrices (CPM) (Raven, 1965)

Procedure

The entire procedure of the test construction is divided into three phases.

Phase I : Preparation and initial trial of the battery.

Phase II : Pilot study by administering the test on a selected sample

Phase III : Main study : a final try out on a large sample

PHASES INVOLVED IN TEST CONSTRUCTION

Phase - I

A total of 191 items were generated from different sources including Wechsler Memory Scale (Wechsler, 1945), PGI Memory Scale (Pershad, 1977), Mini Mental State Examination (Strub & Black, 1977), Brown Peterson Test (Peterson, 1959), Mann-Suiter Visual Memory Screen for Objects (Mann-Suiter, 1984), Picture Scanning (Kamat, 1967), Knox Cube Imitation Test (Arthur, 1947), Benton Visual Retention Test (1974), Complex Figure Test (Osterrieth, 1944), and Cattell's Retentivity Test (Cattell, 1953) and they were grouped into 16 sub tests namely:

(i) Personal Information (5 items)

This test is a measure of remote memory which constitutes recall of past events of personal life.

(ii) Mental Control (5 items)

These verbal automatisms learned by rote in early childhood and frequently used throughout life are normally recalled unthinkingly, effortlessly and accurately.

(iii) Sentence Repetition (5 items)

This measures sequential reproduction of the sentences verbatim.

(iv) Logical Memory (18 facts)

Measures the immediate and delayed recall of the logical material in the form of a passage of 18 facts.

(v) Word Recall Meaningful (10 items)

Recall of 10 meaningful words after 2 minutes following the exposure for 30 seconds.

(vi) Digit Span

This comprises of span for digits forward and backward and measures attention and concentration.

(vii) Brown Peterson Test (10 items)

This test measures short term retention of non-verbal memory in terms of recalling ten non-sense syllables after 2 minutes following the presentation.

(viii) Word Recall Non meaningful (10 items)

Recall of 10 non meaningful words after 2 minutes following the exposure for 30 seconds.

(ix) Delayed response learning (4 items)

This essentially requires the ability to delay previous response in order to arrive at a final solution. There are four sets of fairly simple arithmetic problems each consisting of 2 parts.

(x) Mann-Suiter Visual Memory Screen for Objects (4 items)

It is designed to assess the ability to revisualise pictures of common objects presented in groups.

(xi) Picture Scanning

This test was taken from Binet Kamat Test of intelligence. This is a test of visual scanning and measures attentional ability.

(xii) Knox Cube Imitation Test

This is a test of attention and concentration.

(xiii) Benton Visual Retention Test (10 items)

This test measures visuo spatial perception, visual and verbal conceptualization and immediate memory span. It has 10 designs on 10 cards, each design is exposed for 10 seconds and subject is asked to reproduce the design immediately from memory.

(xiv) Complex Figure Test (Rey Osterreitch)

This measures constructional and memory abilities of the subject examine the functions of specific areas of the brain.

(xv) Paired Associate Learning Test (30 items)

This test measures well learned verbal association and retention of new, unfamiliar verbal material.

(xvi) Cattell's Retentivity Test

Recall of 10 complex and unfamiliar geometrical figures after 2 minutes following the exposure for 30 seconds.

The 16 subtests were given to five judges who were trained clinical psychologists for content validation. There was consensus among all the five judges that,

(i) items were culturally relevant.

(ii) items could be easily comprehended by the target population.

(iii) items represented the hypothesized aspect of memory function.

Phase-II

A pilot study was conducted in order to make certain modifications in the tests of memory and to allow the investigator to get familiarized with the tests. A sample of 219 children were screened with the help of CBQ for behaviour problems. Four children who scored above the cut off point were excluded and remaining 215 children were assessed on CPM. Fifteen children who scored below the 5th percentile were excluded from the study. Remaining 200 children were administered the test of memory for children. A matched sample of 15 epileptic children with two or more years duration of antiepileptic medication, attending neurology OPD NIMHANS Bangalore were selected as the clinical sample on which this test was also administered.

Modifications in the test

The items which significantly discriminated the groups were retained. Items which were too easy and too difficult across 7-11 years were removed. There were tests which were found to be similar; knox cube test was similar to Digit Span Test; Brown Peterson Test was similar to word recall non meaningful test. Hence these tests were deleted from

the battery. Picture scanning was too simple and complex figure test was too difficult for all children, so these two tests were also removed. Consequently the battery was reduced to 147 items in the form of 12 sub tests.

Phase III

In the main study the test battery was administered on 573 normal children and 75 epileptic children to establish the norms and clinical validity of the test.

The tests of memory were re-administered on 38 children from the normal sample after a period of 6 weeks in order to establish the test retest reliability.

RESULTS

Table 1

Correlation between sub tests and total memory score

SI. No.	Sub Tests	Correlation
01.	Personal information	0.52 **
02.	Mental control	0.64 **
03.	Sentence repetition	0.71 **
04.	Story recall (immediate)	0.75 **
05.	Story recall (delayed)	0.77 **
06.	Word recall meaningful	0.53 **
07.	Digit forward	0.27 **
08.	Digit backward	0.36 **
09.	Word recall non meaningful	0.53 **
10.	Delayed response learning	0.40 **
11.	Picture recall	0.51 **
12.	B.V.R.T.	0.50 **
13.	Paired Associate learning	0.53 **
14.	Cattell's Retentivity test	0.78 **

** P < 0.01

Table 2
Percentage of cases within 1, 2, 3 SD
with regard to their total memory scores

SD Limits	AGE				
	7 yrs	8 yrs	9 yrs	10 yrs	11 yrs
Cases within 1 SD	62%	59%	61%	61%	60%
Cases within 2 SD	97%	98%	97%	94%	97%
Cases within 3 SD	98%	100%	100%	100%	100%

Table - 3
Correlation of Memory Data with CPM Scores

Sl. No.	Sub Tests	Correlation
01.	Personal information	0.22 **
02.	Mental control	0.36 **
03.	Sentence repetition	0.24 **
04.	Story recall (immediate)	0.33 **
05.	Story recall (delayed)	0.36 **
06.	Word recall meaningful	0.21 **
07.	Digit forward	0.15 **
08.	Digit backward	0.21 **
09.	Word recall non meaningful	0.22 **
10.	Delayed response learning	0.21 **
11.	Picture recall	0.29 **
12.	B.V.R.T.	0.31 **
13.	Paired Associate learning	0.31 **
14.	Cattell's Retentivity test	0.28 **
15.	Total Memory	0.45 **

* * P < 0.01

RELIABILITY

Table - 4
Test- Retest Reliability Of The Subtests Of Memory

Sl.No.	Subjects	Mean I	Mean II	Correlation
01.	Personal information	3.97	3.84	0.95
02.	Mental Control	10.97	11.10	0.97
03.	Sentence Repetition	7.44	7.10	0.81
04.	Logical memory- story recall immediate story recall delayed	7.55	5.57	0.82
		7.94	8.00	0.88
05.	Word recall meaningful	7.55	6.71	0.96
06.	Digit span- Digit forward Digit backward	5.97	6.18	0.96
		3.81	4.10	0.73
07.	Word recall non meaningful	6.52	6.18	0.97
08.	Picture recall	2.89	2.86	0.91
09.	Delayed response learning	2.57.	2.50	0.51
10.	Benton visual retention test	7.13	6.02	0.72
11.	Paired associate learning	12.76	10.56	0.81
12.	Cattell's retentivity test	6.63	5.94	0.98
	Total memory score	93.76	86.72	0.97

Validity

Content validity of the test was established through the use of consensus amongst five experts on adequacy and representativeness of the sampling domain in the items of sub tests. Construct validity of the scale, in the form of factorial validity has been established.

Clinical Validity

The tests of memory for children has been found effective in differentiating between normal and epileptic children with regard to their memory abilities.

Table - 5
Comparison of mean scores of normal and clinical sample (epileptic) on sub tests of memory

SI. No.	Sub tests	Normal Group (N=573) Mean	Clinical group (N=75) mean	Z Values
01.	Personal information	4.06	2.99	2.75 ***
02.	Mental Control	10.73	9.01	7.38 ***
03.	Sentence Repetition	7.12	3.32	13.06 ***
04.	Story recall immediate	5.53	4.01	4.00 ***
05.	Story recall delayed	5.70	4.28	3.08 **
06.	Word recall meaningful	7.26	7.19	0.28 NS
07.	Digit forward	6.49	4.57	12.60 ***
08.	Digit backward	3.88	2.95	5.79 ***
09.	Word recall nonmeaningful	6.31	5.84	2.06 *
10.	Delayed response learning	2.62	1.67	6.97 ***
11.	Picture recall	2.59	1.73	6.96 ***
12.	B.V.R.T	5.85	4.68	5.42 ***
13.	Cattell's retentivity test	5.90	6.32	1.84 NS
14.	Paired associated learning	10.20	8.10	3.05 **
15.	Total memory score	84.25	65.66	7.76 ***

* $P < .05$, ** $P < .01$, * * * $P < .001$, N.S. - Not significant

Further efforts have been made to find out the clinical validity of the tests in different clinical groups like learning disabilities, head injury, emotional disorders, other organic condition etc., which are known to produce memory problems or in which memory deficit is a major symptom.

Administration and scoring

The tests of memory for children consist of 12 sub tests namely,

1. Personal information
2. Mental control
3. Sentence repetition
4. Logical memory-
 - a. Story recall immediate
 - b. Story recall delayed
5. Word recall meaningful
6. Digit span
 - a. Digit forward
 - b. Digit backward
7. Word recall non meaningful
8. Delayed response learning
9. Picture recall
10. BVRT
11. Paired Associate learning
12. Cattell's Retentivity Test

Conclusions

1. A test battery consisting of twelve subtests for assessing memory process of children has been developed.
2. Norms have been developed for children (for memory abilities of children in the age group of 7 to 11 years)
3. Reliability (test- retest) of the whole battery has been found adequate ranging from 0.51 to 0.97 for different sub tests.
4. Validity in terms of internal consistency among the subtests and correlation of different test scores with total memory score have been calculated that ranges from 0.27 to 0.78.
5. Profile analysis on all subtests for normal children as well as children with epilepsy, indicates adequate clinical application.

TEST OF MEMORY FOR CHILDREN (Barnabas, I and Kapur, M., 2002)

NAME

AGE

SEX

CLASSSECTION.....

RELIGION

SCHOOL

FATHER'S OCCUPATION

MOTHER'S OCCUPATION

DATE

I. Personal Information

Each item of the test was read distinctly and slowly at a steady rate. If the subject failed to understand the item, it was repeated.

1. How many people are there in your family

2. What day is it today

3. When did you first join school.....

4. What is the name of this month

5. In which state are you living.....

II. Mental control

Time taken and errors/omission made by the subject on each of five sub tests were noted down verbatim.

	Time	Error
1. Repeating day of the week forward		
2. Repeating day of the week backward		
3. Repeating months of the year forward		
4. Repeating months of the year backward		
5. Deduct 5's from 60		

III. Sentence repetition

Each of the sentences was presented one by one to the subject for immediate reproduction. Each sentence was read out slowly, distinctly and at a uniform rate during presentation. The subject's recall was either noted verbatim or each of the correctly recalled clauses was ticked. If the subject made any error, omission, in recalling the sentence, then while presenting the subsequent sentences, the subject was reminded once again to be attentive and careful and to reproduce the sentence as exactly as possible.

1. He hoped to win the prize
2. The ink had dried up and I could not write
3. Be good and you will be happy
4. He has gone down to the river to bathe
5. The city that is set on a hill cannot be hid

IV. Logical Memory

Story Recall Immediate

"I am going to read a short story, listen carefully, because when I finish reading I want you to tell me everything that I told you. Do you understand?", after reading the story, say, "Now tell me everything that you can remember of the story, start the beginning of the story and tell me all that happened." The separate items in the story are indicated by slab (/) mark. As the subject retells the story indicate the number of ideas recalled. Immediately following the first trial, the examiner says, "in a little while I'm going to ask you to tell me how much of the story you can still remember. I am going to read the story to you again now so that you will have it fresh in your memory for the next time." Recall following the second reading comes after approximately 20 minutes of testing involving verbal material. Once again the, examiner asks the subject to "Tell everything you can remember."

Rani was a little girl / who lived in a village/ one day/ she took/ her dog/ with her/ to a forest/ there they played / and enjoyed nature/ very much / there suddenly / a strong windstorm arose/ and carried them away/ to an unknown land/ both Rani and her dog were happy/ and thrilled/ to see the wonders/ of the new land.

- a) Immediate
- b) Story recall delayed

V. Word Recall (Meaningful Words)

"I will be showing you a card, on which some meaningful words have been written. Look at them carefully for (30 seconds) after sometime (2

minutes), I will show you a second card from which you have to show me the meaningful words you have already seen." (See page 216 for words)

- 1) 2) 3) 4) 5)
6) 7) 8) 9) 10)

VI. Digit span test

"I will be saying some numbers listen to me carefully, when I finish saying them, you will have to repeat in the same order." The examiner after instructing the child.

- Gives an example. If I say 3, 7 you say 3,7.
- Reads numbers at the rate of one number per second to the child.
- Note whether the immediate response of the child is correct or incorrect. The digit span is the highest number of digits repeated correctly.
- The child is instructed as follows, "I will be saying some numbers, listen to me carefully and repeat them after me in a reversed order, for example if I say 2-5, you have to say 5-2".

The procedure is the same as for digits forward. No digit should be repeated in a series. If a child fails to recall the numbers correctly on both the trials, the test can be discontinued.

DIGIT FORWARD

- | | |
|-------------------------------|-------------------------------|
| 4 - 2 - 6 | 6 - 4 |
| 5 - 7 - 3 | 4 - 1 - 7 |
| 5 - 3 - 8 - 7 | 6 - 1 - 5 - 8 |
| 1 - 6 - 4 - 9 - 5 | 2 - 9 - 7 - 3 - 4 |
| 3 - 4 - 1 - 7 - 9 - 6 | 6 - 1 - 5 - 8 - 3 - 9 |
| 7 - 2 - 5 - 9 - 6 - 8 - 5 | 4 - 7 - 2 - 9 - 1 - 6 - 8 |
| 4 - 7 - 1 - 5 - 3 - 8 - 6 - 2 | 9 - 2 - 5 - 8 - 3 - 1 - 7 - 4 |

DIGIT BACKWARD TEST

- | | |
|---------------------------|---------------------------|
| 5 - 8 | 8 - 2 |
| 7 - 5 - 4 | 1 - 5 - 8 |
| 3 - 6 - 5 - 8 | 9 - 5 - 7 - 3 |
| 1 - 9 - 2 - 7 - 4 | 4 - 8 - 5 - 2 - 9 |
| 3 - 8 - 4 - 9 - 5 - 2 | 6 - 9 - 3 - 5 - 1 - 7 |
| 2 - 9 - 1 - 6 - 8 - 5 - 3 | 5 - 8 - 4 - 1 - 7 - 3 - 6 |

VII. Word Recall (Non Meaningful)

"I will be showing you a card, on which some nonsense syllables have been written, look at them carefully (for 30 seconds), after some time (2 minutes) I will show you a second card from which you have to show me the nonsense (meaningless) syllables you have already seen." (see page 217)

- 1) 2) 3) 4) 5)
6) 7) 8) 9) 10)

VIII. Delayed Response Learning

"I will tell you a small arithmetic problem, you keep the result of it in your mind, use it to solve another problem 10 seconds later."

- | | |
|---|------------|
| 1. $6 + 4 = x$ interval add x to | $5 + 2 =$ |
| 2. $3 + 9 = x$ look at the picture add x to
(any picture can be shown) | $5 + 5 =$ |
| 3. $7 - 3 = x$ interval add x to | $11 - 4 =$ |
| 4. $4 - 4 = x$ look at the picture add x to | $12 - 7 =$ |

IX. Picture Recall

"I am going to show you a row of some pictures. After I cover the pictures, I want you to tell me each time exactly what pictures you saw, start at one end of the row each time and tell me all the pictures you saw. Try to he recall them to me in the same order they were in a row", (see page 215)

The exposure time for each row is as follows

Row 1 : 2 Seconds

Row 2 : 3 Seconds

Row 3 : 4 Seconds

Row 4 : 5 Seconds

1. 2 Seconds 1) 2)

2. 3 Seconds 1) 2) 3)

3. 4 Seconds 1) 2) 3) 4)

4. 5 Seconds 1) 2) 3) 4) 5)

X. Benton Visual Retention Test (Copyrighted)

"One by one I will be showing you some cards, look at them carefully, after some time (10 seconds) I will take back the card and then you will draw on paper the same figures which you have seen" (provide the subject with 5 1/2 x 8 inches paper and pencil with an eraser).

No of correct reproductions

1. Omission
2. Distortion
3. Perserveration
4. Rotation
5. Misplacement
6. Size error

XI. Paired Associate Learning

Tell the subject, "I am going to read to you a list of words two at a time. Listen carefully because I will expect you to remember the words that go together. For example, if the words are "big", and "small", when I say "big" you will have to say "small". After the subject understands the instructions, read the first presentation at the rate of one pair every two seconds. After reading the first presentation, test for recall by presenting the first recall list. Give the first word of a pair and allow Five seconds for a response. If the subject gives a correct response, say, that its right and proceed with the next pair. If the subject gives an incorrect response, say, "No" provide the correct association and proceed to the next pair. After the first recall has been completed, allow a 10 seconds interval and give the second presentation list, proceeding as before".

SI. First Recall		Second Recall		Third Recall	
No.					
1.	Needle... THREAD	Diary ...PEN		Match box... FIRE	
2.	Match box... FIRE	Coin ... PURSE....		Needle... THREAD	
3.	Penknife... ERASER.	Penknife... ERASER..		Diary PEN...	
4.	Cloth.... FLOWER	Needle.. THREAD		Penknife ... ERASER..	
5.	Diary.... PEN...	PostcardSTAMP...		Ribbon....COMB	
6.	Pencil... SPOON...	Pencil ...SPOON		Coin PURSE	
7.	Post card....STAMP...	Ribbon ... COMB		Blade... SCISSORS....	
8.	Ribbon....COMB	Cloth... FLOWER		Pencil... SPOON	
9.	Blade... SCISSORS...	Matchbox... FIRE		Cloth FLOWER	
10.	Coin.... PURSE....	Blade ... SCISSORS.		Postcard....STAMP...	
First Recall		Second Recall		Third Recall	
Easy	Hard	Easy	Hard	Easy	Hard
1	3	1	3	1	4
2	4	2	6	2	5
5	6	4	7	3	7
7	8	5	8	6	8
10	9	9	10	10	9
T =	T =	T =	T =	T =	T =

XII. Cattell's Retentivity Test (copyrighted)

"I will be showing you a card on which some geometrical figures have been drawn. Look at them carefully (for 30 sec) after some time (2 minutes) I will show you a second card from which you have to show me the geometrical figures you have already seen."

- | | | | | |
|----|----|----|----|-----|
| 1) | 2) | 3) | 4) | 5) |
| 6) | 7) | 8) | 9) | 10) |

SCORING

Coloured Progressive Matrices scoring (copyrighted)

Subjects initial response was scored. Each response was scored based on Raven's scoring key. Each correct response was given a score of 1. The summation of set A, AB and B make the total score and the

Sub-test wise Scoring of Memory Data

- | | | |
|-----|--|------------------|
| I. | Personal Information | Maximum Score 5 |
| | Each correct answer (supported by information from the admission register) was given a score of one. | |
| II. | Mental Control | Maximum Score 15 |
| 1. | Naming days of the week forward | |
| | All correct within 15 seconds | 3 |
| | All correct beyond 15 seconds | 2 |
| | With one mistake | 1 |
| | More than one mistake | 0 |
| 2. | Naming days of the week backward | |
| | All correct within 20 seconds | 3 |
| | All correct beyond 20 seconds | 2 |
| | With one mistake | 1 |
| | More than one mistake | 0 |
| 3. | Naming months of the year forward | |
| | All correct within 25 seconds | 3 |
| | All correct beyond 25 seconds | 2 |
| | With one mistake | 1 |
| | More than one mistake | 0 |
| 4. | Naming months of the year backward | |
| | All correct within 45 seconds | 3 |
| | All correct beyond 45 seconds | 2 |
| | With one mistake | 1 |
| | More than one mistake | 0 |

5. Deducting 5' s from 60

All correct within 60 seconds	3
All correct beyond 60 seconds	2
With one mistake	1
More than one mistake	0

III. Sentence Repetition Maximum Score 9

One mark for each clause correctly reproduced. Thus the maximum possible range of scores on each sentence were as follows;

Sentence 1	0	-1
Sentence 2	0	-1
Sentence 3	0	-2
Sentence 4	0	-2
Sentence 5	0	-3

IV. Story Recall Immediate Maximum Score 18

Record verbatim and score according to number of ideas as marked off. The final scoring is the number of ideas correctly reproduced.

V. Word Recall (meaningful) Maximum Score 10

Each meaningful word correctly identified and named was given a score of one.

VI. Digit Span Test Maximum Score 15

a) Digit forward- number of digits in the longest series correctly reproduced.

b) Digit backward- number of digits in the longest series correctly reproduced in reversed order. Summation of scores earned on digit forward and backward was the score for this subtest.

VII. Word Recall Maximum Score 10

Each non meaningful word correctly identified and named was given a score of one.

VIII. Delayed response learning Maximum score 4

Giving correct answer for each problem is given a score of 1.

IX. Picture Recall Maximum Score 4

Recalling the pictures in the same order within a row suggests adequate visual memory.

X. Benton Visual Retention Test Maximum score 10

Designs are first scored as either correct or incorrect. One credit for each correct response is assigned.

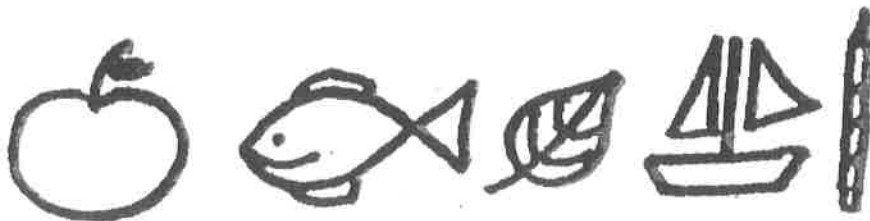
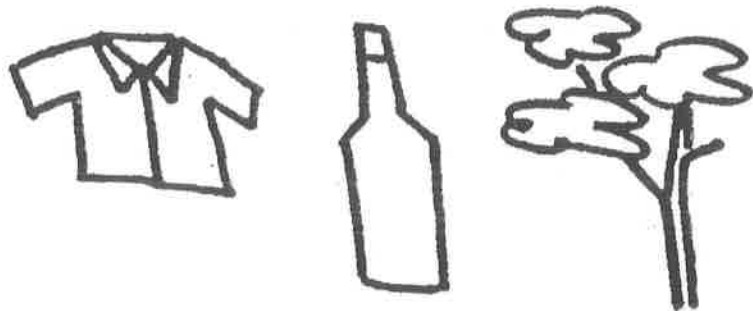
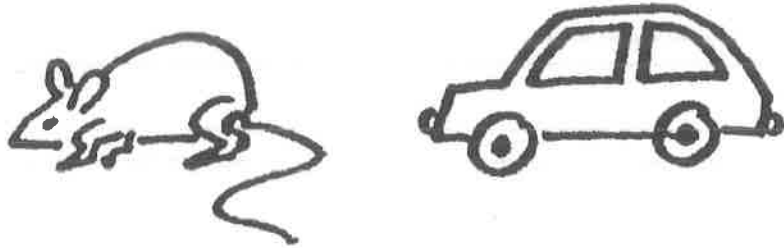
XI. Paired Association Learning Maximum score 21

One credit for each correct response, if given within 5 seconds. Get final score as follows: Add all credits obtained on easy associations and divide score by 2. Add credits on hard association. Total each column separately. Score on entire test is sum of both easy and hard association scores.

XII. Cattell's Retentivity Test Maximum score 10

Each geometrical figure correctly identified and was given a score of one.

IX SUBTEST - PICTURE RECALL



V Subtest - Word Recall

I CARD

MONEY SHOE BIRD BOOK GUN

HAT NAIL MOON FISH KEY

II CARD

DISH FISH SAIL SPOON

MAT MOON HAT SHOE

HOOK NAIL MONEY KEY

BIRD PAIL GUN SUN

HONEY BOOK JAM CURD

VII Subtest - Word Recall

I CARD

DAG ROF MUN CVC PEL
LON MEZ PAE LIF SEG

II CARD

DAG NID ROF TIB
MUN WUL CVC XUM
PEL KAS LON FOL
MEZ VEE PAE TUL
LIF RET SEG HAR

NORMS

6.1 Comparison of Boys and Girls Performance on subtest personal information

N = 573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	df
7 years	56	55	3.41	1.02	3.67	0.98	1.37	109
8 years	58	53	3.50	1.20	4.03	0.80	2.73**	109
9 years	55	70	4.27	0.83	4.41	0.67	1.05	123
10 years	53	55	4.00	0.85	4.30	0.83	1.90	106
11 years	55	63	4.45	0.78	4.42	0.73	0.18	116

** p<0.01

6.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean and Personal Information for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	3.41	1.02	2-5	3.57	
58	8 years	3.50	1.20	0-5	3.60	7.40***
55	9 years	4.27	0.83	2-5	4.27	
53	10 years	4.00	0.85	0-5	3.83	
55	11 years	4.45	0.79	1-5	4.34	

df = 271, *** p<0.001

6.3 Shows the Mean differences for agewise comparison on Personal Information for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-0.14	-3.74 **	-1.22	-3.88***
8 years		3.72 ***	-1.16	-3.93*1**
9 years			2.33**	-0.40
10 years				-2.82**
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

* $p < .05$, ** $p < .01$, *** $p < .001$,

6.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean and Personal Informtion for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	3.67	0.98	1-5	3.74	
53	8 years	4.04	0.81	2-5	3.98	8.22 ***
70	9 years	4.41	0.67	2-5	4.44	
55	10 years	4.31	0.84	2-5	4.25	
63	11 years	4.43	0.73	2-5	4.42	

df = 290, *** $p < 0.001$

6.5 Shows the Mean differences for agewise comparison on Personal Information for girls

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-1.42	-4.80***	-3.00**	-4.39**
8 years		-2.95 **	-1.74	-2.86 **
9 years			1.23	0.15
10 years				-1.10
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290.

* $p < 0.5$ ** $p < .01$, *** $p < .001$.

7.1 Comparison of Boys and Girls Performance on subtest Mental Balance

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	df
7 years	56	55	7.92	2.31	8.76	2.07	1.99*	109
8 years	58	53	8.48	2.71	10.62	2.73	4.13***	109
9 years	55	70	10.80	3.64	10.30	2.90	0.85	123
10 years	53	55	12.41	2.76	12.34	2.54	0.13	106
11 years	55	63	13.50	2.34	12.22	2.24	3.04	116

* $p < 0.5$, *** $p < 0.001$.

7.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Mental Control for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	7.93	2.32	3-15	8.37	
58	8 years	8.48	2.72	0-15	8.75	
55	9 years	10.80	3.64	0-15	10.79	22.28***
53	10 years	12.42	2.77	4-15	11.95	
55	11 years	13.51	2.34	5-15	13.21	

df = 271, *** p<0.001

7.3 Shows the Mean differences for agewise comparison on Mental Control for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-0.73	-4.43 **	-5.82***	-8.29 ***
8 years		-3.86 ***	-5.48***	-8.01 ***
9 years			-2.08*	-4.50 ***
10 years				-2.35*
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 271, * p<0.05, *** p< 0.001.

7.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean of Mental Control for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	8.76	2.07	3-15	9.06	
53	8 years	10.62	2.73	2-15	10.40	
70	9 years	10.30	2.90	3-15	10.43	13.86***
55	10 years	12.35	2.54	7-15	12.11	
63	11 years	12.22	2.25	6-15	12.19	

df = 290, *** p<0.001

7.5 Shows the Mean differences for agewise comparison on Mental Control for girls

Age Group	age group			
	8 years	9 years	10 years	11 years
7 years	-2.48*	2.99**	-5.67***	-6.44***
8 years		0.07	-3.54***	-3.77***
9 years			-3.47***	-3.98***
10 years				-0.18
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290.

* p<.05, ** p< . 01, *** p<.001.

8.1 Comparison of Boys and Girls Performance on subtest Sentence Repetition

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	6.30	2.66	7.09	2.59	1.57	109
8 years	58	53	4.46	2.62	6.37	2.69	3.76***	109
9 years	55	70	7.60	2.29	7.18	2.21	1.02	123
10 years	53	55	8.11	1.82	8.07	1.8	0.11	106
11 years	55	63	8.20	1.88	2.85	2.04	0.94	116

*** p<0.01

8.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Sentence Repetition for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value	DF=270
56	7 years	6.30	2.67	0-9	6.50		
58	8 years	4.47	2.62	0-9	4.58	7.40	P<0.001
55	9 years	7.60	2.29	1-9	7.59		
53	10 years	8.11	1.83	2-9	7.90		
55	11 years	8.20	1.89	2-9	8.06		

df = 271, *** p<0.001

8.3 Shows the Mean differences for agewise comparison on Sentence Repetition for boys.

age group				
Age group	8 years	9 years	10 years	11 years
7 years	443***	-2.42*	-2.75**	-3.23 **
8 years		6.87 ***	-6.87***	-7.55 ***
9 years			-0.67	-1.05
10 years				-0.36
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df=271, * p<.05, ** p< .01, *** p<.001

8.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Sentence Repetition for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	7.09	2.59	0-9	7.56	
53	8 years	6.38	2.69	0-9	6.02	
70	9 years	7.19	2.22	2-9	7.40	5.64***
55	10 years	8.07	1.81	2-9	7.70	
63	11 years	7.86	2.05	3-9	7.82	

df = 290, *** p<0.001

8.5 Shows the Mean differences for agewise comparison on Sentence Repetition for Girls

Age Group	age group			
	8 years	9 years	10 years	11 years
7 years	-3.24**	0.40	-0.27	-0.58
8 years		-3.21 **	-3.92	-4.25
9 years			-0.69	-1.06
10 years				-0.28
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290.

* p<.05, ** p< . 01, *** p<.001,

9.1 Comparison of Boys and Girls Performance on subtest Story Recall Immediate

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	3.80	2.29	4.90	2.79	2.27*	109
8 years	58	53	4.20	3.09	5.00	3.40	1.28	109
9 years	55	70	7.23	3.45	5.60	3.54	2.59	123
10 years	53	55	5.49	2.75	6.09	3.19	1.04	106
11 years	55	63	6.85	3.24	6.11	3.35	1.21	116

* p<0.05

9.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Story Recall Immediate for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	3.80	2.30	0-8	4.61	
58	8 years	4.21	3.09	0-11	4.70	
55	9 years	7.24	3.46	0-14	7.23	9.46***
53	10 years	5.49	2.76	0-11	4.65	
55	11 years	6.85	3.25	0-13	6.31	

df = 271, *** p<0.001

9.3 Shows the Mean differences for agewise comparison on Story Recall Immediate for boys.

Age Group	8 years	9 years	10 years	11 years
7 years	-0.16	-4.60***	-0.05	-2.80 **
8 years		-4.59 ***	0.08	-2.78 **
9 years			4.47***	1.64
10 years				-2.99**
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df=271, * p<.05, ** p< . 01, *** p<.001

9.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Story Recall Immediate for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	4.91	2.80	0-14	5.86	2.73 NS
53	8 years	5.00	3.40	0-12	4.28	
70	9 years	5.60	3.54	2-15	6.03	
55	10 years	6.09	3.19	1-14	5.34	
63	11 years	6.11	3.36	0-15	6.03	

df = 290

9.5 Shows the Mean differences for agewise comparison on Story Recall Immediate for Girls

	Age Group			
	8 years	9 years	10 years	11 years
7 years	2.34	-0.30	0.76	-0.28
8 years		2.88 **	-1.76	-2.94
9 years			1.14	-0.00
10 years				-1.16
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290, ** $p < .01$.

10.1 Comparison of Boys and Girls Performance on subtest Story Recall Delayed

	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	3.05	2.33	4.20	3.17	2.16	109
8 years	58	53	2.37	3.14	5.54	3.97	3.35	109
9 years	55	70	6.63	4.47	6.00	4.23	0.81	123
10 years	53	55	6.13	2.96	6.63	4.21	0.71	106
11 years	55	63	8.21	3.53	7.19	4.03	1.46	116

* $p < 0.5$, ** $p < 0.05$.

10.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Story Recall Delayed for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	3.05	2.34	0-8	3.97	7.40 ***
58	8 years	3.27	3.14	0-10	3.84	
55	9 years	6.64	4.48	0-16	6.63	
53	10 years	6.13	2.97	0-14	5.18	
55	11 years	8.22	3.53	0-15	7.60	

df = 271, *** $p < 0.001$

10.3 Shows the Mean differences for agewise comparison on Story Recall Delayed for boys.

age group				
Age Group	8 years	9 years	10 years	11 years
7 years	0.21	-4.17 ***	-1.68	-5.34 ***
8 years		-4.53 ***	-1.97*	-5.81 ***
9 years			2.24**	-1.56
10 years				-3.89***
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df=271 *p<.05,

** p< .01, *** p<.001.

10.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Story Recall Delayed for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	4.20	3.18	0-13	5.22	
53	8 years	5.55	3.97	0-14	4.78	
70	9 years	6.00	4.24	0-18	6.46	3.51*
55	10 years	6.64	4.22	0-15	5.84	
63	11 years	7.19	4.03	0-17	7.11	

df = 290, *** p<0.001, *p< .05

10.5 Shows the Mean differences for agewise comparison on Story Recall for girls

Age group	8 years	9 years	10 years	11 years
7 years	0.53	-1.78	-0.75	-2.55*
8 years		2.28*	-1.43	-3.21 **
9 years			0.85	-0.96
10 years				-1.76
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df=290, * p<.05, ** p< .01.

11.1 Comparison of Boys and Girls Performance on Word Recall Meaningful

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	df
7 years	56	55	6.92	2.07	7.20	2.34	0.64	109
8 years	58	53	6.74	2.21	7.13	2.22	0.92	109
9 years	55	70	6.60	2.16	7.38	2.26	1.96	123
10 years	53	55	7.67	1.87	8.29	1.70	1.77	106
11 years	55	63	7.38	2.08	7.30	2.18	0.20	116

11.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Word Recall Meaningful for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value	Df=270
56	7 years	6.93	1.19	1-10	7.29		
58	8 years	6.74	0.10	0-10	6.96		
55	9 years	6.60	1.10	1-10	6.59	1.10	NS
53	10 years	7.68	2.10	2-10	7.29		
55	11 years	7.30	2.09	3-10	7.13		

df = 271, NS= Not significant

11.3 Shows the Mean differences for agewise comparison on Word Recall Meaningful for boys.

Age Group	8 years	9 years	10 years	11 years
7 years	0.85	1.72	0.00	0.37
8 years		0.94	-0.75	-0.40
9 years			-1.69	-1.34
10 years				0.40
11 years				

@ Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df=271

11.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Word Recall Meaningful for Girls

N.296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	7.20	2.34	3-10	7.62	
53	8 years	7.13	2.23	2-10	6.81	
70	9 years	7.39	2.26	2-10	7.58	2.22 NS
55	10 years	8.29	1.71	4-10	7.95	
63	11 years	7.30	2.19	0-10	7.26	

df = 290, NS= Not significant

11.5 Shows the Mean differences for agewise comparison on Word Recall Meaningful for Girls

	Age Group			
	8 years	9 years	10 years	11 years
7 years	1.79	0.12	-0.73	0.87
8 years		-1.87	-2.80**	-1.13
9 years			-0.92	0.83
10 years				1.72
11 years				

@ Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290. ** p<.01.

12.1 Comparison of Boys and Girls Performance on subtest Digit Forward

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	6.35	1.29	6.32	1.72	0.10	109
8 years	58	53	6.15	1.30	6.96	0.99	3.62**	109
9 years	55	70	6.41	0.91	6.41	1.08	0.02	123
10 years	53	55	7.11	1.31	6.47	1.69	2.18*	106
11 years	55	63	7.09	1.09	5.77	0.95	6.95***	116

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

12.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Digit Forward for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	6.35	1.30	3-8	6.39	
58	8 years	6.16	1.31	2-8	6.18	
55	9 years	6.42	0.92	4-8	6.41	4.85NS
53	10 years	7.11	1.31	2-8	7.07	
55	11 years	7.09	1.09	4-8	7.06	

df = 271, NS =Not significant

12.3 Shows the Mean differences for agewise comparison on Digit Forward for boys.

Age Group	8 years	9 years	10 years	11 years
7 years	0.96	0.08	-2.51	-2.62
8 years		1.03	-3.51	-3.65
9 years			-2.71	-2.77
10 years				0.03
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df =271

12.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Digit Forward for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	6.33	1.72	1-8	6.39	
53	8 years	6.96	1.00	4-8	6.90	
70	9 years	6.41	1.08	2-8	6.44	5.45**
55	10 years	6.47	1.70	3-8	6.41	
63	11 years	5.78	0.96	3-7	5.77	

df = 290, ** $p < 0.001$

12.5 Shows the Mean differences for agewise comparison on Digit Forward for girls

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-1.80	-0.19	-0.06	2.45*
8 years		1.81	1.93	4.54***
9 years			0.11	2.90**
10 years				2.59**
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290

* p<0.05, ** p< 0.01, *** p<0.001

13.1 Comparison of Boys and Girls Performance on subtest Digit Backward

	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	3.85	1.36	3.81	1.64	0.13	109
8 years	58	53	4.00	1.36	4.33	1.23	1.36	109
9 years	55	70	3.63	1.51	3.40	1.37	0.91	123
10 years	53	55	3.77	1.28	3.78	1.31	0.03	106
11 years	55	63	4.01	1.38	4.23	1.20	0.92	116

13.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Digit Backward for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	3.86	1.37	0-6	4.12	
58	8 years	4.00	1.36	0-7	4.16	
55	9 years	3.64	1.52	0-5	3.63	2.00 NS
53	10 years	3.77	1.28	0-6	3.49	
55	11 years	4.02	1.38	0-6	3.83	

df = 271, NS = Not significant

13.3 Shows the Mean differences for agewise comparison on Digit Backward for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-0.15	1.83	2.09*	1.00
8 years		2.04*	2.34*	1.19
9 years			0.51	-0.77
10 years				-1.31
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df =271, * p< 0.05

13.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Digit Backward for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	3.82	1.65	0-7	4.12	3.90**
53	8 years	4.34	1.24	0-7	4.11	
70	9 years	3.40	1.38	0-6	3.53	
55	10 years	3.78	1.32	2-6	3.54	
63	11 years	4.24	1.20	2-6	4.21	

df = 290, ** p<0.001

13.5 Shows the Mean differences for agewise comparison on Digit Backward for girls

	Age Group			
	8 years	9 years	10 years	11 years
7 years	0.03	2.41*	2.04*	-0.35
8 years		2.23*	2.22*	-0.40
9 years			-0.02	-2.89**
10 years				-2.68**
11 years				

Tests on all pairwise comparisons using Fisher's Protected LSD procedures.

df = 290.

* p<0.05, ** p<0.01

14.1 Comparison of Boys and Girls Performance on subtest Word Recall Meaningful

	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	5.16	1.89	6.03	2.17	2.26*	109
8 years	58	53	5.37	1.78	6.45	1.87	3.08**	109
9 years	55	70	5.87	1.99	6.52	1.77	1.94	123
10 years	53	55	6.58	1.75	7.36	1.72	2.32*	106
11 years	55	63	6.67	1.81	7.00	2.02	0.91	116

* p<0.05, ** p<0.01

14.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Word Recall Meaningful for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	5.16	1.90	0-9	5.42	2.55NS
58	8 years	5.38	1.79	1-10	5.53	
55	9 years	5.87	1.99	0-10	5.87	
53	10 years	6.58	1.76	3-10	6.31	
53	11 years	6.67	1.82	0-10	6.49	

df = 271, NS = Not significant

14.3 Shows the Mean differences for agewise comparison on Word Recall Meaningful for boys.

age group	8 years	9 years	10 years	11 years
7 years	-0.34	1.23	-2.18*	-2.78**
8 years		0.94	-2.00*	-2.60**
9 years			-1.20	-1.76
10 years				-0.51
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures, $df = 271$, $*p < .05$, $**p < 0.01$

14.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Word Recall Meaningful for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	6.04	2.18	0-10	6.27	
53	8 years	6.45	1.88	2-9	6.27	
70	9 years	6.53	1.78	2-10	6.63	2.41 NS
55	10 years	7.36	1.73	2-10	7.17	
63	11 years	7.00	2.02	0-10	6.98	

$df = 290$, NS = Not significant

14.5 Shows the Mean differences for agewise comparison on Word Recall Meaningful for girls

Age Group				
	8 years	9 years	10 years	11 years
7 years	0.01	1.03	-2.20*	-1.90
8 years		-0.99	-2.46*	-1.96*
9 years			-14.6	-1.02
10 years				0.53
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

$df = 290$. * $p < 0.05$

15.1 Comparison of Boys and Girls Performance on subtest Delayed Response Learning

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	of
7 years	56	55	2.28	1.03	2.12	1.18	0.74	109
8 years	58	53	2.25	1.25	2.56	1.20	1.31	109
9 years	55	70	2.45	1.21	2.56	1.03	1.00	123
10 years	53	55	3.28	0.94	2.60	1.32	3.06***	106
11 years	55	63	3.21	0.95	2.79	1.15	2.15*	116

15.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Delayed Response Learning for Boys

N=271	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	2.29	1.04	0-4	2.26	
58	8 years	2.26	1.25	0-4	2.24	
55	9 years	2.45	1.21	0-4	2.45	2.01NS
53	10 years	3.28	0.95	1-4	3.30	
55	11 years	3.22	0.96	1-4	3.23	

15.3 Shows the Mean differences for agewise comparison on Delayed Response Learning for Boys

	Age Group			
	8 years	9 years	10 years	11 years
7 years	0.08	-0.88	-4.28***	-4.19***
8 years		-1.00	-4.59***	-4.49***
9 years			-3.87***	-3.66***
10 years				0.34
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 271, *** $p < .001$

15.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Delayed Response Learning for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	2.13	1.19	0-4	2.37	
53	8 years	2.57	1.20	0-4	2.38	
70	9 years	2.66	1.03	0-4	2.76	2.01NS
55	10 years	2.60	1.33	0-4	2.40	
63	11 years	2.79	1.15	0-4	2.77	

df = 290, NS = Not significant

15.5 Shows the Mean differences for agewise comparison on Delayed Response Learning for Girls

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-0.04	-1.89	-0.15	-1.81
8 years		-1.73	-0.11	-1.79
9 years			1.62	-0.02
10 years				-1.68
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290

16.1 Comparison of Boys and Girls Performance on subtest Picture Recall

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	2.39	1.18	2.38	1.06	0.05	109
8 years	58	53	2.03	1.15	2.47	1.08	2.05*	109
9 years	55	70	2.74	0.94	2.32	1.90	2.23*	123
10 years	53	55	3.15	0.81	3.07	0.87	0.47	106
11 years	55	63	2.96	0.85	2.49	0.99	2.73*	116

16.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Picture Recall for Boys

N=271	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	2.39	1.19	0-4	2.56	
58	8 years	2.03	1.15	0-4	2.13	
55	9 years	2.75	0.95	1-4	2.74	
53	10 years	3.15	0.82	1-4	2.97	
55	11 years	2.96	0.86	0-4	2.85	

df = 271, * p<0.05

16.3 Shows the Mean differences for age-wise comparison on Delayed Response Learning for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	2.26*	-0.92	-1.86	-1.36
8 years		3.18**	-3.98***	-3.55***
9 years			-1.15***	-0.54
10 years				0.64
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df =271, * p<0.05, ** p<0.01, *** p< 0.001

16.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Picture Recall for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	2.38	1.06	1-4	2.57	
53	8 years	2.47	1.08	0-4	2.33	
70	9 years	2.33	1.10	0-4	2.41	2.84***
55	10 years	3.07	0.88	0-4	2.92	
63	11 years	2.49	1.00	0-4	2.47	

df = 290, *** p<0.001.

16.5 Shows the Mean differences for age-wise comparison on Picture Recall for Girls.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	1.13	0.85	-1.61	0.49
8 years		-0.45	-3.05**	-0.77
9 years			-2.60	-0.34
10 years				2.33*
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290, *p< .05, ** p<.01.

17.1 Comparison of Boys and Girls Performance on subtest Benton Visual Retention Test

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	df
7 years	56	55	4.28	1.54	4.89	1.78	1.91	109
8 years	58	53	5.10	1.65	5.28	1.97	0.52	109
9 years	55	70	6.01	1.73	5.24	1.71	2.49*	123
10 years	53	55	7.09	1.39	7.18	1.36	0.33	106
11 years	55	63	6.65	1.65	6.82	1.15	0.65	116

17.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Benton Visual Retention Test for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	4.29	1.55	1-8	4.56	
58	8 years	5.10	1.65	1-8	5.27	
55	9 years	6.02	1.74	3-9	6.01	12.40***
53	10 years	7.09	1.39	4-9	6.80	
55	11 years	6.65	1.66	3-10	6.46	

df = 271, *** p<0.001.

17.3 Shows the Mean differences for agewise comparison on Benton Visual Retention Test for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-2.38*	-4.65***	-6.37***	-5.70***
8 years		-2.45***	-4.58***	-3.75***
9 years			-2.48***	-1.47
10 years				1.10
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df =271, * p<0.05 , *** p<0 .001

17.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Benton Visual Retention * Test for girls.

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	4.89	1.78	0-8	4.95	
53	8 years	5.28	1.97	0-9	5.23	
70	9 years	5.24	1.71	1-9	5.27	20.45***
55	10 years	7.18	1.36	4-9	7.12	
63	11 years	6.83	1.16	4-10	6.82	

df = 290, *** p<0.001

17.5 Shows the Mean differences for agewise comparison on Benton Visual Retention Test for Girls.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-0.79	-1.06	-6.28***	-5.94***
8 years		-0.12	-6.09***	-5.16***
9 years			-5.97***	-5.43***
10 years				-1.01
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290, *** p<0.001.

18.1 Comparison of Boys and Girls Performance on subtest Paired Associate Learning

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	4.58	1.84	5.01	2.32	1.07	109
8 years	58	53	5.39	1.63	5.39	2.16	0.00	109
9 years	55	70	6.00	2.23	6.14	1.83	0.39	123
10 years	53	55	6.07	1.65	6.89	1.46	2.72**	106
11 years	55	63	7.07	1.54	6.33	1.65	2.49*	116

18.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Paired Associate Learning for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	5.13	4.30	0.0-16.5	5.63	
58	8 years	9.09	5.92	0.0-21.0	9.40	
55	9 years	9.24	4.42	0.0-17.0	9.23	14.31***
53	10 years	9.85	5.13	0.0-21.0	9.32	
55	11 years	13.67	4.77	3.5-20.5	13.33	

df = 271, *** p<0.001

18.3 Shows the Mean, differences for agewise comparison on Paired Associate Learning for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-2.00	2.99**	-1.88	-5.03***
8 years		-1.11	-0.19	-3.40***
9 years			0.85	-2.43*
10 years				3.33***
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 271, * $p < .05$, ** $p < .01$, *** $p < .001$

18.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Paired Associate Learning for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	6.55	5.43	0-18.5	7.76	
53	8 years	11.58	6.15	0-20	10.67	
70	9 years	10.90	6.03	0-21	11.45	7.44***
55	10 years	11.89	6.11	0-21	10.94	
63	11 years	13.60	5.02	1.5-21.0	13.51	

df = 290, *** $p < 0.001$

18.5 Shows the Mean differences for agewise comparison on Paired Associate Learning for Girls.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	0.83	-2.67**	-2.90**	-2.44*
8 years		3.44***	-4.15***	-3.43***
9 years			-0.69	0.07
10 years				0.78
11 years				

@ Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

19.1 Comparison of Boys and Girls Performance on subtest Cattell's Retentivity Test

N=573	Boys	Girls	Mean	S.D	Mean	S.D	t-value
7 years	56	55	5.13	4.30	6.54	5.43	1.51
8 years	58	53	9.09	5.91	11.58	6.15	2.17
9 years	55	70	9.32	4.42	10.90	6.03	1.71
10 years	53	55	9.84	5.12	11.89	6.10	1.87
11 years	55	63	13.67	4.76	13.60	5.02	0.07

** $p < 0.05$

19.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean Cattell's Retentivity Test for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	4.59	1.85	0-8	4.95	
58	8 years	5.40	1.63	0-8	5.62	
55	9 years	6.08	1.65	2-9	5.69	7.05***
53	10 years	4.00	0.85	0-5	3.83	
55	11 years	7.07	1.55	4-10	6.82	

df = 271, *** p<0.001

19.3 Shows the Mean differences for agewise comparison on Cattell's Retentivity Test for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-4.05	-3.68* **	-3.36***	-7.39 ***
8 years		0.18	0.07	-3.96 ***
9 years			-0.09	-4.28***
10 years				-.4.20***
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df=271, *** p<.001.

19.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Cattell's Retentivity Test for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	5.02	2.32	0-10	5.42	
53	8 years	5.40	2.16	0-9	5.09	
70	9 years	6.14	1.84	0-9	6.32	6.76***
55	10 years	6.89	1.46	3-9	6.57	
63	11 years	6.33	1.66	0-9	6.30	

df = 290, *** p<0.001

19.5 Shows the Mean differences for agewise comparison on Cattell's Retentivity Test for Girls.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-2.42***	-3.60***	-2.65***	-5.28***
8 years		-0.41***	-0.24	-2.65**
9 years			0.47	-2.07*
10 years				-2.42*
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290, * p<.05, **p< .01, *** p< .001

20.1 Comparison of Boys and Girls Performance on Total Memory Score

	Boys	Girls	Mean	S.D	Mean	S.D	t-value	Df
7 years	56	55	64.49	15.61	72.98	17.88	2.35*	109
8 years	58	53	70.09	19.38	83.77	21.71	3.50**	109
9 years	55	70	85.52	20.16	84.50	18.86	0.29	123
10 years	53	55	90.75	14.02	95.00	17.63	1.38	106
11 years	55	63	99.98	16.13	94.17	15.53	1.98*	116

20.2 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Total Memory Score for Boys

N=277	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
56	7 years	65.49	15.61	28.5-96.5	70.27	
58	8 years	70.09	19.38	13.0-102.0	73.03	
55	9 years	85.53	20.16	38.5-125.0	18.06**	
53	10 years	90.75	14.02	47.0-123.0	85.80	
55	11 years	99.98	16.13	61.0-126.5	96.79	

df = 271, *** p<0.001

20.3 Shows the Mean differences for agewise comparison on Total Memory Score for boys.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-0.88	-4.66***	-4.23***	-7.61***
8 years		-3.94***	-3.66***	-7.16***
9 years			-0.09	-3.52***
10 years				-3.44***
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df =271 *** p< 0.001

20.4 Shows the Mean, S.Ds, F-value, Ranges and Adjusted Mean on Total Memory Score for Girls

N=296	Age Group	Mean	S.D	Range	Adjusted Mean	F-value
55	7 years	72.98	17.88	39-127	78.98	
53	8 years	83.77	21.72	46-117	79.30	
70	9 years	84.50	18.87	52-122	87.23	6.76***
55	10 years	95.00	17.64	55-125	90.34	
63	11 years	94.17	15.54	53-126	93.71	

df = 290, *** p<0.001

20.5 Shows the Mean differences for agewise comparison on Total Memory Score for Girls.

	Age Group			
	8 years	9 years	10 years	11 years
7 years	-0.08	-2.62**	-3.08**	-4.42***
8 years		-2.38*	-3.33***	-4.40***
9 years				-2.13**
10 years				-1.04
11 years				

Tests on all pair wise comparisons using Fisher's Protected LSD procedures.

df = 290, * $p < .05$, ** $p < .01$, *** $p < .001$

PERCENTILE NORMS
Percentile Norms for each subtest for the total sample of the age groups 7-11 year

Subtest	PR	0	10	20	30	40	50	60	70	80	90	100
Ages												
I. Personal Information	7	0	2	-	-	3	-	-	-	4	-	5
	8	0	2	-	3	-	-	-	4	-	-	5
	9	0	3	-	-	-	4	-	-	-	-	5
	10	0	3	-	-	-	-	4	-	-	-	5
	11	0	3	-	-	4	-	-	-	-	-	5
II. Mental Control	7	0	-	6	7	-	-	-	-	9	10	12
	8	0	6	7	-	-	9	10	-	-	12	15
	9	0	6	-	9	10	-	-	12	13	-	15
	10	0	9	10	11	-	12	-	-	-	-	15
	11	0	9	10	-	12	13	-	-	-	-	15
III. Sentence Repetition	7	-	3	4	5	6	8	-	-	-	-	9
	8	0	2	3	4	-	5	-	6	-	-	9
	9	0	3	5	6	7	0	0	0	0	0	9
	10	0	5	7	-	-	-	-	-	-	-	9
	11	0	4	7	-	-	-	-	-	-	-	9
IV. Story Recall Immediate	7	0	0	1	-	2	3	4	5	6	8	9
	8	0	0	2	2	3	4	5	6	8	9	10
	9	0	2	3	4	5	6	7	8	10	11	12
	10	0	2	3	4	5	-	6	7	9	10	11
	11	0	2	3	5	4	6	7	8	10	11	12
V. Story Recall Delayed	7	0	0	1	-	2	3	4	5	6	8	9
	8	-	-	0	2	3	4	5	7	8	10	11
	9	-	0	2	4	5	6	7	9	10	12	13
	10	0	2	3	4	5	6	8	9	-	10	13
	11	0	3	4	5	6	8	-	10	11	12	14

Subtest	PR	0	10	20	30	40	50	60	70	80	90	100
Ages												
VI. Word Recall	7	-	4	5	6	-	7	8	-	9	-	10
Meaningful	8	0	4	-	5	6	7	-	8	9	-	10
	9	0	4	-	5	6	7	8	-	9	-	10
	10	0	4	-	5	6	7	8	-	9	-	10
	11	-	4	5	6	7	-	8	-	9	-	10
VII. Digit	7	0	4	5	6	-	-	-	7	-	-	8
Forward	8	0	5	-	6	-	-	-	-	7	-	8
	9	0	5	-	-	6	-	-	-	-	7	8
	10	0	4	5	-	-	-	-	7	-	-	8
	11	0	5	-	-	-	6	-	7	-	-	8
VIII. Digit												
Backward	7	0	2	-	3	-	-	4	-	-	5	6
	8	0	-	3	-	-	4	-	-	-	5	6
	9	9	0	2	3	-	-	4	-	-	-	5
	10	0	2	3	-	-	-	4	-	-	-	5
	11	0	2	3	-	-	8	-	-	-	5	6
IX. Word Recall												
Nonmeaningful	7	0	3	4	-	-	5	6	-	7	8	9
	8	0	-	4	-	5	-	6	7	8	-	9
	9	0	4	-	5	-	6	-	7	8	-	9
	10	0	5	-	-	6	7	8	-	-	9	10
	11	0	-	5	6	-	7	-	-	8	9	10
X. Delayed												
Response												
Learning	7	0	-	1	-	-	-	2	-	3	-	4
	8	0	-	1	-	-	2	-	3	-	-	4
	9	0	1	-	-	2	-	-	3	-	-	4
	10	0	1	-	2	-	3	-	-	-	-	4
	11	0	-	-	2	-	3	-	-	-	-	4

Subtest	PR	0	10	20	30	40	50	60	70	80	90	100
Ages												
XI Picture Recall	7	0	-	1	-	2	-	-	-	2	-	4
	8	0	-	1	-	-	2	-	-	3	-	4
	9	0	-	1	-	2	-	-	-	3	-	4
	10	0	2	-	-	-	-	3	-	-	-	4
	11	0	1	-	2	-	-	-	3	-	-	4
XII Bentons	7	0	2	3	-	4	-	5	-	6	-	7
visual	8	0	3	-	4	-	5	-	6	-	7	8
Retention	9	0	3	4	-	5	-	6	-	7	-	8
Test	10	0	5	6	-	7	-	-	-	8	-	9
	11	0	5	-	-	-	-	-	7	8	-	9
XIII Palred												
Associate	7	0	0	1	2	3	5	5	7	10	13	15
Learning	8	0	3	5	7	8	11	13	-	15	17	18
	9	0	3	5	6	8	9	12	13	14	16	18
	10	0	5	5	8	9	11	12	14	16	19	9
	11	0	5	8	10	12	14	15	17	18	19	19
XIV Cattell's												
retentivity test	7	0	1	-	4	-	-	5	-	6	7	8
	8	0	3	-	4	-	-	5	6	-	7	8
	9	0	4	-	5	-	6	-	7	-	8	9
	10	0	4	5	-	6	-	-	7	8	-	9
	11	0	-	5	-	6	-	-	7	8	-	9
XV Total	7	0	46	53	58	63	68	73	78	85	94	99
Memory	8	0	47	54	63	72	76	82	89	97	106	110
	9	0	58	66	74	80	85	91	97	103	109	115
	10	0	72	80	85	89	93	97	102	108	114	118
	11	0	74	80	90	95	99	102	107	110	117	122

22.1 Shows the correlation of memory Data of boys with C.P.M. Scores

Subtests	Scores	p-values
Personal Information	0.31***	P<0.001
Mental Control	0.44***	P<0.001
Sentence Repetition	0.30**	P<0.01
Story Recall Immediate	0.37***	P<0.001
Story Recall Delayed	0.45***	P<0.001
Word Recall Meaningful	0.23*	P<0.05
Digit Span Forward	0.19*	P<0.05
Digit Span Backward	0.16	N.S.
Word Recall Non Meaningful	0.29**	P<0.01
Delayed Response Learning	0.19*	P<0.05
Picture Recall	0.33***	P<0.001
Bentons Visual Retention Test	0.43***	P<0.001
Cattell's Retentivity Test	0.36***	P<0.001
Paired Associate Learning	0.29**	P<0.01
Total Memory Score	0.51***	P<0.001

N= 277, df=275 NS= not significant.

22.2 Shows the correlation of memory Data of Girls with C.P.M. Scores

Subtests	Scores	p-values
Personal Information	0.14	N.S
Mental Control	0.27**	P<0.01
Sentence Repetition	0.20*	P<0.05
Story Recall Immediate	0.29**	P<0.01
Story Recall Delayed	0.30**	P<0.01
Word Recall Meaningful	0.22**	P<0.01
Digit Span Forward	0.09	N.S
Digit Span Backward	0.26**	P<0.01
Word Recall Non Meaningful	0.19*	P<0.05
Delayed Response Learning	0.24**	P<0.01
Picture Recall	0.25**	P<0.01
Bentons Visual Retention Test	0.19**	P<0.05
Cattell's Retentivity Test	0.27**	P<0.01
Paired Associate Learning	0.31**	P<0.01
Total Memory Score	0.42***	P<0.001

N = 296, df=294 NS= Not significant.

22.3 Shows the Correlation of Memory with C.P.M. Scores

Subtests	Scores	p-values
Personal Information	0.22**	P<0.01
Mental Control	0.36**	P<0.01
Sentence Repetition	0.24**	P<0.01
Story Recall Immediate	0.33**	P<0.01
Story Recall Delayed	0.36**	P<0.01
Word Recall Meaningful	0.21**	P<0.01
Digit Span Forward	0.15**	P<0.01
Digit Span Backward	0.21**	P<0.01
Word Recall Non Meaningful	0.22**	P<0.01
Delayed Response Learning	0.21**	P<0.01
Picture Recall	0.29**	P<0.01
Bentons Visual Retention Test	0.31**	P<0.01
Cattell's Retentivity Test	0.31**	P<0.01
Paired Associate Learning	0.28**	P<0.01
Total Memory Score	0.45**	P<0.01

N= 573, df=571

22.4 Shows the Correlation between Subscales and Total Score with C.P.M. Scores for the Clinical Sample

Subtests	Scores	p-values
Personal Information	0.26*	P<0.05
Mental Control	0.24*	P<0.05
Sentence Repetition	0.11	N.S
Story Recall Immediate	0.07	N.S
Story Recall Delayed	0.05	N.S
Word Recall Meaningful	0.09	N.S
Digit Span Forward	0.16	N.S
Digit Span Backward	0.24*	P<0.05
Word Recall Non Meaningful	0.31*	P<0.05
Delayed Response Learning	0.30**	P<0.05
Picture Recall	0.00	N.S
Bentons Visual Retention Test	0.23*	P<0.05
Cattell's Retentivity Test	0.14	N.S
Paired Associate Learning	0.05	N.S
Total Memory Score	0.30	N.S

N= 75, df=73, NS= Not significant.

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Chapter 12

CONCLUDING REMARKS

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It is imperative that we point out to the deficiencies which may be encountered in this monograph.

1. First and foremost, these tests represent the work done only at NIMHANS. There may be several excellent tests developed elsewhere by other eminent workers. This monograph is meant to represent the work in the Department of Clinical Psychology in the area of Child and Adolescent Mental Health.
2. These do not cover the entire gamut of tests developed including objective and projective tests of personality. It is hoped that such a monograph will be put together at a later date.
3. No claim has been made that the tests have adequate standardisation. The work has been carried out under constraints of manpower, time and resources.
4. The tests which are copyrighted such as WISC, MISIC, BVRT, DTVMI, Cattell's Retentivity and BGT tests have not been included in the illustrations. They have to be purchased from the respective agencies/authors dealing with psychological tests.
5. In all the tests developmental trends are evident be it attention, memory or learning problems. For example, comparison of profiles on various tests at different age (grade) levels and their intercorrelations would form a fascinating area of study as well as

research. This is yet to be explored by the authors and being saved for future publications.

6. In the Indian setting it has been observed that children from slum, rural or government schools may show deficit profiles similar to those with SLD. But these are due to poor school system rather than defects inherent in the child.
7. It is recommended that those who wish to use the tests should undergo brief training at NIMHANS in the use of these tests.
8. In this monograph not only assessment but also remediation is described. This is not a convention which is followed in the books on assessment. It has been done especially for use in the clinical setting. It is hoped that as there are hardly any remedial packages available in the Indian context, this monograph would be of some help.
9. The approach is holistic and developmental and takes into account the psychosocial context in which the child faces a variety of problems.
10. Lastly, the monograph only provides a frame work and makes no claims of comprehensive coverage of the topics.

This being only the beginning of our efforts, we would like to see great deal of work to be carried out in the future by us and others a like. These are some suggestions for further work.

- (i) The DPCL, especially temperament subsection needs to be replicated across age and gender in normal children.
- (ii) The clinical validation of the specific learning disability test developed for older children needs to be established.

- (iii) SLD material in the regional languages are required especially in the provincial areas of the different states. The remedial work in the regional Indian languages may be very different than what is carried out for English Language.
- (iv) Tests of memory may be standardised for different clinical groups and the disabled groups. Remedial work for memory needs to be evolved and evaluated for children across ages from a developmental perspective.
- (v) There is a need to study the relation between the memory profile and information processing across ages, from a developmental perspective.
- (vi) The utility of NIMHANS sentence completion test (NSCA) and My experience scale in assessing adjustment in different clinical conditions/paediatric population can be further studied.
- (vii) The resilience test need to be administered in a large sample of children.
